Clinical Study Protocol

PHASE 2 DOSE-RANGING AND INTERCEPTION STUDY OF LINVOSELTAMAB IN PATIENTS WITH HIGH-RISK MONOCLONAL GAMMOPATHY OF UNDETERMINED SIGNIFICANCE OR NON-HIGH-RISK SMOLDERING MULTIPLE MYELOMA

Compound:	Linvoseltamab (REGN5458)
Study Name:	LINKER-MGUS1
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Medical Director/Clinical Scientist:	Shawn Sarkaria, MD Medical Director, Clinical Sciences Hematology Clinical Development James Drew, MA Director, Clinical Sciences Hematology Clinical Development Regeneron Pharmaceuticals, Inc. 777 Old Saw Mill River Road Tarrytown, NY 10591

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LIST OF ABBREVIATIONS AND DEFINITIONS OF TERMS

ADA	Antidrug antibody
ADL	Activities of daily living
AE	Adverse event
AEI	Adverse event of interest
AESI	Adverse event of special interest
ALT	Alanine aminotransferase
aPTT	Activated partial thromboplastin time
AST	Aspartate aminotransferase
ASTCT	American Society for Transplantation and Cellular Therapy
AxMP	Auxiliary Medicinal Product
BAFF	B-cell activation factor
BCMA	B-cell maturation antigen
β-hCG	β-human chorionic gonadotropin
BiPAP	Bilevel positive airway pressure
BLyS	B lymphocyte stimulator
BMPC	Bone marrow plasma cell
BOIN	Bayesian optimal interval
BP	Biochemical progression
BUN	Blood urea nitrogen
C1D1	Cycle 1 day 1
C3D1	Cycle 3 day 1
CAR	Chimeric antigen receptor
CARTOX-10	CAR T-cell therapy-associated toxicity – 10-point neurological assessment
CD3	Cluster of differentiation 3
CD4	Cluster of differentiation 4
CD8	Cluster of differentiation 8
CD25	Cluster of differentiation 25
CD38	Cluster of differentiation 38
CD69	Cluster of differentiation 69
C_{eoi}	Concentration at the end of infusion
CI	Confidence interval
C_{max}	Maximum (peak) concentration
CMV	Cytomegalovirus
CNS	Central nervous system

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COVID-19	Coronavirus disease 19
CPAP	Continuous positive airway pressure
CR	Complete response
CRAB	Hyper**C**alcemia, **R**enal insufficiency, **A**nemia, or lytic **B**one lesions
CRF	Case report form (electronic or paper)
CRO	Contract research organization
CRP	C-reactive protein
CRS	Cytokine release syndrome
CSR	Clinical study report
CT	Computed tomography
CTC	Circulating tumor cell
CTCAE	Common Terminology Criteria for Adverse Events
CTFG	Clinical Trial Facilitation Group
C_{trough}	Lowest concentration in a dosing interval
CYP450	Cytochrome P450
DL0	Dose level 0
DL1	Dose level 1
DL2	Dose level 2
DNA	Deoxyribonucleic acid
DOR	Duration of response
EBV	Epstein-Barr virus
EC	Ethics Committee
ECG	Electrocardiogram
Echo	Echocardiogram
eCOA	Electronic Clinical Outcome Assessment Consortium
ECOG	Eastern Cooperative Oncology Group
EDC	Electronic data capture
EOI	End of infusion
EOT	End of treatment
EORTC QLQ-C30	European Organization for Research and Treatment of Cancer Quality of Life Questionnaire-Core 30
EORTC QLQ-MY20	European Organization for Research and Treatment of Cancer Quality of Life Questionnaire-Multiple Myeloma Module 20
EQ-5D-5L	EuroQol 5 Dimensions 5 Levels
EQ-VAS	EuroQol Visual Analog Scale
EU	European Union
FACIT Item GP5	Functional Assessment of Chronic Illness Therapy Item GP5

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FAS	Full analysis set
FBR	Future biomedical research
FDA	Food and Drug Administration
FDG	^{18}F-fluorodeoxyglucose
FIH	First-in-human
FISH	Fluorescent in situ hybridization
FLC	Free light chain
FSH	Follicle-stimulating hormone
GCP	Good Clinical Practice
G-CSF	Granulocyte colony-stimulating factor
GFR	Glomerular filtration rate
GPRC5D	G-protein-coupled receptor, class C, group 5, member D
HBsAg	Hepatitis B surface antigen
HBV	Hepatitis B virus
HCV	Hepatitis C virus
HHV-6	Human herpesvirus 6
HIV	Human immunodeficiency virus
HLA-DR	Human leukocyte antigen-DR isotype
HR-MGUS	High-risk monoclonal gammopathy of undetermined significance
HRQoL	Health-related quality of life
IB	Investigator's brochure
ICANS	Immune effector cell-associated neurotoxicity syndrome
ICE	Immune effector cell-associated encephalopathy
ICF	Informed consent form
ICH	International Council for Harmonisation
IDSA	Infectious Diseases Society of America
IEC	Institutional Ethics Committee
IFN	Interferon
IgA	Immunoglobulin A
IgD	Immunoglobulin D
IgE	Immunoglobulin E
IgG	Immunoglobulin G
IgG1	Immunoglobulin G subclass 1
IgG2	Immunoglobulin G subclass 2
IgG3	Immunoglobulin G subclass 3
IgG4	Immunoglobulin G subclass 4
IgM	Immunoglobulin M

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IHC	Immunohistochemistry
IL-2	Interleukin-2
IL-6	Interleukin-6
IL-10	Interleukin-10
IMP	Investigational medicinal product
IMWG	International Myeloma Working Group
INR	International normalized ratio
IRR	Infusion-related reaction
IUD	Intrauterine device
IUS	Intrauterine hormone-releasing system
IV	Intravenous
LAM	Lactational amenorrhea method
LDH	Lactate dehydrogenase
MDRD	Modification of diet in renal disease
MedDRA	Medical Dictionary for Regulatory Activities
MGUS	Monoclonal gammopathy of undetermined significance
MM	Multiple myeloma
MR	Minimal response
MRD	Minimal residual disease
MRI	Magnetic resonance imaging
MS	Mass spectrometry
MUGA	Multigated acquisition scan
N/A	Not applicable
NCCN	National Comprehensive Cancer Network
NCI-CTCAE	National Cancer Institute-Common Terminology Criteria for Adverse Events
NGS	Next-generation sequencing
NHR-SMM	Non-high-risk smoldering multiple myeloma
NK	Natural killer
NYHA	New York Heart Association
ORR	Overall response rate
PCR	Polymerase chain reaction
PCSV	Potentially clinically significant value
PD	Progressive disease
PET/CT	Positron emissions tomography/computed tomography
PFS	Progression-free survival
PJP	*Pneumocystis jirovecii* pneumonia
PK	Pharmacokinetic

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PO	Oral
PR	Partial response
PRO	Patient-reported outcome
PT	Prothrombin time
PTT	Partial thromboplastin time
Q1	Lower quartile
Q3	Third quartile
QW	Weekly
Q2W	Every 2 weeks
Q4W	Every 4 weeks
Q12W	Every 12 weeks
Q24W	Every 24 weeks
QLQ-C30	Quality of Life Questionnaire-Core 30
QLQ-MY20	Quality of Life Questionnaire-Multiple Myeloma Module 20
QoL	Quality of life
QTcF	Heart rate-corrected QT interval (Fridericia's correction formula)
RANKL	Receptor activator of nuclear factor kappa-B ligand
RBC	Red blood cell
RBQM	Risk-Based Quality Management
Regeneron	Regeneron Pharmaceuticals, Inc.
RNA	Ribonucleic acid
RRMM	Relapsed/refractory multiple myeloma
SAE	Serious adverse event
SAF	Safety analysis set
SAP	Statistical analysis plan
SAS	Statistical Analysis System
sBCMA	Soluble B-cell maturation antigen
sCR	Stringent complete response
SD	Stable disease
SDR	Source data review
SDV	Source data verification
SEQ	Sequencing
sFLC	serum free light chain
SI	International System of Units
SIFE	Serum immunofixation electrophoresis
SLAMF7	Signaling lymphocytic activation molecule family member 7

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SliM	Greater than or equal to **S**ixty percent clonal plasma cells in the bone marrow, involved/uninvolved free **Li**ght chain ratio of ≥100 with the involved free light chain being ≥100 mg/L, **M**agnetic resonance imaging with > 1 focal lesion
SMM	Smoldering multiple myeloma
SMQ	Standardized Medical Dictionary for Regulatory Activities (MedDRA) query
SOC	System organ class
SPEP	Serum protein electrophoresis
SSET	Study safety evaluation team
SSRE	Symptomatic skeletal-related event
SUSAR	Suspected unexpected serious adverse reaction
TEAE	Treatment-emergent adverse event
trSAE	Treatment-related serious adverse event
TLS	Tumor lysis syndrome
TNF	Tumor necrosis factor
UIFE	Urine immunofixation electrophoresis
ULN	Upper limit of normal
UPEP	Urine protein electrophoresis
US	United States
VGPR	Very good partial response
W1D1	Week 1 day 1
WBC	White blood cell
WES	Whole-exome sequencing
WGS	Whole-genome sequencing
WOCBP	Women of child-bearing potential

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CLINICAL STUDY PROTOCOL SYNOPSIS

Title	Phase 2 Dose-Ranging and Interception Study of Linvoseltamab in Patients with High-Risk Monoclonal Gammopathy of Undetermined Significance or Non-High-Risk Smoldering Multiple Myeloma
Site Location(s)	Approximately 40 global sites
Objective(s)	
Primary:	Safety Run-In (Part 1): To evaluate the safety and tolerability of linvoseltamab in ascending dose-level cohorts and select the highest dose level for investigation in the expansion part Expansion (Part 2): To evaluate the ability of linvoseltamab to induce complete responses (CR) by dosing regimen
Secondary:	The secondary objectives of the study are applicable to both safety run-in (part 1) and expansion (part 2) unless otherwise specified: • To characterize the safety of linvoseltamab • To evaluate the proportion of participants that achieve minimal residual disease (MRD) negativity • To evaluate the proportion of participants with sustained MRD negativity • To evaluate the overall response rate (ORR) • To evaluate the duration of response (DOR) • To evaluate biochemical progression-free survival (PFS) • To characterize the pharmacokinetic (PK) properties of linvoseltamab • To assess the immunogenicity of linvoseltamab
Study Design	This is a phase 2, open-label, dose-ranging, proof-of-concept study designed to evaluate the safety and efficacy of linvoseltamab monotherapy in patients with high-risk monoclonal gammopathy of undetermined significance (HR-MGUS) or non-high-risk smoldering multiple myeloma (NHR-SMM). Because linvoseltamab will be evaluated for the first time in this population, enrollment will begin with a safety run-in and dose-escalation component (part 1) to assess the suitability of the step-up schedule and select the highest dose level (≤100 mg) appropriate for expansion. After preliminary safety and tolerability are established in part 1, a dose-ranging and expansion component (part 2) will open to further characterize the safety and evaluate the clinical activity of linvoseltamab when administered according to a randomly assigned dosing regimen (up to 2 dose levels and 2 dosing schedules).

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	Safety Run-in (Part 1): Sequential cohorts of 3 to 6 participants will be enrolled according to a Bayesian optimal interval (BOIN) dose-escalation design. Tolerability will be assessed by the incidence of adverse events of interest (AEIs) during a 35-day safety observation period. Each participant will start IV linvoseltamab treatment with a 3-week step-up schedule consisting of weekly dose increments of 1 mg (week 1), 4 mg (week 2), and 25 mg (week 3). After completion of the step-up regimen, the assigned full dose of linvoseltamab will be given over 28-day treatment cycles for a fixed duration of 6 cycles. Dosing will occur once weekly (QW) for the first cycle and then transition to once every 4 weeks (Q4W) for the next 5 cycles. The initial dose level evaluated will be 50 mg (dose level 1 [DL1]), and the highest dose level will not exceed 100 mg (dose level 2 [DL2]). Depending upon the safety assessment of the initial dose level tested, additional patients may be enrolled at a higher or lower dose level to determine a tolerable full dose of linvoseltamab for the dose expansion portion of the study. Modified full dose levels will be implemented if emerging safety issues warrant a smaller increment of dose escalation following a completed dose level. **Expansion (Part 2):** Once acceptable safety and tolerability are demonstrated, a minimum of 80 participants will be randomized in a 1:1:1:1 ratio across 4 dosing regimens comprising 2 full doses (25 mg [low dose] or the highest tolerated full dose in part 1 [high dose]) given according to 2 dosing schedules (schedule 1 or schedule 2) to identify a dosing regimen that minimizes the burden of treatment and optimizes safety and efficacy. Each dosing regimen will begin with the same 3-week step-up schedule established in the safety run-in, followed by a fixed number of 28-day treatment cycles with full dose linvoseltamab. Schedule 1 mirrors the dosing schedule used in part 1: QW for the first cycle and then Q4W for the next 5 cycles. Schedule 2 consists of biweekly dosing (once every 2 weeks; Q2W) for a total of 3 cycles. Randomization will be stratified according to number of 20-2-20 risk factors (0 or 1). Over the entire treatment period, starting with the first infusion of linvoseltamab, all participants enrolled in part 1 or part 2 will be monitored in aggregate for cumulative toxicity that meets criteria for enrollment pause and/or safety review (cumulative rate of trigger events).
Study Duration	For each participant, the study will span the following 3 periods: **Screening Period:** Screening procedures may occur over a period of up to 28 days for each study participant, ending on the day preceding the planned first dose of linvoseltamab. **Treatment Period:** The treatment period will start on the day when the first step-up dose of linvoseltamab is administered and continues until completion of the end-of-treatment (EOT) visit or, if the EOT visit does not occur, 30 days after the last dose of linvoseltamab is administered. Treatment may continue

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	through the last scheduled dose in the assigned dosing regimen or until fulfillment of a protocol-defined criterion for study drug discontinuation. **Follow-up (up to 5 years after last dose):** • **Post-treatment follow-up:** Post-treatment follow-up will start after the EOT visit and involves ongoing collection of safety and efficacy data. Safety monitoring will continue for up to 90 days after the last linvoseltamab dose. Participants with an ongoing biochemical response will continue to undergo routine efficacy assessments until confirmed biochemical progression, start of non-protocol cancer therapy, or death. • **Long-term follow-up:** Long-term follow-up will begin after post-treatment follow-up to collect survival status as well as other relevant long-term outcomes (eg, progression to MM, subsequent non-protocol cancer therapy) and will continue for up to 5 years after the last dose of linvoseltamab, unless death or study termination by the sponsor occurs earlier. Participants may be followed remotely for survival and subsequent treatment history.
End of Study Definition	An individual participant will reach the end of study whenever one of the following events occurs: completion of 5 years of follow-up after the last dose of study drug, death, loss to follow-up, or withdrawal of consent. The study/trial will end on the date of the last participant's last visit for any reason listed above or study termination by the sponsor, whichever occurs earlier.
Population	
Sample Size:	Approximately 104 participants including 6 to 24 participants in part 1 (safety run-in) and a minimum of 80 participants in part 2 (expansion).
Target Population:	Adult patients diagnosed with HR-MGUS or NHR-SMM: **HR-MGUS** defined as <10% bone marrow plasma cells (BMPCs), <3 g/dL serum M-protein, and at least 2 of the following 3 criteria: abnormal serum free light chain (sFLC) ratio of <0.26 or >1.65, serum M-protein ≥1.5 g/dL, or non-IgG M-protein. In the case of light-chain MGUS, involved to uninvolved sFLC ratio >10 **NHR-SMM** defined as 1 or more of the following: • Serum M-protein ≥3 g/dL • Urinary M-protein ≥500 mg per 24 hours • BMPCs ≥10% **AND**

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No evidence of myeloma-defining events or other related conditions (SLiM CRAB*)

*SLiM (greater than or equal to **S**ixty percent clonal plasma cells in the bone marrow, involved/uninvolved free **Li**ght chain ratio of ≥100 with the involved free light chain (FLC) being ≥100 mg/L, **M**agnetic resonance imaging with >1 focal lesion) CRAB (hyper**C**alcemia, **R**enal insufficiency, **A**nemia, or lytic **B**one lesions)

No evidence of high-risk SMM defined as presence of any two of the following:

- Serum M-protein >2 g/dL
- Involved to uninvolved sFLC ratio >20
- BMPCs >20%

OR

- Total score of >9 points using the following scoring system:
  - FLC ratio
    - >10–25 = 2 points
    - >25–40 = 3 points
    - >40 = 5 points
  - Serum M-protein (g/dL)
    - >1.5–3 = 3 points
    - >3 = 4 points
  - BMPC%
    - >15–20 = 2 points
    - >20–30 = 3 points
    - >30–40 = 5 points
    - >40 = 6 points
  - Fluorescence in situ hybridization (FISH) abnormality (t(4;14), t(14;16), 1q gain, or del(13q)) = 2 points

Treatment(s)

Study Drug Linvoseltamab

Dose/Route/Schedule: Each participant will start intravenous (IV) linvoseltamab treatment with a 3-week step-up schedule consisting of weekly dose increments of 1 mg (week 1), 4 mg (week 2), and 25 mg (week 3).

A range of full doses from 25 to 100 mg will be explored according to the following fixed-duration dosing schedules:

- **Schedule 1 (Part 1 and Part 2):** Assigned dose given QW for the first cycle followed by Q4W for 5 cycles
- **Schedule 2 (Part 2):** Assigned dose given Q2W for 3 cycles

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Endpoint(s)	
Primary:	
Safety Run-In (Part 1)	• Frequency of AEI assessed during the safety observation period • Frequency and severity of treatment-emergent adverse events (TEAEs) assessed during the safety observation period
Expansion (Part 2)	• Achievement of CR as determined by the investigator using International Myeloma Working Group (IMWG) response criteria
Secondary:	The secondary endpoints of the study are applicable to both safety run-in (part 1) and expansion (part 2) unless otherwise specified: • Frequency and severity of TEAEs/serious adverse events (SAEs) and laboratory abnormalities • MRD negativity among participants that achieve a response of CR • Sustained MRD negativity on an annual basis for up to 3 years after achievement of CR • Overall response of partial response (PR) or better as determined by the investigator using IMWG response criteria • DOR as determined by the investigator using IMWG criteria • Biochemical PFS as determined by the investigator using IMWG criteria • Concentration of linvoseltamab in serum over time • Incidence and titer of antidrug antibodies (ADAs) to linvoseltamab over time
Procedures and Assessments	Screening: Medical/oncology history, demographics, complete physical examination, height, β2-microglobulin, echocardiogram (Echo) or multigated acquisition scan (cardiac ejection fraction; MUGA), viral serologies (human immunodeficiency virus [HIV], hepatitis B virus [HBV], hepatitis C virus [HCV]), coagulation parameters (prothrombin time [PT]/international normalized ratio [INR] and activated partial thromboplastin time [aPTT]/partial thromboplastin time [PTT]), and whole-body imaging. Safety: Vital signs, weight, physical examination, brief neurologic examination, electrocardiogram (ECG), Eastern Cooperative Oncology Group (ECOG) performance status, laboratory assessments, adverse events (AEs), and concomitant medications. Efficacy: Serum protein electrophoresis and immunofixation (SPEP/SIFE), 24-hour urine collection for urine protein electrophoresis and immunofixation

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	(UPEP/UIFE), serum FLC testing, immunoglobulin (Ig) quantification (IgA, IgM, IgG, IgD, IgE), and bone marrow aspirate/biopsy. Additional biomarker assessments and patient-reported outcome (PRO) measurements. Blood samples for drug concentration analysis and ADA detection in serum.
Statistical Plan	The statistical plan may be revised during the study to accommodate amendments to the clinical study protocol if necessary.
Statistical Hypothesis:	This is an exploratory study to evaluate the safety and preliminary clinical activity of linvoseltamab, and therefore, no formal statistical hypothesis testing is planned.
Justification of Sample Size:	Up to 24 participants will be enrolled during the safety run-in part of the study, and a minimum of 80 participants will be enrolled in the expansion part. Results from a minimum of 20 participants per treatment arm, treated with the selected dosing regimen and having at least 1 post-baseline response assessment, can estimate the CR rate with reasonable precision and explore preliminary clinical activity, safety, PK properties, and biomarker changes associated with the dosing regimen.
Efficacy Analyses	The full analysis set (FAS) will include all participants who received at least 1 dose (any amount) of study drug and had at least 1 post-baseline response assessment. Efficacy endpoints will be analyzed using the FAS. Participants will be classified according to the dosing regimen assigned at study entry. **Primary efficacy analyses:** The primary endpoint for the expansion portion (part 2) is achievement of CR as determined by the investigator using IMWG criteria. The CR rate is defined as the proportion of participants with a best response of CR by IMWG criteria. For the primary analysis, participants with inevaluable responses will be considered as non-responders. The point estimate and exact 95% confidence interval (CI) will be provided using the Clopper-Pearson method. **Secondary efficacy analyses:** The rate of MRD negativity is defined as the proportion of participants who achieve post-baseline MRD negativity (assessed using bone marrow aspirate at $\leq 10^{-5}$ sensitivity) among participants with CR or better based on FAS. MRD negativity rate at 12 months after the first linvoseltamab dose will be summarized. ORR is defined as the proportion of treated participants who achieve a best overall response of PR or better. The ORR and the corresponding 95% CI will be calculated using the Clopper-Pearson method. DOR is defined for responders (participants who achieve PR or better) only. It is the time from the date of the first documented response (at least PR) until

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	the date of first documented biochemical progression or death due to any cause, whichever occurs first. For participants who are alive and have not experienced biochemical progression at the time of analysis, DOR will be censored at their latest available adequate disease assessments. Biochemical PFS is defined as the time from first dose date until the first date of biochemical progression or death due to any cause, whichever occurs first. For participants who are alive and have not experienced biochemical progression at the time of analysis, biochemical PFS will be censored at their latest available adequate disease assessments. The Kaplan-Meier curve and estimated median (or an appropriate percentile if the median is not reached) along with its 95% CI will be provided for each of the secondary time-to-event endpoints described above.
Safety Analyses	All AEs reported in this study will be coded using the Medical Dictionary for Regulatory Activities (MedDRA®). Summary of TEAEs, deaths and other SAEs, and TEAEs leading to permanent treatment discontinuation will be summarized. The safety analysis set (SAF) will include all participants who received any amount of study drug. Other safety, including vital signs (temperature, pulse rate, pulse oximetry [on dosing dates], blood pressure, and respiration rate) and laboratory tests will be summarized by baseline and change from baseline to each scheduled assessment time point with descriptive statistics.
Interim Analysis	No interim analysis will be performed.

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1. INTRODUCTION

1.1. Overview of Plasma Cell Dyscrasias

MGUS is an asymptomatic condition that precedes the development of MM (Kyle, 2006) (Landgren, 2009). It is characterized by the detection of a monoclonal immunoglobulin (M-protein) in the peripheral blood and/or urine that is produced by clonal plasma cells in the bone marrow (Landgren, 2011). Population-based screening studies have shown that MGUS can emerge early in adulthood, and the prevalence increases with age, affecting roughly 3 to 5% of the population aged 50 years or older and up to 10% over 80 years of age (Kyle, 2018) (Landgren, 2017) (Murray, 2019) (El-Khoury, 2022) (Love, 2022). Long-term follow-up studies estimate that the average risk of progression from MGUS to MM is around 1% annually and approaches 2 to 3% annually for patients with high-risk features such as elevated M-protein (≥1.5 g/dL), non-IgG isotype, abnormal FLC ratio, immunoparesis, and evolving changes in laboratory values (Kyle, 2006) (Dispenzieri, 2010) (Cowan, 2023) (Landgren, 2019).

Risk of progression to MM generally correlates with measures of increasing tumor burden such as a higher percentage of clonal plasma cells in the bone marrow or higher amount of M-protein in the blood or urine. These routinely available biomarkers have been used to define a more advanced precursor condition that lies between MGUS and MM known as SMM (Kyle, 2007). A recent population-based screening study estimated the prevalence to be 0.5% in individuals older than 40 years of age (Rögnvaldsson, 2021). The risk of progression to MM is much higher among patients with SMM—around 10% per year for the first 5 years. More refined risk stratification methods such as the 20-2-20 model (Mayo criteria) use thresholds for M-protein (>2 g/dL), serum FLC ratio (>20), and BMPC infiltration (>20%) to divide SMM into low-risk SMM, intermediate-risk SMM, and high-risk SMM (Lakshman, 2018) (Mateos, 2020). Low-risk SMM behaves similarly to MGUS, whereas high-risk SMM has a >50% risk of progression within 2 years of diagnosis (Rajkumar, 2015). This variability in risk reflects the heterogeneous biology of premalignant plasma cell disorders and the need to further improve strategies to accurately identify patients with progressive myeloma precursor conditions (Landgren, 2021).

Uncontrolled proliferation of abnormal monoclonal plasma cells in the bone marrow eventually leads to the development of end-organ damage manifested as hypercalcemia, renal insufficiency, anemia, or lytic bone lesions (CRAB features), which are the classic myeloma-defining events. MM is the second most common hematologic malignancy in the US, and new diagnoses are projected to rise to over 35,000 in 2023 (American Cancer Society, 2023). Similar demographic trends are reported in Europe where the estimated incidence is 4.5 to 6.0 per 100,000 annually (Dimopoulos, 2021). The last decade has welcomed the development of highly effective plasma cell-directed therapies that have significantly improved outcomes for patients with MM; however, despite meaningful advances in care, MM remains incurable, hampered by the development of drug resistance and clonal evolution leading to eventual disease progression. This reality has introduced new challenges related to the clinical management of multiply relapsed disease and the associated prolonged psychosocial and financial burden to patients and their families (Hulin, 2017) (Bhattacharya, 2021).

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1.2. Clinical Management and Opportunity for Earlier Intervention

Most MM patients are diagnosed at the onset of symptomatic disease without a previously identified precursor condition because there is no established screening approach for MGUS. Cases of MGUS or SMM are generally diagnosed incidentally during the medical evaluation of an unrelated condition (eg, anemia, elevated total protein, peripheral neuropathy, or renal insufficiency). Even when early detection occurs, deferral of treatment until the presence of end-organ damage (ie, progression to MM) has long been the standard of care, although this paradigm is changing as research continues to uncover the biologic determinants that influence the risk of progression to MM.

At present, it is recognized that many SMM patients have early MM where all the biological ingredients for malignant transformation are present, and further observation will inevitably lead to the development of CRAB features (Rajkumar, 2015). In 2014, the definition of MM was expanded to include myeloma-defining biomarkers (>60% clonal BMPCs, FLC ratio ≥100, >1 focal lesion on MRI) that incorporated an ultra-high-risk subgroup of patients previously classified as SMM (Rajkumar, 2014). However, only about 10% of SMM patients were impacted, leaving roughly a third of patients with biologic malignancy in the SMM category. The justification for this change was predicated on the rapid time to progression to MM (median time approximately 12 months), the desire to prevent morbidity from end-organ damage, and the availability of safer and more effective treatments. As emerging technologies improve the ability to comprehensively interrogate the underlying biology of aberrant plasma cells and differentiate progressive versus stable myeloma precursor conditions (Bolli, 2018) (Oben, 2021), it is anticipated that earlier treatment will be offered to more patients with a significant risk of developing MM in their lifetime.

Knowing that MM is a serious, incurable cancer with significant morbidity, efforts to intervene earlier are becoming more attractive with the introduction of novel therapies, including recent immunotherapies like bispecific antibodies that have demonstrated clinical activity far exceeding drugs of the past. Not only could earlier treatment delay progression to MM and prevent complications from end-organ damage, but it could alter the natural history of the underlying plasma cell clone and potentially eliminate the risk of ever developing MM (Musto, 2021) (Rajkumar, 2012). One explanation why curative therapy has remained elusive in MM may be related to the fact that therapy is initiated when the disease is genomically more complex and clinically more advanced. Thus, to test this hypothesis, clinical trials are needed to demonstrate the benefit of early therapy in MGUS and non-high-risk SMM.

Further rationale for linvoseltamab treatment in patients with HR-MGUS or NHR-SMM is described in Section 3.2.3. Background information about linvoseltamab and the overall development program can be found in the Investigator's Brochure.

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2. STUDY OBJECTIVES AND ENDPOINTS

The primary, secondary, and exploratory objectives and endpoints of the study are shown in Table 1.

Table 1: Study Objectives and Endpoints

Objective	Endpoint
Primary	
Safety Run-In (Part 1): To evaluate the safety and tolerability of linvoseltamab in ascending dose-level cohorts and select the highest dose level for investigation in the expansion part	• Frequency of AEI assessed during the safety observation period • Frequency and severity of TEAEs assessed during the safety observation period
Expansion (Part 2): To evaluate the ability of linvoseltamab to induce CR by dosing regimen	Achievement of CR as determined by the investigator using IMWG response criteria
Secondary *Applicable to both safety run-in (part 1) and expansion (part 2) unless otherwise specified*	
To characterize the safety of linvoseltamab	Frequency and severity of TEAEs/SAEs and laboratory abnormalities
To evaluate the proportion of participants that achieve MRD negativity	MRD negativity among participants that achieve a response of CR
To evaluate the proportion of participants with sustained MRD negativity	Sustained MRD negativity on an annual basis for up to 3 years after achievement of CR
To evaluate the ORR	Overall response of PR or better as determined by the investigator using IMWG response criteria
To evaluate the DOR	DOR as determined by the investigator using IMWG criteria
To evaluate biochemical PFS	Biochemical PFS as determined by the investigator using IMWG criteria
To characterize the PK properties of linvoseltamab	Concentration of linvoseltamab in serum over time
To assess the immunogenicity of linvoseltamab	Incidence and titer of ADAs to linvoseltamab over time

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Objective	Endpoint
Exploratory	
To evaluate the time to progression to MM	Time from date of first linvoseltamab administration to date of any myeloma-defining event per SLiM-CRAB diagnostic criteria (Appendix 2)
To evaluate clinical PFS	Time from date of first linvoseltamab administration to date of progression to MM or death, whichever happens first
To evaluate changes in sBCMA levels in serum	Concentration of total serum sBCMA over time
To detect serum M-protein by MS and correlate with clinical response characteristics	Concentration of M-protein in serum by MS over time
To evaluate CTCs	CTC enumeration and phenotypic characterization over time
To explore markers related to T-cell activation/exhaustion and immune modulation	• Concentrations of serum cytokines such as IL-2, IFN-gamma, IL-10, IL-6, and TNF-alpha during treatment • Proportion of cells expressing antigens such as but not limited to CD69, HLA-DR, and other T-cell activation/exhaustion markers in bone marrow and/or blood during the study
To characterize the effects of linvoseltamab on various immune cells	Numeric and phenotypic changes in B, T, NK, and monocyte cells in bone marrow and/or peripheral blood as required during treatment with linvoseltamab
To evaluate the genomic determinants of HR-MGUS or NHR-SMM and progression to MM	Incidence and numbers of complex structure variants in plasma cells at baseline and over time measured through whole-exome DNA and RNA sequencing of neoplastic plasma cells and bone marrow immune cells
To characterize bone marrow T-cell clonality and gene expression and identify associations with clinical outcomes	• Sequence of the T-cell receptor in T cells from bone marrow • Gene expression profile of bone marrow T cells at baseline and during treatment
To analyze the phenotype and spatial localization of immune cells in bone marrow core biopsies and associate with clinical outcomes	• Spatial transcriptomics on bone marrow biopsy samples • Imaging mass cytometry on bone marrow biopsy samples • IHC or immunofluorescence on immune cells from bone marrow biopsy samples

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Objective	Endpoint
To evaluate plasma cell surface antigens and their relationship to clinical response and progression to MM	Expression of BCMA and other surface antigens on plasma cells from bone marrow biopsies
To evaluate the effects of linvoseltamab on polyclonal immunoglobulins	Concentrations of total IgG, total IgA, total IgM, total IgD, and total IgE over time
To evaluate the effects of linvoseltamab on patient-reported HRQoL, functioning, and symptoms	• Change from baseline in patient-reported symptoms, functioning, and global health status/QoL per EORTC QLQ-C30 and EORTC QLQ-MY20 during study participation • Change from baseline in patient-reported general health status per EQ-5D-5L during study participation
To evaluate treatment effects of linvoseltamab on patient-reported overall impact of treatment toxicity as measured by the FACIT Item GP5	Change over time in patient-reported overall impact of treatment toxicity per the FACIT Item GP5
To evaluate the impact of linvoseltamab on SSREs	Time to first SSRE, defined as the time from start of treatment to the first instance of one of the following events: Pathological fracture Spinal cord compression Surgery or radiation to the bone
To evaluate healthcare resource use attributed to hospitalizations in participants receiving linvoseltamab	Number and duration of hospitalizations (including general ward, intensive care unit, and emergency room visits) from start of study treatment to 90 days after the last dose of linvoseltamab
To evaluate modulation of the gut microbiome by linvoseltamab and its association with response and/or resistance to therapy	Bacterial taxa evaluation on stool samples before the start of treatment and at 1, 3, 6, and 12 months after the first linvoseltamab dose
To evaluate the effects of linvoseltamab on immunity to common pathogens	Antibody titers against pneumococcus and diphtheria before and after treatment with linvoseltamab

ADA=antidrug antibody; AEI=adverse event(s) of interest; BCMA=B-cell maturation antigen; CR=complete response; CRAB=hyperCalcemia, Renal insufficiency, Anemia, or lytic Bone lesions; CRS=cytokine release syndrome; CTC=circulating tumor cell; DNA=deoxyribonucleic acid; DOR=duration of response; EORTC QLQ-C30=European Organization for Research and Treatment of Cancer Quality of Life Questionnaire-Core 30; EORTC QLQ-MY20=European Organization for Research and Treatment of Cancer Quality of Life Questionnaire -Multiple Myeloma Module 20; EQ-5D-5L=EuroQol 5 Dimensions 5 Levels; FACIT Item GP5=Functional Assessment of Chronic Illness Therapy Item GP5; HR-MGUS=high-risk monoclonal gammopathy of undetermined significance; ICANS=immune effector cell-associated neurotoxicity syndrome; IgA=immunoglobulin A; IgD=immunoglobulin D; IgE=immunoglobulin E; IgG=immunoglobulin G; IMWG=International Myeloma Working Group; NHR-SMM=non-high-risk smoldering multiple myeloma MM=multiple myeloma; MRD=minimal residual disease; MS=mass spectrometry; NK=natural killer; ORR=overall response rate; PFS=progression-free survival; PK=pharmacokinetic; PR=partial response; SAE=serious adverse event; SLiM=greater than or equal to Sixty percent clonal plasma cells in the bone marrow, involved/uninvolved free Light chain ratio ≥100 with the involved FLC ≥100 mg/L, MRI with >1 focal lesion; SSRE=symptomatic skeletal-related event; TEAE=treatment-emergent adverse event

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3. HYPOTHESIS AND RATIONALE

3.1. Hypothesis

It is hypothesized that BCMA-targeted elimination of premalignant plasma cells with linvoseltamab in patients with HR-MGUS or NHR-SMM can induce deep and durable responses, sparing patients the morbidity associated with progression to MM.

3.2. Rationale

3.2.1. Rationale for Study Design

This is a phase 2, open-label, dose-ranging, proof-of-concept study designed to evaluate the safety and efficacy of linvoseltamab monotherapy in patients with HR-MGUS or NHR-SMM. Because linvoseltamab will be evaluated for the first time in this population, enrollment will begin with a safety run-in and dose escalation part (part 1) to assess the suitability of the step-up schedule and select the highest dose level (≤100 mg) appropriate for expansion. After preliminary safety and tolerability are established in part 1, a dose-ranging and expansion part (part 2) will open to further characterize the safety and evaluate the preliminary clinical activity of linvoseltamab when administered according to a randomly assigned dosing regimen (up to 2 dose levels and 2 dosing schedules). The goal is to generate data to guide future clinical development and selection of a dosing regimen for possible phase 3 studies in patients with HR-MGUS or NHR-SMM.

3.2.1.1. Rationale for Therapeutic Interception

Cancer interception refers to a form of secondary prevention intended to stop or delay a premalignant condition from developing into a clinically significant and life-threatening malignancy (Blackburn, 2011) (Narchi, 2021). Interception strategies applied at the time of HR-MGUS or NHR-SMM may offer a window of opportunity to halt the development of future MM. This concept is comparable to solid tumor cancers where removing a localized breast lump or colon polyp is preferable to initiating therapy at the time of symptomatic disease when the cancer is more biologically complex and can more readily adapt to therapeutic interventions. In recent years, two norm-challenging phase 3 studies in patients with high-risk SMM showed that early treatment with lenalidomide and dexamethasone or lenalidomide alone conferred clinically significant benefit compared to observation. Both studies showed an improvement in PFS, and one also demonstrated an improvement in overall survival (Mateos, 2013). These results were observed even though the majority of patients achieved no better than a PR (Mateos, 2022). The question remains whether the achievement of deeper remissions with novel agents can change the natural history of premalignant plasma cell disorders and prevent the development of MM along with the toxicity, secondary malignancy, and clonal selection that occur with traditional MM therapy.

BCMA-directed bispecific antibodies such as linvoseltamab induce T-cell–mediated depletion of plasma cells and have demonstrated impressive monotherapy activity in RRMM. Their use in earlier stages of disease development could result in eradication of the premalignant tumor clone and prevent progression to MM (Roccaro, 2015). This study will evaluate a therapeutic interception strategy using immunotherapy in the premalignant stages of plasma cell dyscrasias.

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3.2.2. Rationale for B-Cell Maturation Antigen-Directed Therapy

HR-MGUS and NHR-SMM represent premalignant stages of plasma cell dyscrasias, and safe, well-tolerated treatment strategies may prevent severe morbidity associated with progression to MM.

BCMA, a member of the TNF superfamily, promotes survival and proliferation of plasma cells through signal transduction of the BAFF/BLyS and a proliferation-inducing ligand. The selective expression of BCMA on mature B cells and plasma cells (neoplastic cells, in particular) has made it an attractive therapeutic target for the management of plasma cell disorders, specifically MM. Although there are no formal studies investigating BCMA expression in the bone marrow of patients with MGUS or SMM, it is known that soluble BCMA levels are elevated in patients with MGUS/SMM who progress to MM (Visram, 2021). This background knowledge provides motivation to evaluate BCMA-targeted therapy in patients with HR-MGUS or NHR-SMM.

There are a variety of approaches in use to target BCMA: antibody drug conjugates, BCMAxCD3 bispecific monoclonal antibodies or other T-cell engagers, and CAR T cells. BCMAxCD3 bispecific monoclonal antibodies like linvoseltamab are attractive therapeutic agents given their off-the-shelf availability, encouraging efficacy in MM, and manageable safety profile.

3.2.3. Rationale for Linvoseltamab Treatment in Patients with High-Risk Monoclonal Gammopathy of Undetermined Significance or Non-High-Risk Smoldering Multiple Myeloma

HR-MGUS and NHR-SMM are asymptomatic precursor conditions that are generally more indolent and less biologically transformed than high-risk SMM. Interception before the onset of frank malignancy represents an attractive strategy to actively intervene when patients are healthier without evidence of end-organ damage and potentially more responsive to therapy, which may lead to better overall outcomes. Early therapy with lenalidomide has been shown to be highly effective at preventing progression from high-risk SMM to MM (Lonial, 2020) (Mateos, 2013), although analogous data are still lacking for NHR-SMM and HR-MGUS. Daratumumab monotherapy has demonstrated good tolerability and promising response rates in NHR-SMM and HR-MGUS, but long-term follow-up data remain immature (Landgren, 2020) (Nadeem, 2021).

Linvoseltamab (REGN5458) is a BCMAxCD3 bispecific antibody that binds to BCMA on plasma cells and to CD3 on T cells. In vitro experiments with linvoseltamab have demonstrated activation of Jurkat T lymphocytes exclusively in the presence of human BCMA-expressing cells and target-mediated cytotoxicity in a range of MM cell lines and primary MM cells with variable BCMA cell surface expression. Additional details regarding preclinical data are provided in the linvoseltamab IB. Phase 1 results from RRMM indicate that linvoseltamab is both effective and tolerable with a manageable safety profile. CRS is the most common TEAE. Most CRS events are grade 1 and occur shortly after administration of the first or second step-up dose, typically resolving within 24 hours. The overall incidence of grade ≥3 treatment-related AEs was 42.7% with the most common being anemia and neutropenia (Regeneron Pharmaceuticals, 2022). Encouraging efficacy has been reported in participants that received a median of 5 prior lines of therapy with 89% being triple-class refractory or worse. Most responses occurred within the first month of treatment and deepened over time. Among responders across all dose groups, 86% (n=32/37) achieved a VGPR or better, 43% (n=16/37) achieved a CR, and 40% of evaluable CR patients (n=4/10) achieved MRD negativity. The probability of an ongoing response at 8 months was 90.2%. Detailed

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information about the reported depth, duration, and durability of responses is provided in the linvoseltamab IB.

Based on the above background and ongoing development of aggressive multi-agent treatment regimens in high-risk SMM (Kazandjian, 2021) (Mateos, 2021), the available data suggest that single-agent linvoseltamab may reasonably yield similar or better efficacy outcomes and less toxicity compared to other emerging treatments, especially when the length of treatment is limited to 3-6 months. Therefore, the safety and efficacy of linvoseltamab in patients with HR-MGUS or NHR-SMM will be investigated. Since this is the first time that linvoseltamab will be evaluated in this patient population, the study will begin with a safety run-in and dose-escalation cohort of 6 to 24 participants, followed by an expansion cohort of 80 participants to further optimize the dosing regimen. The hypothesis is that treatment with a potent BCMAxCD3 bispecific antibody for a brief duration in patients with HR-MGUS or NHR-SMM may lead to elimination of the premalignant clone and long-term freedom from progression to MM.

3.2.4. Rationale for Linvoseltamab Dosing Regimen

Since this is the first study to treat MGUS/SMM patients with linvoseltamab, the rationale for the proposed dosing regimens is informed by the current biological and clinical understanding of MGUS/SMM as well as data from the FIH study (R5458-ONC-1826) that investigated linvoseltamab monotherapy in participants with RRMM. Additional information supporting this dose rationale is provided in the relevant sections of the linvoseltamab IB.

3.2.4.1. Rationale for Step-up Schedule and Inpatient Monitoring

Linvoseltamab, like other T-cell–redirecting therapies, can induce CRS during the first few weeks of treatment, and conditioning the T-cell response with stepwise dose escalation (step-up dosing) has been shown to reduce the risk of severe CRS. Across the higher full-dose levels (96 to 800 mg) studied, a step-up schedule consisting of an initial 5 mg dose followed a week later by an intermediate 25 mg dose adequately mitigated the risk of CRS in conjunction with premedications and inpatient monitoring for 24 hours after the EOI. As of 30 Sep 2022, clinical data from Study R5458-ONC-1826 indicate an overall favorable CRS profile characterized by a low rate of grade 2 or greater CRS (grade 2: 22/218 [10.1%]; grade 3: 3/218 [1.4%]) and relatively rapid onset and resolution of CRS events. The median time to onset of CRS was 11.0 hours from the start of infusion, and the median duration was 15.6 hours. No cases of grade ≥2 CRS began after approximately 24 hours from the EOI, and only 2 of 218 participants (0.9%) experienced grade ≥2 CRS after receiving the first full dose (both grade 2). These results support the adequacy of precautionary inpatient monitoring for 24 hours after each step-up dose.

While patients with MGUS/SMM have been reported to have impaired immunity (Paiva, 2016) (Zavidij, 2020), they have relatively intact immune systems compared to RRMM patients (Nakamura, 2020), which could predispose them to more frequent and/or severe CRS. However, any heightened risk of CRS is largely theoretical and may be counterbalanced by the lower tumor burden, since linvoseltamab-induced T-cell activation relies on the presence of BCMA on target cells (Hay, 2017) (DiLillo, 2021). Given the uncertain contributions of these competing factors and the low clinical urgency to achieve an immediate therapeutic response, a more gradual step-up schedule serves as an additional safeguard that may further mitigate the risk of CRS. The current study will therefore implement a step-up schedule that splits the 5 mg dose (initial step-up dose of

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linvoseltamab in RRMM) into 1 mg at week 1 and 4 mg at week 2, followed by 25 mg at week 3, before reaching the full dose of linvoseltamab in addition to the established premedication process and required inpatient monitoring after each step-up dose.

3.2.4.2. Rationale for Selection of Full Dose and Dosing Schedule

Tumor burden is a key classifier and risk determinant of plasma cell disorders. Accordingly, HR-MGUS/NHR-SMM patients have reduced levels of established tumor burden biomarkers such as BMPCs and sBCMA compared to RRMM patients (Ghermezi, 2017) (Visram, 2021). MGUS and SMM patients, even those who progress to MM, have been reported to have lower baseline sBCMA concentrations (MGUS: median baseline sBCMA 106.5 ng/mL with IQR: 78.9–227.3 ng/mL; SMM: median baseline sBCMA 162.7 ng/mL with IQR: 90.5–275.8 ng/mL) compared to RRMM patients (Visram, 2021) (Chen, 2019) (Girgis, 2021).

Given the lower tumor burden in patients with MGUS/SMM compared to RRMM, it is hypothesized that less intensive and limited-duration dosing regimens may provide adequate linvoseltamab exposures to achieve meaningful clinical responses. Supporting evidence from RRMM participants in Study R5458-ONC-1826 suggests that participants with lower tumor burden (sBCMA ≤200 ng/mL and/or BMPC ≤33%) responded favorably to lower full doses (≤96 mg) compared to those that received the 200 mg full dose (refer to linvoseltamab IB for detailed analysis), which is optimized for all patients, including those with higher tumor burden. A less intensive regimen is also desirable to avoid overtreating otherwise healthy patients and to minimize the potential risks of prolonged immunosuppression. Most responses in RRMM were observed within 4 to 8 weeks of starting treatment, and even when treatment was interrupted or stopped, responses continued to deepen over time.

The optimal duration and frequency of treatment are unknown. Therefore, following completion of the step-up regimen, 2 different fixed-duration dosing schedules will be explored: schedule 1 consists of QW dosing for 4 weeks followed by Q4W dosing for 20 weeks (9 doses), whereas schedule 2 consists of Q2W dosing for 12 weeks (6 doses) (see Section 6.1 for full description of study design). Both dosing schedules are predicted to maintain a median linvoseltamab C_{trough} above where CRs have been observed (3–6 mg QW or ~1 mg/L) in patients with RRMM across a range of dose levels. Although the durability of responses after ending treatment is difficult to predict in the MGUS/SMM population, the less deranged hematopoietic systems in these patients, and the fact that some RRMM patients have demonstrated sustained responses for more than 1 year off treatment, support the hypothesis that deep remissions and freedom from progression to MM are achievable.

Additionally, to evaluate a sufficiently broad exposure range for dose optimization, 2 different full doses of linvoseltamab will be explored: a low-dose level of 25 mg and a high-dose level based on the highest tolerated dose (not to exceed 100 mg) identified in the safety run-in and dose escalation component (part 1) of the study. The 25 mg dose level was selected based on simulations of linvoseltamab exposure using a population PK model generated from Study R5458-ONC-1826 data aimed at differentiating drug exposures between the expected high and low doses to enable the identification of any exposure-response relationship (detailed analysis provided in linvoseltamab IB). Importantly, the predicted systemic linvoseltamab concentrations at the 25 mg dose level using the proposed dosing schedules provide exposures similar to those observed with the lower dose levels at which clinical efficacy was observed in RRMM patients.

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3.2.5. Rationale for Exploratory Objectives

In light of the differences between MM and HR-MGUS/NHR-SMM, interrogating plasma cell, T cell, and other immune biology may inform selection of a tumor-specific dosing regimen and assist in interpreting efficacy, tolerability, and safety results.

The exploratory analysis plan in HR-MGUS and NHR-SMM will include evaluation of the mechanism of action of linvoseltamab to induce T-cell activation and plasma cell depletion in bone marrow. Given that the patient population (HR-MGUS and NHR-SMM) and the dosing level and schedule are different with regard to RRMM, the dose-dependent engagement of T cells in this population may inform identification of an appropriate dose regimen. In peripheral blood, linvoseltamab may induce changes in T and B cells that are important to inform dose selection and to characterize tolerability in this patient population.

In this study, detection and phenotyping of circulating plasma cells will be performed at baseline and post-treatment to inform plasma/tumor cell depletion anticipated by linvoseltamab treatment. Monitoring circulating plasma cells, BCMA expression, and specific cytogenetic changes such as chromosomal translocations t(4;14), t(14;16), del(17p), and del(13q), copy number changes (1q gain) and mutations (TP53 and KRAS) have been reported for patients with SMM (Ndacayisaba, 2022). As characterization of BCMA expression in bone marrow in patients with HR-MGUS or NHR-SMM has not been reported, analysis of bone marrow core biopsies for BCMA expression will be performed. Genetic lesions and transcriptional changes are known to be associated with differences in clinical response to therapy and will also be characterized as part of the study. The gut microbiome has also been shown to be associated with disease progression and clinical response in a number of malignancies, including MM. Consequently, how the patients' gut microbiome associates with response and/or resistance to therapy will be examined at numerous time points throughout the study. Lasty, the impact of treatment with linvoseltamab on immune recovery remains poorly understood and important to characterize in the HR-MGUS and NHR-SMM population. This will be investigated by examining total immunoglobulin concentrations and antibody titers against common pathogens before and after treatment with linvoseltamab.

3.2.5.1. Symptomatic Skeletal-Related Events

SSREs may include fracture, spinal-cord compression, and/or surgery or radiation to the bone; these events may result in an increased participant morbidity with negative impacts to quality of life. While all of these types of SSREs can be observed with progression to MM, evidence suggests that the risks of fracture and spinal cord compression are present in early myelomagenesis and may precede clinical progression to MM. Patients with MGUS have been shown to have microarchitectural deficits at various skeletal sites, which may contribute to increased skeletal fragility and subsequent risk of fracture (Stein, 2019). Large, population-based epidemiological studies have demonstrated an increased risk of fracture in these patients with a 29% increased rate of fracture among patients with MGUS (n=8395) compared to matched controls (n=30,851) (Rögnvaldsson, 2021). The risk of fracture has been demonstrated at both axial (including vertebral) and distal skeletal regions with a significantly higher risk in axial locations (Kristinsson, 2010). The risk of fracture has not been formally evaluated within population-based SMM studies. However, given that SMM typically represents a more advanced state within the spectrum of precursors to MM with a generally higher rate of progression to MM, the relative rate

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of fractures is expected to be high. Surgery and/or radiation to the bone play an important role in the treatment of osteolytic lesions, prevention of fracture, and bone pain palliation. Altogether, these disease-related events occurring pre- and/or post-progression to MM comprise patient-relevant outcomes that are expected to correlate closely with QoL.

3.2.6. Rationale for Patient-Reported Outcomes

Both MGUS and SMM are largely considered to be asymptomatic with observation regarded as the standard of care for patients living with these conditions. As disease-related symptom burden in these patient populations is expected to be low prior to progression to MM, demonstrating maintenance of QoL with limited impact on overall symptoms is a high priority in this patient population. The cancer-specific 30-item EORTC QLQ-C30 and the 20-item MM-specific EORTC QLQ-MY20 have been validated previously for use together within MM and have also been tested among patients with SMM. It is likely that these questionnaires would cover similar concepts among patients with MGUS. The EQ-5D-5L is a generic standardized health status measure that assesses QoL across 5 dimensions of health (mobility, self-care, usual activities, pain/discomfort, and anxiety/depression),and will be especially useful for contextualizing QoL findings in this largely asymptomatic population with other disease areas. Furthermore, data generated from the EQ-5D-5L in this study will be useful for future potential health technology assessments. The FACIT Item GP5 provides a global measure of impact related to burden associated with treatment toxicity.

3.3. Risk-Benefit

3.3.1. Risk Assessment of Linvoseltamab

Important risks of clinical significance, rationale for risks, and associated mitigation strategies are presented in Table 2. Important risks include identified and potential risks (see the linvoseltamab IB for additional details).

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Table 2: Risks Assessment of Linvoseltamab

Important Risks of Clinical Significance	Rationale for Risk	Mitigation Strategy
CRS/IRR/Systemic Hypersensitivity Reactions	There is significant overlap in the clinical presentation of CRS, IRRs, and systemic hypersensitivity. The symptomatology associated with these entities is consistent with an underlying mechanism of exaggerated immune response and release of inflammatory cytokines. The mechanism of action of T-cell–engaging immunotherapy such as linvoseltamab is non-physiologic T-cell activation with the occurrence of cytokine release. Clinical effects of these can range from mild symptoms, such as fever, to a severe inflammatory syndrome, which may be life threatening or lead to death.	Participants at high risk of hypersensitivity reactions are excluded (Exclusion Criterion #7). Medications and equipment to manage medical emergencies must be available for immediate use (Section 8.3.1 and Section 8.3.2). Stepwise increase in dose (step-up period) prior to reaching full dose is used (Section 6.1.1, Section 6.1.2, and Section 6.1.3.1). Infusion of linvoseltamab in a monitored setting and subsequent inpatient monitoring following administration of step-up doses for early detection and management of CRS/IRRs and hypersensitivity is required (Section 6.1.1 and Section 8.3). Premedication with dexamethasone, antihistamines, and acetaminophen (paracetamol) is required for all participants from step-up doses through the first full dose (Section 8.1.2). Guidance to investigators on drug interruption and/or termination, grading, and management of symptoms of CRS/IRRs and hypersensitivity reactions is provided in Section 8.3.1 and Section 8.3.2.
ICANS	CNS symptoms such as confusion and aphasia have been observed with CAR T-cell therapy and other bispecific antibodies and were previously considered part of CRS. These symptoms are now considered to be a separate syndrome called ICANS. Cytokines may be implicated in the pathophysiology (Lee, 2019).	Participants with CNS pathology are excluded (Exclusion Criterion #14). Guidance for early identification of neurologic events and monitoring is provided in Section 8.3.3. Guidance on management of neurological toxicity including study drug administration is provided in Table 11.

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Important Risks of Clinical Significance	Rationale for Risk	Mitigation Strategy
Serious Infections	Based on the mechanism of action, treatment with linvoseltamab may result in pronounced B-cell depletion and hypogammaglobulinemia. The expected primary pharmacology of B-cell depletion may result in increased risk of infections, including opportunistic infections and viral infections (including HBV and CMV).	Participants with known active infections are excluded (Exclusion Criteria #5 and #6). Immunoglobulin levels will be checked at baseline, and levels will be monitored periodically throughout the trial (Section 9.1). Total lymphocyte counts and T-lymphocyte subsets will be checked at baseline and monitored periodically throughout the trial (Section 9.1). Consideration of prophylaxis for viral infections (herpesviruses) and fungal infection and IV immunoglobulin for participants with severe hypogammaglobulinemia is recommended (Section 8.3.4).
TLS	TLS is an oncologic emergency triggered by the rapid release of intracellular material from lysing malignant cells. TLS has been reported in virtually every cancer type. The highest risk of TLS is seen in large-volume, highly metabolic malignancies such as B-cell acute lymphoblastic leukemia and Burkitt's lymphoma, whereas solid tumors and slow-growing hematologic malignancies (such as MM) carry lower risks (Wilson, 2014).	Participants with hypersensitivity to both allopurinol and rasburicase are excluded (Exclusion Criterion #8). Section 8.3.5 provides guidance on prophylaxis, monitoring, and management of TLS.
Embryo-Fetal Toxicity	Reproductive and developmental toxicology studies have not been conducted; therefore, the effects of linvoseltamab on reproduction and embryo-fetal development are unknown.	Pregnancy testing is performed for WOCBP (Inclusion Criterion #77.b). Pregnant or breastfeeding women are excluded (Inclusion Criterion #77.b). Contraception is required to prevent pregnancy in study participants or their sexual partners (Inclusion Criterion #77.b).

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Important Risks of Clinical Significance	Rationale for Risk	Mitigation Strategy
Clinical consequences of immunogenicity: ADAs	All protein therapeutics, including linvoseltamab, have the potential to induce an immunogenic response (formation of antibodies against the study drug) following administration.	Anti-linvoseltamab antibody status will be assessed during the study (Table 16). Evaluation of clinical consequences of immunogenicity will be completed (Section 2).

ADAs=antidrug antibodies; CAR=chimeric antigen receptor; CMV=cytomegalovirus; CNS=central nervous system; CRS=cytokine release syndrome; HBV=hepatitis B virus; ICANS=immune effector cell-associated neurotoxicity syndrome; IRR=infusion-related reaction; IV=intravenous; MM=multiple myeloma; TLS=tumor lysis syndrome; WOCBP=women of child-bearing potential

More information about the known and expected benefits and risks of linvoseltamab may be found in the most recent version of the linvoseltamab IB.

3.3.2. COVID-19 Considerations

Recognizing that an event of unforeseen circumstances (eg, COVID-19 or other national/international crisis) may have an impact on the conduct of clinical trials, the sponsor does not intend to screen any participants in this study until the impact of the emergency is deemed manageable and no longer interfering with the conduct of studies at individual sites and participants can safely participate in this study.

The continuity of clinical study conduct dictated by the schedule of events and oversight may require implementation of temporary or alternative mechanisms. Examples of such mechanisms may include but are not limited to any of the following: phone contact, virtual visits, telemedicine visits, online meetings, noninvasive remote monitoring devices, use of local clinic or laboratory locations, and home visits by skilled staff. Any study data collected using an alternative mechanism will be documented.

For information regarding the permitted timing of COVID-19 vaccinations, see Section 7.2.2 Exclusion Criteria, criterion #15 and Section 8.7.1.

3.3.3. Benefit Assessment of Linvoseltamab

Linvoseltamab has demonstrated promising antitumor activity in heavily pretreated patients with RRMM. Responses, including ≥VGPR, were observed at all dose levels studied in the FIH study, and participants with lower tumor burden appeared to benefit similarly across a wide range of doses (Section 3.2.4.2). Additional data are needed to better understand the clinical benefit of linvoseltamab in participants with HR-MGUS or NHR-SMM. See the most recent version of the linvoseltamab IB for additional details.

3.3.4. Overall Benefit-Risk Conclusion

The available nonclinical data and preliminary clinical data from the ongoing FIH R5458-ONC-1826 study support linvoseltamab as an effective therapeutic for the treatment of RRMM with a manageable safety profile. The linvoseltamab IB provides a risk-benefit statement with respect to the overall development program.

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Considering the multiple safety measures taken to minimize important risks of linvoseltamab, and the anticipated benefit to study participants with HR-MGUS or NHR-SMM, clinical development of linvoseltamab is justified in this patient population.

4. ENDPOINTS

See Table 1 for endpoints.

5. STUDY VARIABLES

5.1. Demographic and Baseline Characteristics

Baseline characteristics will include standard demography (eg, age, race, etc), physical examination findings (including height and weight), and medical history including oncologic history (including prior cancer-related therapies, radiotherapies, and surgeries) for each participant.

5.2. Efficacy Variables

Efficacy variables include, but are not limited to, the following:

- Best overall response of CR, as determined by the investigator using IMWG criteria (Appendix 1)
- Best overall response of PR or better, as determined by the investigator using IMWG criteria (Appendix 1)
- Biochemical PFS, as determined by the investigator using IMWG criteria (Appendix 1)
- DOR, as determined by the investigator using IMWG criteria (Appendix 1)
- MRD negative status (at 10^{-5})

Biochemical progression is defined as an increase in serum or urine monoclonal protein according to the PD response category of the IMWG criteria (Appendix 1).

5.3. Safety Variables

Safety variables include, but are not limited to, the following:

- Vital signs, as described in Section 9.2.3.2
- Physical examination results, as described in Section 9.2.3.3
- Neurologic examination results, as described in Section 9.2.3.4
- Clinical laboratory results, as described in Section 9.2.3.6
- AESI, as described in Section 10.1.3 and Section 10.2.3
- AEs, as described in Section 10.2.1
- SAEs, as described in Section 10.2.2

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5.4. Pharmacokinetic Variables

The PK variable is the concentration of total linvoseltamab in serum over time. These sampling time points are specified in Table 16.

5.5. Immunogenicity Variables

The immunogenicity variables are ADA status and titer at nominal sampling time/visit. Samples in this study will be collected at the clinic visits specified in Table 16. Samples positive in the REGN5458 ADA assay will be further characterized for ADA titers.

5.6. Pharmacodynamic and Other Biomarker Variables

Biomarker variables include the percentages of T-, B-, and NK-cell subsets; monocytes; and CTCs over time in peripheral blood. Target engagement by linvoseltamab includes evaluation of the percentages of activated T- and NK-cell subsets in peripheral blood over time assessed by markers such as CD69 and CD25 as well as induction of serum cytokines such as IL-2, IL-6, IL-10, IFN-gamma, and TNF-alpha. Treatment may impact normal plasma cell survival; therefore, the serum concentrations of total polyclonal IgG, IgA, IgE, and IgM (excluding the clonal subtype) and antibody titers against common pathogens such as pneumococcus (*Streptococcus pneumonia*) and diphtheria (*Corynebacterium diphtheriae*) will be evaluated over time. Other biomarkers related to plasma cell disorders or the mechanism of action of linvoseltamab may be evaluated, including sBCMA. Exploratory biomarkers are described in Section 9.2.7.

5.7. Patient-Reported Outcome Variables

PRO variables include patient-reported symptoms, functioning, and general health status/QoL as measured by the validated instruments EORTC QLQ-C30 (Aaronson, 1993b), EORTC QLQ-MY20 (Cocks, 2007) (Stead, 1999), and EQ-5D-5L. Symptomatic AEs and global side effects will be measured by the FACIT Item GP5.

5.7.1. Variables from European Organization for Research and Treatment of Cancer Quality of Life Questionnaire – Core 30

The EORTC QLQ-C30 scale scores will be calculated, and missing data will be handled per developer instructions using the EORTC QLQ-C30 Scoring Manual (Fayers, 2001). The instrument yields the following scales:

- Global health status / QoL with 2 items
- Functional scales
  - Physical functioning with 5 items
  - Role functioning with 2 items
  - Emotional functioning with 4 items
  - Cognitive functioning with 2 items
  - Social functioning with 2 items

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- Symptom scales / items
  - Fatigue with 3 items
  - Nausea and vomiting with 2 items
  - Pain with 2 items
  - Dyspnea, insomnia, appetite loss, constipation, diarrhea with 1 item each
- Financial impact with 1 item

Change from baseline in EORTC QLQ-C30 scale scores (transformed 0 to 100 scores) will be categorized as improvement/stable/deterioration using a 10-point threshold value based on available literature (Osoba, 1998) (Kvam, 2010).

5.7.2. Variables from European Organization for Research and Treatment of Cancer Quality of Life Questionnaire – Multiple Myeloma Module 20

The EORTC QLQ-MY20 instrument yields the following scales:

- Disease symptoms with 6 items
- Side effects of treatment with 10 items
- Future perspective with 3 items
- Body image with 1 item

Change from baseline in EORTC QLQ-MY20 scale scores (transformed 0 to 100 scores) will be categorized as improvement/stable/deterioration using clinically meaningful thresholds identified based on literature (Sully, 2019).

5.7.3. Variables from EuroQol 5 Dimensions 5 Levels

EQ-5D-5L variables include:

- 5 dimensions of mobility, self-care, usual activities, pain/discomfort, and anxiety/depression
- EQ-5D-VAS

Change from baseline in item scores (raw scores) of the 5 dimensions and EQ-VAS will be classified as improvement/stable/deterioration based on literature (EuroQol., 2017).

5.7.4. Variables from Functional Assessment of Chronic Illness Therapy Item GP5

FACIT Item GP5 is a single-item ("I am bothered by side effects of treatment") summary measure of patient-reported overall impact of treatment toxicity measured and scored on a 5-point Likert-type scale (0=Not at all, 1=A little bit, 2=Somewhat, 3=Quite a bit, 4=Very much).

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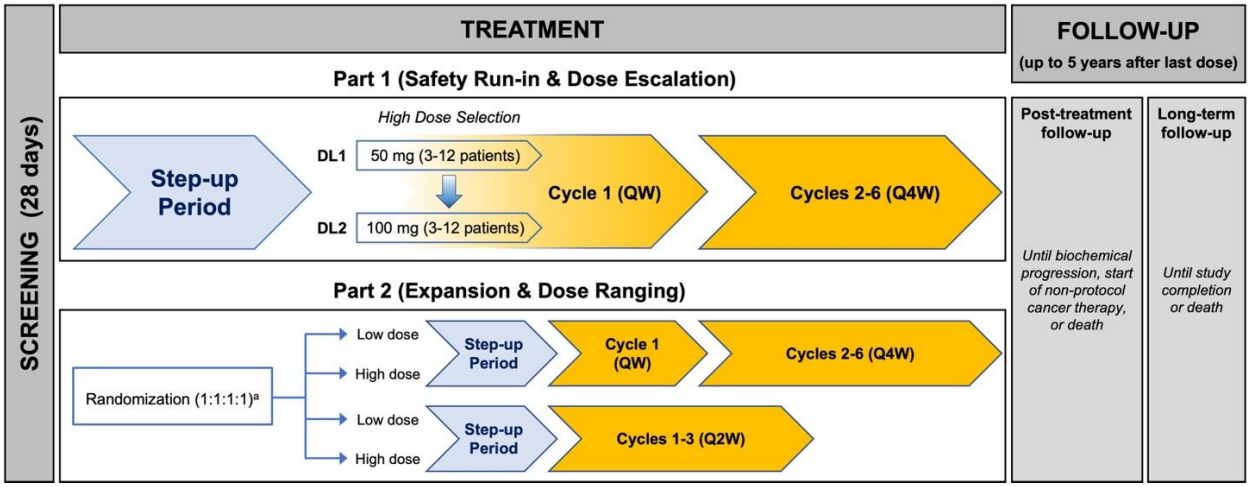
6. STUDY DESIGN

6.1. Study Description and Duration

This is a phase 2, open-label, dose-ranging, proof-of-concept study designed to evaluate the safety and efficacy of linvoseltamab monotherapy in patients with HR-MGUS or NHR-SMM. Because linvoseltamab will be evaluated for the first time in this population, enrollment will begin with a safety run-in and dose-escalation component (part 1) to assess the suitability of the step-up schedule and select the highest dose level (≤100 mg) appropriate for expansion (see Figure 1). After preliminary safety and tolerability are established in part 1, a dose-ranging and expansion component (part 2) will open to further characterize the safety and evaluate the preliminary clinical activity of linvoseltamab when administered according to a randomly assigned dosing regimen (up to 2 dose levels and 2 dosing schedules).

Figure 1: Overview of Study Design



[a] Randomization will be stratified by number of 20-2-20 risk factors (0 or 1).
QW=once weekly; Q2W=once every 2 weeks; Q4W=once every 4 weeks

6.1.1. Description of Study Cohorts and Dose Escalation

Safety Run-In (Part 1): Sequential cohorts of 3 to 6 participants will be enrolled according to a BOIN dose-escalation design (Section 6.1.3.2). Tolerability will be assessed by the incidence of AEI during a 35-day safety observation period. Each participant will start IV linvoseltamab treatment with a 3-week step-up schedule consisting of weekly dose increments of 1 mg (week 1), 4 mg (week 2), and 25 mg (week 3). After completion of the step-up regimen, the assigned full dose of linvoseltamab will be given over 28-day treatment cycles for a fixed duration of 6 cycles. Dosing will occur QW for the first cycle and then transition to Q4W for the next 5 cycles (Figure 2). The initial dose level evaluated will be 50 mg (DL1), and the highest dose level will not exceed 100 mg (DL2) (part 1 in Figure 1).

Participants enrolled in the first dose level must start treatment with linvoseltamab in a staggered fashion (no less than 48 hours apart) to allow sufficient observation for acute toxicities and limit risk to other participants if such events are detected. Modified step-up dosing regimens may be explored if unacceptable toxicity is encountered. If a change is deemed necessary, revisions to the step-up dosing regimen and rationale will be described in a protocol amendment. Modified full

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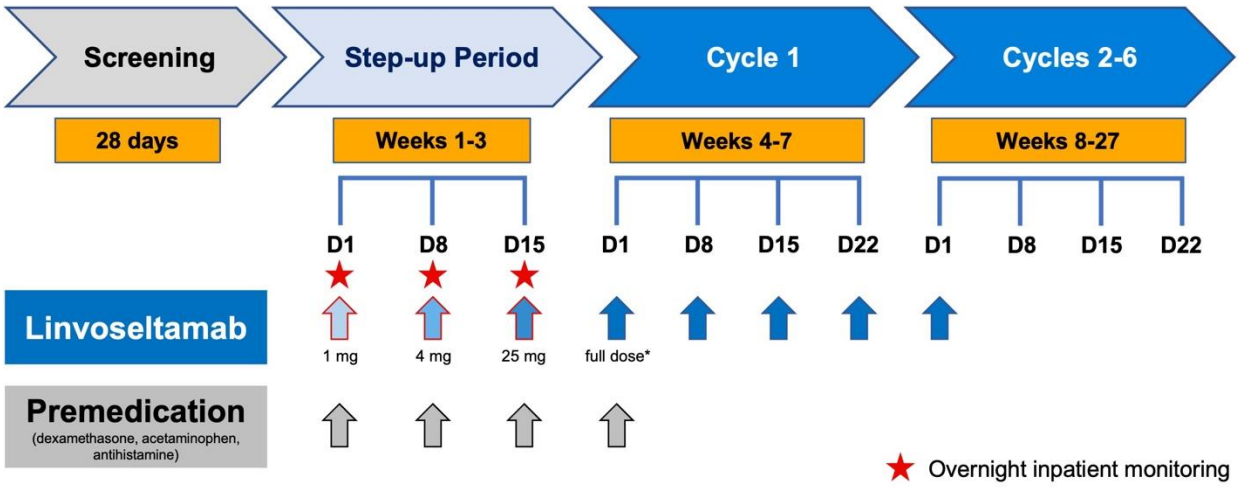
dose levels (listed in Appendix 6) will be implemented if emerging safety issues warrant a smaller increment of dose escalation following a completed dose level.

Expansion (Part 2): Once acceptable safety and tolerability are demonstrated, a minimum of 80 participants will be randomized in a 1:1:1:1 ratio across 4 dosing regimens, comprising 2 full doses (25 mg [low dose] or the highest tolerated full dose in part 1 [high dose]) given according to 2 dosing schedules (schedule 1 or schedule 2) to identify a dosing regimen that minimizes the burden of treatment and optimizes safety and efficacy (part 2 in Figure 1). Each dosing regimen will begin with the same 3-week step-up schedule established in the safety run-in, followed by a fixed number of 28-day treatment cycles with full dose linvoseltamab. Schedule 1 mirrors the dosing schedule used in part 1: QW for the first cycle and then Q4W for the next 5 cycles (Figure 2). Schedule 2 consists of biweekly dosing (Q2W) for a total of 3 cycles (Figure 3).

Randomization will be stratified according to number of 20-2-20 risk factors (0 or 1) as specified in Section 7.2.2, exclusion criterion #1. Participants must begin treatment according to the assigned arm within 3 days of randomization.

Over the entire treatment period, starting with the first infusion of linvoseltamab, all participants enrolled in part 1 or part 2 will be monitored in aggregate for cumulative toxicity that meets criteria for enrollment pause and/or safety review (cumulative rate of trigger events) (Section 6.1.3.5).

Figure 2: Study Dosing Schema – Schedule 1 (Part 1 & Part 2)



*Assigned full dose will be according to the dose escalation rules in part 1 or randomization in part 2.
D=day

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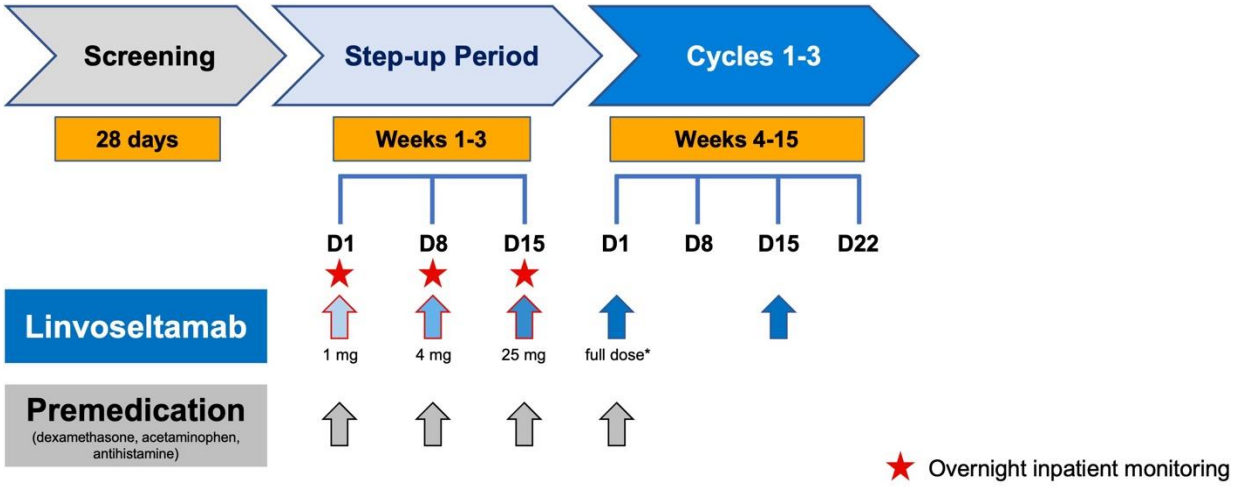
Figure 3: Study Dosing Schema – Schedule 2 (Part 2)



*Assigned full dose will be according to randomization in part 2.
D=day

6.1.2. Study Duration

For each participant, the study will span the following 3 periods:

- **Screening Period:** Screening procedures may occur over a period of up to 28 days for each study participant, ending on the day preceding the planned first dose of linvoseltamab.
- **Treatment Period:** The treatment period will start on the day when the first step-up dose of linvoseltamab is administered and continues until completion of the EOT visit or, if the EOT visit does not occur, 30 days after the last dose of linvoseltamab is administered. Treatment may continue through the last scheduled dose in the assigned dosing regimen or until fulfillment of a protocol-defined criterion for study drug discontinuation.
- **Follow-up (up to 5 years after last dose):**
  - **Post-treatment follow-up:** Post-treatment follow-up will start after the EOT visit and involves the ongoing collection of safety and efficacy data. Safety monitoring will continue for up to 90 days after the last linvoseltamab dose. Participants with an ongoing biochemical response will continue to undergo routine efficacy assessments until confirmed biochemical progression, start of non-protocol cancer therapy, or death.
  - **Long-term follow-up:** Long-term follow-up will begin after post-treatment follow-up to collect survival status as well as other relevant long-term outcomes (eg, progression to MM, subsequent non-protocol cancer therapy) and will continue for up to 5 years after the last dose of linvoseltamab, unless death or study termination by the sponsor occurs earlier. Participants may be followed remotely for survival and subsequent treatment history.

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6.1.3. Dose Finding and Study Safety and Stopping Rules

6.1.3.1. Safety Observation Period Prior to Dose Escalation and Expansion

The safety run-in (part 1) will evaluate the suitability of a 3-dose step-up regimen (1, 4, and 25 mg) and the tolerability of at least 1 full dose of linvoseltamab. Each participant included in part 1 will be observed for dose-limiting AEI over a 35-day safety observation period that begins on the date of their first linvoseltamab infusion. A participant will be considered dose evaluable if they (1) receive at least 1 full dose of linvoseltamab and complete the 35-day safety observation period or (2) experience an AEI (defined below) during the 35-day safety observation period.

An AEI is a toxicity potentially related to study treatment that may preclude dose escalation or expansion according to the BOIN design decision rules (Section 6.1.3.2). Occurrence of the following TEAEs will constitute an AEI unless there is a clear alternative explanation:

- Grade 3 AEs including CRS and ICANS with the following exceptions:
  - Isolated laboratory abnormalities without clinical consequence
  - Laboratory abnormalities associated with grade ≤2 CRS that improve within 72 hours
- Grade ≥4 AEs

Any grade ≥4 toxicity will prompt an enrollment pause pending full review of available clinical data. If study treatment is considered a potential contributing factor, then the SSET will reassess the linvoseltamab dosing regimen (step-up and full doses). Alternative dosing regimens (eg, lower initial dose, smaller step-up increments) may be explored depending on the totality of the available data. If a change is deemed necessary, revisions to the dosing regimen and rationale will be described in a protocol amendment.

Dose escalation and expansion decisions will be contingent on the above criteria and SSET agreement. The SSET will evaluate all available data collected during the 35-day safety observation period, including safety data from inevaluable participants. Toxicities observed beyond the 35-day safety observation period will also be considered. Additionally, trigger event thresholds (Section 6.1.3.5) will be continuously monitored throughout the entire duration of study treatment.

6.1.3.2. Dose Escalation Rules

This study will utilize a BOIN design in conjunction with supplemental rules limiting the maximum dose increment between dose levels (Section 6.1.3.3 and Appendix 6) to guide dose-escalation decisions during the safety run-in part. The starting full dose of linvoseltamab will be 50 mg, designated as DL1. Participants will be enrolled and treated in cohorts of 3 to 6 participants at a time. If DL1 is tolerated as per BOIN, the full dose will be escalated to DL2 of 100 mg. If DL1 is not tolerated, the full dose will be de-escalated to DL0 of 25 mg. A maximum of 12 evaluable participants may be enrolled at any dose level. Data from all evaluable participants will be considered.

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As shown in Table 3, the BOIN design stipulates the following rules:

- De-escalate the linvoseltamab dose to a lower dose level if the observed event rate at the current dose is ≥0.298.
- Escalate to a higher dose level if the observed event rate at the current dose is ≤0.197 and the next higher dose level has not been eliminated. An eliminated dose is a dose level that is associated with unacceptable toxicity. The specific dose level will be determined by the rules in Table 3.
- Remain at the current dose in all other situations (observed event rate at the current dose is > 0.197 but <0.298). If 12 evaluable participants have already been evaluated at the current regimen, the current dose may be declared to be acceptable for expansion. Otherwise, enroll the next group of participants at this dose level.
- Eliminate the dose due to unacceptable toxicity. If the posterior probability of high toxicity rate at a certain dose is large, then the BOIN design eliminates the corresponding dose and other higher doses from consideration in the trial (ie, at dose level j, if P(True toxicity rate >25% | Data) > 0.80, then dose j and higher will be eliminated).

A minimum of 3 evaluable participants at a given dose level are needed to apply these rules, unless a participant experiences an AEI during the 35-day safety observation period that, by itself, would already meet criteria for de-escalation of the dose level.

Table 3: Decision Rule at the Current Dose with Target Toxicity Rate of 25%*

Action**	**Number of evaluable participants treated at the current dose**											
	1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**
Escalate if # of events ≤	0	0	0	0	0	1	1	1	1	1	2	2
De-escalate if # of events ≥	1	1	1	2	2	2	3	3	3	3	4	4
Eliminate if # of events ≥	N/A	N/A	2	2	2	3	3	3	4	4	4	5

* The event rate for a dose level is calculated by the number of participants who experience an AEI divided by the number of evaluable participants in the dose level.

** If none of the criteria for the actions shown in this table are met, the dose will remain at the current level, and more participants may be added if maximum enrollment has not been reached. The SSET may decide to keep the dose unchanged even if escalation is permitted in view of clinical considerations other than the number of AEI.

N/A=not applicable

6.1.3.3. Dose Levels

Rationale for the starting the full dose of 50 mg (DL1) is provided in Section 3.2.4.2. Based on the BOIN statistical design described in Section 6.1.3.2, the primary dose levels for the full dose will be as follows:

- DL2 = 100 mg
- DL1 = 50 mg

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- DL0 = 25 mg

The maximum dose increment will be limited to 50% (eg, 50 to 75 mg) if either of the following occur in a given dose level cohort:

- 1 participant experiences an AEI during the safety observation period

  or

- 2 or more participants experience a grade 2 AE during the safety observation period with the following exceptions:
  - AEs attributable to a clear alternative etiology
  - Grade 2 fatigue or asthenia
  - Grade 2 nausea, vomiting, constipation, or diarrhea lasting <48 hours (with symptomatic treatment)
  - Grade 2 headache lasting <24 hours (with symptomatic treatment)
  - Grade 2 hypertension
  - Grade 2 hematologic toxicity
  - Clinically significant grade 2 electrolyte abnormality lasting <48 hours (with appropriate medical management)
  - Clinically insignificant laboratory abnormalities of any grade

If recurrent grade 2 toxicity is observed during administration of the step-up doses (eg, grade 2 CRS) before reaching the full dose, then the SSET will pause dose escalation and consider revisions to the step-up regimen. Any proposed change will be introduced by protocol amendment.

The maximum dose increment will be limited to 30% (eg, 50 to 65 mg) if 2 participants experience an AEI out of 11 or 12 dose-evaluable patients during the safety observation period.

See Appendix 6 for all modified full dose levels.

6.1.3.4. Study Safety Evaluation

The SSET is responsible for overseeing the safety run-in period and start of expansion. Formal meetings will be conducted regularly including before a decision to escalate the full dose, at the end of the safety run-in, and when required on an ad hoc basis. Members of the SSET will include representatives from the sponsor's clinical study team (the medical monitor, study biostatistician, and global patient safety lead) and the clinical trial principal investigators. Information reviewed by the SSET will be comprehensive and include participant demographics, medical/oncology history, frequency and severity of AEs, clinical and specialty laboratory data, and available clinical PK or pharmacodynamic data. A final decision on whether or not to proceed with dose escalation or expansion (part 2) will ensue after thorough discussion and be contingent upon the consensus judgment of the SSET.

The sponsor will continue to monitor cumulative safety data throughout the entire treatment period for all participants (safety run-in and expansion).

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6.1.3.5. Conditions for Enrollment Pause and Safety Review

All participants will be monitored for trigger events over the entire treatment period, starting with the first infusion of linvoseltamab.

The trigger events will include all cases of grade ≥3 CRS, grade ≥2 ICANS, and grade ≥3 treatment-related AEs (except grade 3 or 4 lymphopenia and isolated grade 3 laboratory abnormalities without clinical consequence).

A stopping bound is set based on the lower bound of the exact 1-sided 80% CI calculated using the Clopper-Pearson method. Thresholds have been defined for the cumulative rate of all trigger events and for specific AEs. For all trigger events, if the lower bound exceeds 20%, then enrollment will be paused. Threshold examples are shown in Table 4.

Table 4: Stopping Boundaries to Pause Enrollment for Cumulative Rate of All Trigger Events (Part 1 and Part 2)

Number of participants treated	Number of participants with trigger events leading to pause
2 to 4	≥ 2
5 to 7	≥ 3
8 to 11	≥ 4
12 to 15	≥ 5
16 to 20	≥ 6

For grade ≥3 treatment-related infection and grade ≥4 non-hematologic AEs, if the lower bound exceeds 10%, then enrollment will be paused (Table 5).

Table 5: Stopping Boundaries to Pause Enrollment for Cumulative Rate of Treatment-Related Grade ≥3 Infection and Grade ≥4 Non-Hematologic AEs

Number of participants treated	Number of participants with trigger events leading to pause
3 to 8	≥ 2
9 to 15	≥ 3
16 to 20	≥ 4

Enrollment will also be immediately paused for any of the following:

- Grade 4 CRS or ICANS
- Treatment-related death

During a study pause, accrual will be suspended, and relevant regulatory authorities will be notified. The sponsor will review all available data for participants treated with linvoseltamab and re-evaluate the dosing regimen(s) and dose selection. Remedial actions such as dose reduction will be considered based on the overall risk-benefit assessment.

6.1.3.6. Individual Patient Stopping Rules

Reasons for permanent discontinuation of study drug are listed in Section 8.2.3.1.

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6.1.4. End of Study Definition

An individual participant will reach the end of study whenever one of the following events occurs: completion of 5 years of follow-up after the last dose of study drug, death, loss to follow-up, or withdrawal of consent.

The study/trial will end on the date of the last participant's last visit for any reason listed above or study termination by the sponsor, whichever occurs earlier.

6.2. Planned Interim Analyses

No interim analysis will be performed.

6.3. Study Committees

A formal committee is not required. The sponsor may elect to obtain independent reviews of the efficacy data. The process for such reviews would be set forth in a separate charter.

The recommendation to stop the study prematurely due to safety prior to the end of study will be made by the SSET (Section 6.1.3.4). A final determination will be made by senior leaders in the sponsor's organization.

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7. SELECTION, WITHDRAWAL, AND REPLACEMENT OF PARTICIPANTS

7.1. Number of Participants Planned

Approximately 104 participants including 6 to 24 participants in part 1 (safety run-in and dose escalation) and a minimum of 80 participants in part 2 (expansion).

7.2. Study Population

The study population is adult patients with HR-MGUS or NHR-SMM.

7.2.1. Inclusion Criteria

A participant must meet the following criteria to be eligible for inclusion in the study:

1. Age ≥18 years at the time of signing the informed consent or ≥legal age of adulthood based on local regulations, whichever is older.
2. HR-MGUS or NHR-SMM defined as:
   - HR-MGUS defined as BMPCs <10%, <3 g/dL serum M-protein, and at least 2 of the following 3 criteria: abnormal serum FLC ratio of <0.26 or >1.65, serum M-protein ≥1.5 g/dL, or non-IgG M-protein. In the case of light-chain MGUS, involved-to-uninvolved sFLC ratio >10.
   - NHR-SMM defined as 1 or more of the following:
     - serum M-protein ≥3 g/dL
     - urinary M-protein ≥500 mg per 24 hours
     - BMPCs ≥10%

     AND

     No evidence of high-risk criteria or myeloma-defining events specified in exclusion criterion #1.
3. ECOG performance status ≤1
4. Adequate hematologic function:
   a. Absolute neutrophil count ≥1500/µL ($1.5x10^9$/L). No G-CSF is permitted to reach this value during the screening period.
   b. Platelets ≥75,000/µL ($75x10^9$/L) (not permissible to transfuse a participant within 2 weeks prior to screening to reach this level)
   c. Clinically stable anemia that is clearly attributable to a condition other than MM (eg, thalassemia, iron deficiency) and that has not required transfusions in the prior year may be considered for inclusion after discussion with the medical monitor.
5. Adequate hepatic function:
   a. Total bilirubin ≤2 mg/dL (except known Gilbert syndrome, in which case indirect bilirubin should be ≤5 mg/dL)

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b. Transaminases (ALT/AST) ≤3x ULN

6. Estimated GFR ≥30 mL/min/1.73 m^2 by the MDRD equation (Appendix 4).
7. Contraception

a. Men: A male participant must agree to use condoms or practice sexual abstinence[†, ‡] unless vasectomized prior to administering the first step-up dose of linvoseltamab, during the study, and for at least 6 months after the last dose. Sperm donation is prohibited during the same period. Vasectomy success must be confirmed by semen analysis.

[†] Sexual abstinence is considered a highly effective method of contraception only if defined as refraining from heterosexual intercourse during the entire period of risk associated with the study regimen. The reliability of sexual abstinence needs to be evaluated in relation to the duration of the clinical trial and the preferred and usual lifestyle of the participant.

[‡] Periodic abstinence (calendar, symptothermal, post-ovulation methods), withdrawal (coitus interruptus), spermicides only, and LAM are not acceptable methods of contraception. Female condom and male condom should not be used together.

b. Women: A female participant must not be pregnant or breastfeeding and must have a negative pregnancy test at screening (serum β-hCG).

WOCBP must agree to undergo pregnancy testing and must practice highly effective contraception* prior to administering the first step-up dose of linvoseltamab, during the study, and for at least 6 months after the last dose. Egg donation (ova, oocytes) for the purposes of assisted reproduction is prohibited during the same period.

– *WOCBP are defined as women who are fertile following menarche until becoming postmenopausal unless permanently sterile. Permanent sterilization methods include hysterectomy, bilateral salpingectomy, and bilateral oophorectomy. A postmenopausal state is defined as no menses for 12 months without an alternative medical cause. An FSH level in the postmenopausal range may be used to confirm a postmenopausal state in women not using hormonal contraception or hormonal replacement therapy. However, in the absence of 12 months of amenorrhea, a single FSH measurement is insufficient to determine the occurrence of a postmenopausal state. The above definitions are according to the CTFG guidance.

* Highly effective contraceptive measures include:

- Stable use of combined (estrogen and progestogen containing) hormonal contraception (oral, intravaginal, transdermal) or progestogen-only hormonal contraception (oral, injectable, implantable) associated with inhibition of ovulation initiated 2 or more menstrual cycles prior to screening;
  - IUD or IUS;
  - Bilateral tubal ligation (also called bilateral tubal occlusion);
  - Vasectomized partner (provided that the male vasectomized partner is the sole sexual partner of the WOCBP study participant and that the

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vasectomized partner has obtained a medical assessment of surgical success); and/or

- Sexual abstinence.[†, ‡]

Pregnancy testing and contraception are not required for women who are postmenopausal or permanently sterile.

8. Capable of giving voluntary written informed consent as described in the protocol.

7.2.2. Exclusion Criteria

A participant who meets any of the following criteria will be excluded from the study:

1. High-risk SMM according to the 20-2-20 risk stratification model (Mayo criteria) defined as the presence of any 2 of the following:
   - Serum M-protein >2 g/dL
   - Involved-to-uninvolved serum FLC ratio >20
   - BMPCs >20%

   OR

   - Total score of >9 points using the following scoring system:
     - FLC ratio
       - >10–25 = 2 points
       - >25–40 = 3 points
       - >40 = 5 points
     - Serum M-protein (g/dL)
       - >1.5–3 = 3 points
       - >3 = 4 points
     - BMPC%
       - >15–20 = 2 points
       - >20–30 = 3 points
       - >30–40 = 5 points
       - >40 = 6 points
     - FISH abnormality (t(4;14), t(14;16), 1q gain, or del(13q)) = 2 points
2. Evidence of any of the following myeloma-defining events (SLiM-CRAB) attributable to the underlying plasma cell dyscrasia:
   - Hypercalcemia: corrected serum calcium >1 mg/dL above the ULN or >11 mg/dL
   - Renal insufficiency: GFR <40 mL/min by MDRD equation or serum creatinine >2 mg/dL (Appendix 4)
   - Anemia: hemoglobin >2 g/dL below the lower limit of normal or <10 g/dL

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Transfusion support or concurrent treatment with erythropoietin-stimulating agents is not permitted.

- ≥1 osteolytic bone lesion detected by low-dose CT or FDG-PET/CT (x-ray is not acceptable)
- BMPCs ≥60%
- Serum involved/uninvolved FLC ratio ≥100
- Whole body MRI with more than 1 focal lesion (≥5 mm in size)

3. Diagnosis of systemic light-chain amyloidosis, Waldenström macroglobulinemia (lymphoplasmacytic lymphoma), soft tissue plasmacytoma, or symptomatic MM
4. Clinically significant cardiac or vascular disease within 3 months of study enrollment including but not limited to myocardial infarction, unstable angina, coronary revascularization (eg, coronary artery bypass graft, percutaneous coronary intervention), peripheral artery revascularization, left ventricular ejection fraction <40% by Echo or MUGA scan, NYHA class III or IV heart failure, transient ischemic attack, stroke, thromboembolic event (eg, pulmonary embolism), or unstable cardiac arrhythmia.
5. Any infection requiring hospitalization or treatment with IV anti-infectives within 28 days of the first dose of linvoseltamab.
6. Uncontrolled HIV, HBV, or HCV infection; other uncontrolled infection; or unexplained signs of infection (eg, fever of unknown origin, rising WBC count):
   a. Participants with HIV who have controlled infection (undetectable viral load and CD4 count above 350 cells/microliter either spontaneously or on a stable antiviral regimen) are permitted.
   b. Participants with HBV (HBsAg-positive) who have controlled infection (serum HBV DNA PCR that is below the limit of detection AND receiving antiviral therapy for HBV) are permitted. Participants with occult HBV infection (negative HBsAg and positive HBV DNA) observed during the screening period should be evaluated by a specialist for start of antiviral treatment and may enroll if HBV DNA becomes negative.
   c. Participants who are HCV antibody positive (HCV antibody-positive) who have controlled infection (undetectable HCV RNA by PCR either spontaneously or in response to a successful prior course of anti-HCV therapy) are permitted.
7. History of severe allergic reaction attributed to compounds with a similar chemical or biologic composition as the study drug or excipient.
8. Known hypersensitivity to both allopurinol and rasburicase.
9. Participants who are committed to an institution by virtue of an order issued either by the judicial or the administrative authorities.
10. Any severe acute or chronic medical condition that could impair the ability of the participant to take part in the study or interfere with the interpretation of study results (eg, presence of systemic infection unless anti-infective therapy is employed, or if the participant is unable to comply with the study procedures.

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11. Diagnosis or treatment of another malignancy within 3 years of enrollment with the exception of completely resected basal cell carcinoma or squamous cell carcinoma of the skin, a definitively managed in situ malignancy, or low-risk prostate cancer after curative therapy. Ongoing hormonal therapy for a history of either prostate cancer or breast cancer is acceptable provided there has been no evidence of disease progression for at least 3 years prior to enrollment.
12. Prior exposure to approved or investigational treatments for SMM or MM (including but not limited to conventional chemotherapies, immunomodulatory drugs, proteasome inhibitors, anti-CD38 antibodies). Concurrent use of bisphosphonates or the RANKL inhibitor denosumab is not permitted; however, prior antiresorptives given for the treatment of osteoporosis are permissible.
13. Ongoing corticosteroid treatment with a dose >10 mg prednisone or equivalent per day at the time of screening.
14. History of progressive multifocal leukoencephalopathy, neurodegenerative condition, CNS movement disorder, or seizure within 12 months prior to study enrollment.
15. Live or live attenuated vaccination within 28 days of planned start of study drug; vaccination within 1 week of planned start of study drug including anticipated completion of vaccination series within 1 week of study drug initiation.
16. History of allogeneic stem cell transplantation.

7.3. Premature Withdrawal from the Study

A participant has the right to withdraw from the study at any time, for any reason, and without repercussion.

The investigator and/or sponsor have the right to withdraw a participant from the study if it is no longer in the interest of the participant to continue in the study or if the participant's continuation in the study places the scientific outcome of the study at risk (eg, if a participant does not or cannot follow study procedures). An excessive rate of withdrawals would render the study uninterpretable; therefore, unnecessary withdrawal of participants should be avoided.

Prior to withdrawal, participants should be asked to complete the EOT visit, as described in Section 9.1.3.

Rules for discontinuation of study treatment (permanent or temporary) are discussed in Section 8.2.

7.4. Replacement of Participants

Participants prematurely withdrawn from the study can be replaced, if needed, to ensure an adequate number of evaluable participants. The medical/study director in cooperation with the study biostatistician will decide whether to replace a withdrawn participant.

Participants enrolled and withdrawn from the study at any time prior to administering the first step-up dose of linvoseltamab or prior to having at least one postbaseline response assessment will not be considered evaluable for efficacy assessment and will be replaced with another participant. All

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participants who have received a dose of study drug and have at least one postbaseline response assessment will be included in an as-treated analysis.

8. STUDY TREATMENTS

The study interventions administered in the study are outlined in Table 6 for IMP, Table 7 for premedications as AxMP, and Table 9 for rescue medications as AxMP.

8.1. Investigational and Reference Treatments

8.1.1. Linvoseltamab Investigational Treatment

A flat dose of linvoseltamab will be administered by IV infusion and will not be dependent on participant weight or body surface area. REGN5458 is a synonym for linvoseltamab. See the pharmacy manual for further details.

A pharmacist or other qualified individual will be identified at each site to prepare linvoseltamab for administration. Instructions on dose preparation are provided in the pharmacy manual.

Table 6: Study Intervention

Intervention Name	Linvoseltamab (REGN5458)
Dose Formulation	Sterile solution
Unit Dose Strength(s)	10 mg/mL
Dosage Level(s)	See Section 6.1.1
Route of Administration	IV infusion
Use	Experimental
Medicinal Product Type	IMP
Sourcing	Provided centrally by the sponsor
Packaging and Labeling	Linvoseltamab will be provided in sterile single-use vials. Each vial will be labeled as required per country requirements.

IMP=investigational medicinal product; IV=intravenous

Each participant will start IV linvoseltamab treatment with a 3-week step-up schedule: 1 mg at week 1, 4 mg at week 2, and 25 mg at week 3. The subsequent fixed-duration dosing regimen will begin on C1D1 according to the assigned full dose of linvoseltamab and schedule of administration:

- Schedule 1: QW for the first cycle followed by Q4W for 5 cycles (part 1 and part 2)
- Schedule 2: Q2W for 3 cycles (part 2 only)

The step-up doses and first full dose of linvoseltamab will be administered as continuous infusions over 4 hours. If the first full dose is adequately tolerated without CRS or IRR events of any grade, then subsequent linvoseltamab infusions may be reduced to 1 hour. If the 1-hour infusion time is tolerated, then subsequent infusions may be reduced to 30 minutes.

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Inpatient monitoring is required for the first 24 hours after each step-up dose as indicated in the schedules of events (Table 14 and Table 15). Additional inpatient monitoring can be implemented at the discretion of the investigator.

The safety run-in and dose escalation part of the study (part 1) will focus on the safety and tolerability of linvoseltamab from the start of the step-up period through administration of at least 1 full dose during a 35-day safety observation period. Based on review of all data gathered during the safety run-in, the recommendations for inpatient monitoring may be updated.

8.1.2. Premedication

Premedication with dexamethasone, an antihistamine, and paracetamol (acetaminophen) is required prior to linvoseltamab infusions during the step-up period and first full dose according to Table 7.

Table 7: Auxiliary Medicinal Product: Premedications for Linvoseltamab (Step-up Period and First Full Dose)

Intervention Name	Dexamethasone	Antihistamine (Diphenhydramine or equivalent)	Acetaminophen (Paracetamol)
Sourcing	Site	Site	Site
Formulation	Determined by site	Determined by site	Determined by site
Unit Strength(s)			
Packaging and Labeling			
Dosage Level(s)	40 mg[a]	25 mg	650 to 1000 mg
Route of Administration	IV (1 to 3 hours prior to dose)	IV or PO (30 to 60 minutes prior to dose)	PO (30 to 60 minutes prior to dose)
Use	Premedication	Premedication	Premedication
Medicinal Product Type	AxMP	AxMP	AxMP

[a] Dose may be reduced to 20 mg for participants ≥75 years of age or those who experience toxicities attributable to corticosteroids following the first dose of linvoseltamab.
AxMP=auxiliary medicinal product; IV=intravenous; PO=oral

If the participant experiences IRR and/or CRS of any grade upon administration of the first full dose of linvoseltamab, premedication should continue until the full dose is tolerated without signs of IRR and/or CRS.

If the full dose is adequately tolerated without CRS or IRR of any grade, the investigator may continue treatment without premedication or with a reduced dose of dexamethasone (eg, 10 mg).

If linvoseltamab is adequately tolerated with a tapered dose of dexamethasone, then discontinuation prior to subsequent treatment administration is encouraged.

When corticosteroids are discontinued, the antihistamine and acetaminophen may also be discontinued at the investigator's discretion.

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8.2. Dose Modification and Study Treatment Discontinuation Rules

8.2.1. Linvoseltamab Dose Modification

Changes in dose (ie, dose reductions) are generally not allowed (see exceptions in Table 11). Delayed dosing of linvoseltamab is recommended in accordance with the below guidance or after consultation with the sponsor.

For clinically significant grade 3 hematologic toxicity:

- Treatment should be paused in the following situations: grade 3 anemia, grade 3 thrombocytopenia with bleeding, grade 3 febrile neutropenia, or grade 3 neutropenia with documented infection.
- Treatment (if paused) can be restarted when toxicity improves to grade ≤2 and, if applicable, bleeding or infection is adequately controlled.

For clinically significant grade 4 hematologic toxicity:

- Treatment should be temporarily paused.
- Treatment can be restarted when toxicity improves to grade ≤2.
- Treatment should be permanently discontinued if toxicity does not resolve to grade <4 within 2 weeks of last infusion or to grade 2 within 8 weeks of last infusion with appropriate supportive measures.

Participants who experience grade 3 non-hematologic toxicity not otherwise specified in this protocol should pause treatment until the toxicity improves to grade 1 or baseline. Exceptions to this guidance are permitted when laboratory abnormalities are considered clinically insignificant, or when health conditions unrelated to study treatment are stable and manageable with standard interventions. Study treatment should be permanently discontinued in the event of grade 4 non-hematologic toxicity (see limited exceptions in Section 8.2.3.1).

For participants anticipated to restart therapy after a prolonged treatment interruption (>28 days), the sponsor should be consulted to discuss the participant's individual risk factors, such as prior history of CRS, before resuming treatment. If treatment is resumed at a dose less than the previously received full dose, the need for post-infusion inpatient monitoring should also be discussed with the sponsor. See Table 8 for guidelines on treatment resumption after a dosing delay.

Table 8: Guidance on Treatment Resumption After Dosing Delay

Time since last dose	Last dose administered	Action
Step-up period treatment interruption (before full dose administration on C1D1)[a,b]		
≤14 days	1 mg	Proceed to 4 mg
	4 mg	Proceed to 25 mg
	25 mg	Proceed to full dose
>14 days to ≤28 days	1 mg	Repeat 1 mg, then escalate as tolerated

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		4 mg	Repeat 4 mg, then escalate as tolerated
		25 mg	Repeat 25 mg, then escalate if applicable
>28 days		Any	Permanently discontinue study treatment
Full dose treatment interruption (after full dose administration on C1D1)			
≤28 days		Any	Resume full dose according to treatment schedule
>28 days to ≤60 days	Cycle 1	25 mg	Repeat 4 mg, then re-escalate to full dose[b]
		≥50 mg	Repeat 25 mg, then re-escalate to full dose[b]
	Cycle 2+	25 mg	Repeat 4 mg, then re-escalate to full dose[b]
		≥50 mg	Resume full dose according to treatment schedule
>60 days		Any	Permanently discontinue study treatment

[a] Treatment resumption at a lower dose may be appropriate if the dosing delay is caused by CRS or ICANS. See Table 12 or Table 13 for guidance related to CRS or ICANS management, respectively.
[b] Administer premedications prior to linvoseltamab infusion (Section 8.1.2) and monitor accordingly. All repeat doses given during the step-up period require post-infusion inpatient monitoring (Section 8.1.1).

The investigator should carefully review with each participant the benefits and risks of restarting versus permanently discontinuing linvoseltamab after a dosing delay >28 days due to an AE. If treatment is resumed >28 days after the administration of at least 1 full dose, premedications are recommended per Section 8.1.2. The infusion duration and associated monitoring as well as the sample collection schedule should follow the procedures specified for the corresponding step-up dose or first full dose detailed in Section 8.1.1 and the Schedule of Events (Section 9.1). Repeat efficacy labs and biomarker samples are not required. Participants who experience a treatment interruption of >60 days must permanently discontinue treatment with linvoseltamab.

8.2.2. Dosing Schedule after Treatment Interruption

All participants must receive each of the incremental step-up doses during the step-up period before receiving the full dose on C1D1. Repeat step-up doses are permitted during the step-up period to manage tolerability limited by CRS/IRR (Table 11). Additional step-up doses should be recorded as unscheduled visits and follow the same assessment schedule as the preceding step-up dose. The pharmacogenomics sample and efficacy labs do not need to be repeated for the W1D1 step-up dose. If a treatment interruption occurs after the start of full-dose therapy on C1D1, dosing should resume where the participant left off (ie, the last scheduled visit where dosing did not occur). When treatment must resume at a lower dose per the treatment resumption guidelines in Table 8, then administration of the lower dose should be recorded as an unscheduled visit, and re-escalation to the full dose should occur at the scheduled visit when treatment was first interrupted. Participants who recover from AEs that result in treatment delays are not bound by the dosing windows specified in the Schedule of Events when treatment is resumed.

8.2.3. Study Drug Discontinuation

Participants who permanently discontinue treatment with study drug should be encouraged to remain in the study for ongoing follow-up and applicable assessments (Section 9.1.3). Those who agree and do not withdraw from the study will be asked to return to the clinic for all remaining study visits per the follow-up schedule.

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8.2.3.1. Reasons for Permanent Discontinuation of Study Drug

Study drug dosing will be permanently stopped in the event of:

- Confirmed biochemical progression
- Start of an alternative anticancer treatment
- Evidence of pregnancy
- Serious or severe allergic reactions considered related to study drug
- Specific types of liver dysfunction; eg, Hy's law is met except in the context of CRS (FDA, 2009)
- Participant withdraws consent
- Any medical condition that the investigator or sponsor determines may jeopardize the participant's safety if the participant continues in the study or continues treatment with study drug
- The investigator determines it is in the best interest of the participant
- Participant non-compliance (eg, not complying with protocol-required visits, assessments, and dosing instructions)
- Tuberculosis or uncontrolled opportunistic infection (including but not limited to CMV or PJP unresponsive to therapy)
- Diagnosis of another malignancy during the study except non-melanoma skin cancer curable with local therapy or in situ cancers
- Any grade 4 toxicity

  Exceptions:

  - Isolated laboratory abnormalities without clinical consequence
  - Clinically significant laboratory abnormalities such as those associated with CRS that improve to grade <4 within 72 hours after onset
  - Hematologic laboratory abnormalities (Section 8.2.1)

8.3. Safety Monitoring and Management of Specific Toxicities and Acute Reactions

Multiple measures will be taken to ensure the safety of the participants in this trial. This includes stringent eligibility criteria, close clinical safety monitoring according to the schedule of assessments, and requirement for hospitalization.

Upon occurrence of a toxicity, the investigator, with input from the sponsor, as needed, will assess the etiology of the toxicity. As a general principle, IRR and CRS are the most commonly encountered AEs during the step-up doses of linvoseltamab and may warrant temporary delays or dose modifications. Specific requirements on when to delay or modify the dose and the discontinuation criteria for linvoseltamab are provided in Section 8.2. Detailed guidance is provided below for specific AEs that are anticipated, such as acute allergic/hypersensitivity

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reactions, IRR/CRS, and ICANS. Guidance for the anticipated B-cell depletion and associated risk of infection as well as the potential need for TLS prophylaxis is also provided.

8.3.1. Acute Intravenous Infusion Reactions

Emergency equipment and medication for the treatment of infusion reactions must be available for immediate use. All infusion reactions must be reported as AEs (as defined in Section 10.2.4) and graded using the grading scales as instructed in Section 10.2.5.

8.3.1.1. Interruption of the Intravenous Infusion

The infusion should be interrupted if any of the following AEs are observed:

- Sustained/severe cough
- Rigors/chills
- Rash, pruritus (itching)
- Urticaria (hives, welts, wheals)
- Diaphoresis (sweating)
- Hypotension
- Dyspnea (shortness of breath)
- Vomiting
- Flushing

The reaction(s) should be treated symptomatically, and the infusion may be restarted at 50% of the original rate.

If investigators feel there is a medical need for treatment or discontinuation of the infusion other than described above, they should use clinical judgment to provide the appropriate response according to typical clinical practice.

8.3.1.2. Termination of the Intravenous Infusion

The linvoseltamab infusion should be terminated and NOT restarted if any of the following AEs occur:

- Anaphylaxis*
- Laryngeal/pharyngeal edema
- Severe bronchospasm
- Chest pain
- Seizure
- Severe hypotension
- Other neurological symptoms (confusion, loss of consciousness, paresthesia, paralysis, etc)

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- Any other symptom or sign that, in the opinion of the investigator, warrants termination of the IV infusion

*Consider anaphylaxis if the following is observed (Sampson, 2006): acute onset of an illness (minutes to several hours) with involvement of the skin, mucosal tissue, or both (eg, generalized hives, pruritus or flushing, swollen lips-tongue-uvula) and AT LEAST ONE of the following:

- Respiratory compromise (eg, dyspnea, wheeze-bronchospasm, stridor, reduced peak expiratory flow, hypoxemia)
- Reduced blood pressure or associated symptoms of end-organ dysfunction (eg, hypotonia [collapse], syncope, incontinence)

8.3.2. Grading and Management of Infusion-Related Reaction and Cytokine Release Syndrome

An acute IRR is defined as any AE that occurs less than 6 hours from the start of the infusion, or within 2 hours after completion of the infusion (whichever is later), and is associated with typical signs and symptoms including, but not limited to, flushing, tachycardia, hypotension, dyspnea, bronchospasm, back pain, fever, urticaria, edema, nausea, and rashes. IRRs must be reported as AEs (Section 10.1.2) and graded according to the NCI-CTCAE v5.0 (Section 10.2.5).

CRS is a disorder characterized by fever, tachypnea, headache, tachycardia, hypotension, rash, and/or hypoxia; in this study, it is defined as such an event that occurs 6 or more hours from the start of the infusion or more than 2 hours after completion of the infusion (whichever is later). Emergency equipment and medication for the treatment of these potential adverse effects (eg, anticytokine therapy [such as tocilizumab], antihistamines, bronchodilators, IV saline, corticosteroids, acetaminophen, and/or epinephrine) must be available for immediate use (Table 9). Grade ≥2 CRS must be reported as an AESI (Section 10.1.3) and graded according to the criteria adapted from (Lee, 2019) (Table 10).

Note: 6 hours as a time criterion to distinguish between CRS and other IRRs is established for reporting purposes only and is not a validated cutoff to distinguish between CRS and other types of IRRs. In any case, the judgment of the investigator should be used, and if symptoms and/or laboratory evaluation suggest development of CRS before 6 hours from the start of the infusion, the participant should be treated, and the event should be reported as CRS.

IRRs and CRS have been observed in participants treated with linvoseltamab (see Section 3.2.4.2 and the linvoseltamab IB).

An extensive clinical and laboratory monitoring plan is in place to allow for early identification of participants who are more likely to develop CRS, and this allows for early initiation of increased participant monitoring and therapeutic intervention.

Participants who develop symptoms consistent with severe CRS, including but not limited to persistent fevers, neurologic disorders (including mental status changes, obtundation, and seizures), or clinical signs of toxicity (hypotension requiring at least 1 IV vasoactive pressor or hypoxia [pO2 < 90%]), should be considered for pharmacologic intervention with anticytokine therapy (eg, tocilizumab and/or corticosteroids), and the sponsor should be contacted to discuss management (Davila, 2014) (Grupp, 2013). Additional guidance regarding the use of anticytokine therapy is provided in Table 12.

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CRS and IRR management guidelines, including clinical evaluation and monitoring, are detailed in Table 10, Table 11, and Table 12.

Emergency equipment and medication for the treatment of infusion reactions must be available for immediate use.

Table 9: Auxiliary Medicinal Product: Rescue Medications

Intervention Name	Tocilizumab	Dexamethasone (Corticosteroid)	Others (Section 8.3.2)
Sourcing	Site	Site	Site
Formulation	Determined by site	Determined by site	Determined by site
Unit Strength(s)			
Packaging and Labeling			
Dosage Level(s)	4 to 8 mg/kg; dose not to exceed 800 mg	20 mg per day in divided doses	According to label
Route of Administration	Infusion	Infusion or oral	According to label
Use	Rescue medication	Rescue medication	Rescue medication
Medicinal Product Type	AxMP	AxMP	AxMP

AxMP=auxiliary medicinal product

Table 10: Cytokine Release Syndrome Toxicity Grading

CRS Parameter	**Grade 1**	**Grade 2**	**Grade 3**	**Grade 4**
Fever[1]	Temperature ≥38°C	Temperature ≥38°C	Temperature ≥38°C	Temperature ≥38°C
With				
Hypotension	None	Not requiring vasopressors	Requiring vasopressors with or without vasopressin	Requiring multiple vasopressors (excluding vasopressin)
And/Or[2]				
Hypoxia	None	Requiring low-flow nasal cannula[3] or blow-by	Requiring high-flow nasal cannula[3], facemask, non-rebreather mask, or Venturi mask or blow-by	Requiring positive pressure (eg, CPAP, BiPAP, intubation, and mechanical ventilation)

[1] Fever is defined as temperature ≥ 38°C not attributable to any other cause. In participants who have CRS and then receive antipyretic or anticytokine therapy such as tocilizumab or steroids, fever is no longer required to grade subsequent CRS severity. In this case, CRS grading is driven by hypotension and/or hypoxia.

[2] CRS grade is determined by the more severe event: hypotension or hypoxia not attributable to any other cause. For example, a participant with a temperature of 39.5°C, hypotension requiring 1 vasopressor, and hypoxia requiring low-flow nasal cannula is classified as grade 3 CRS.

[3] Low-flow nasal cannula is defined as oxygen delivered at ≤6 L/minute. Low flow also includes blow-by oxygen delivery, sometimes used in pediatrics. High-flow nasal cannula is defined as oxygen delivered at >6 L/minute.

BiPAP=bilevel positive airway pressure, CPAP=continuous positive airway pressure, CRS=cytokine release syndrome

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Note: Organ toxicities associated with CRS may be graded according to CTCAE v5.0, but they do not influence CRS grading.
Adapted from ASTCT CRS Consensus Grading (Lee, 2019).

Table 11: Management Guidance and Linvoseltamab Dosing in Setting of Cytokine-Mediated Toxicities & Infusion Related Reactions

Event	Grade	Action with Linvoseltamab	Management Guide	Rechallenge
Acute IRR*	2	Temporarily pause study drug	Treat symptomatically: Supportive care, antipyretics (eg, acetaminophen [paracetamol]), meperidine.	No dose modification required. May resume when clinical symptoms are grade 1 or baseline. Consider an infusion rate of ≤50% of previous rate (up to 6 hr total). If the rate is reduced and tolerated, for subsequent infusions, the infusion rate may be increased to 100%.
	3	Temporarily pause study drug	Treat symptomatically: Supportive care, antipyretics (eg, acetaminophen [paracetamol]), meperidine. Consider corticosteroids. Vigilant supportive care is needed. The investigator should consider a possible CRS diagnosis and need for applicable CRS treatment.	May resume treatment when clinical symptoms are grade 1 or baseline. Reduce dose by 50% from the previous dose with infusion rate of ≤50% of previous rate (up to 6 hr total). If the reduced dose is tolerated with no signs/symptoms of grade ≥3 IRR, the participant may return to the previous dose with ≤50% rate of infusion. The infusion rate may be increased to 100% if tolerable and, if tolerated, then proceed to the full dose. If grade 3 IRR recurs with subsequent doses, permanently discontinue treatment. Continue dosing if no recurrence of toxicity.

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Event	Grade	Action with Linvoseltamab	Management Guide	Rechallenge
	4	Permanently discontinue study drug	Treat symptomatically: Supportive care, antipyretics (eg, acetaminophen [paracetamol]) and meperidine. Also consider corticosteroids. Vigilant supportive care is needed. The investigator should consider a possible CRS diagnosis and need for applicable CRS treatment.	Not applicable.
CRS*	1	Temporarily pause study drug	Vigilant supportive care (ie, assess for infection; treat febrile neutropenia if present, monitor fluid balance, antipyretics/ analgesics as needed); monitor cardiac and other organ functions closely; symptomatic management of organ toxicities. Initiate corticosteroids for participants with organ toxicities (listed in Table 12) excluding neurotoxicity. Consider corticosteroids and/or anticytokine therapy (eg, tocilizumab) as per institutional/consensus guidelines. Tocilizumab may be used for participants with comorbidities, older age >65 years, or symptoms refractory to supportive measures (Reagan, 2021).	No dose modification required. May resume when clinical symptoms are grade 1 or baseline. Consider decreasing the infusion rate to ≤50% of previous rate (up to 6 hr total). If the rate is reduced and tolerated, for subsequent infusions, the infusion rate may be increased to 100%.
	2	Temporarily pause study drug	Vigilant supportive care as per grade 1: Administer anticytokine therapy (eg, tocilizumab) as per institutional/consensus guidelines (Reagan, 2021).	No dose modification required. During the step-up dosing period, dose modification may be allowed after discussion with the

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Event	Grade	Action with Linvoseltamab	Management Guide	Rechallenge
			Initiate corticosteroids for participants with organ toxicities (listed in Table 12) excluding neurotoxicity.	sponsor, including repeating or delaying a step-up dose. May resume when symptoms of CRS are grade ≤1 or baseline. Consider hospitalization and consider decreasing the infusion rate to ≤50% of previous rate (up to 6 hr total). If the rate is reduced and tolerated, for subsequent infusions the infusion rate may be increased to 100%.
	3	Temporarily pause study drug	Vigilant supportive care as per grade 2: Administer anticytokine therapy (eg, tocilizumab) and corticosteroids.	May resume linvoseltamab when symptoms of CRS are grade ≤1 or baseline. Hospitalize the participant and reduce dose by 50% from the previous dose with infusion rate of ≤50% of previous rate (up to 6 hr total). Consider prophylactic anticytokine therapy (eg, tocilizumab). If the reduced dose is tolerated with no signs/symptoms of grade ≥2 CRS, the participant may return to the previous dose with ≤50% rate of infusion. The infusion rate may be increased to 100% if tolerable. If the reduced dose of linvoseltamab is tolerated, then proceed to the full dose. If grade 3 CRS recurs with subsequent doses, permanently discontinue treatment.

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Event	Grade	Action with Linvoseltamab	Management Guide	Rechallenge
	4	Permanently discontinue study drug	Vigilant supportive care as per grade 3.	Not applicable.
Transaminase elevations (grade ≥3) associated with CRS*	≥3	Temporarily pause study drug	Supportive care as needed, monitor closely. Initiate corticosteroids immediately (see Table 12). Study drug will be temporarily withheld, and subsequent study drug treatment administrations will not resume until laboratory parameters are trending toward normal.	Resume linvoseltamab only when symptoms of CRS are grade ≤1 or baseline. No dose adjustment is required if the event is clinically uncomplicated and there is evidence of normalization within 7 days. Otherwise, reduce dose by 50%. Infusion rate adjustments should follow the guidance provided for each grade of CRS listed above.

*CRS will be graded according to criteria specified in (Lee, 2019). All other events will be graded according to criteria specified in NCI-CTCAE v5.0. Consistent with these criteria, grade 1 IRRs require no infusion interruption or intervention.

Note: If subsequent administration is tolerated, participants who received a reduced dose may have their dose increased at subsequent administration but no higher than the full dose, based on the clinical judgment of the investigator in consultation with the sponsor.

CRS=cytokine release syndrome; IRR=infusion-related reaction

Table 12: Guidelines for Administration of Anticytokine Therapy (eg, Tocilizumab) and Corticosteroids in Setting of Cytokine Release Syndrome

Event	Anticytokine Therapy Dose
Left ventricular ejection fraction <40% by echocardiogram Creatinine >2.5-fold higher than the most recent level prior to linvoseltamab activated PTT >2x ULN Clinically significant bleeding Creatine kinase >5 x ULN for longer than 2 days ALT/AST increase >5 x ULN	For example, 4 to 8 mg/kg tocilizumab infused over 1 hour, dose not to exceed 800 mg 20 mg dexamethasone per day in divided doses

Guidelines adapted from (Brudno, 2016). Note: This is not an exhaustive list.

ALT=alanine aminotransferase; AST=aspartate aminotransferase; PTT=partial thromboplastin time; ULN=upper limit of normal

8.3.3. ICANS

8.3.3.1. Study Definition of ICANS

ICANS is a stereotypic disorder characterized by a pathologic process involving the CNS following any immune therapy that results in the activation or engagement of endogenous or

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infused T cells and/or other immune effector cells. Symptoms or signs can be progressive and may include aphasia, altered level of consciousness, impairment of cognitive skills, motor weakness, seizures, and cerebral edema. ICANS may have features that overlap with other encephalopathies but has the more specific characteristic (when severe) of an awake participant who is mute and does not respond verbally or physically to an examiner (Lee, 2019).

Other neurological symptoms and/or signs may occur and possibly be attributable to immune effector cell–engaging therapies. As isolated signs/symptoms, these are not considered ICANS and are considered under other neurologic events. For example, headache or tremor with no decrease in ICE score (Appendix 5) or no decreased level of consciousness, disorientation, or other stereotypic symptoms or signs of ICANS would not be considered ICANS. Neurologic symptoms may be observed in association with other pathological processes, including the examples listed below. Neurologic symptoms considered wholly explained by other processes like these would not be considered ICANS.

- Sundowning or similar process, especially if associated with infection, dehydration, hepatic dysfunction, symptomatic hypertension/hypotension, electrolyte imbalance, or other metabolic abnormality
- Adverse reaction to immunosuppressive therapy, cytotoxic drug therapy, or other concomitant medication
- Underlying disease process (eg, a pre-existing seizure disorder)

The CARTOX-10–based assessment tool proposed by the ASTCT for grading neurotoxicity, termed the ICE score, is described in Section 9.2.3.4 and Appendix 5. The ICE score should be used to grade all cases of ICANS in this trial.

8.3.3.2. Management of ICANS

If a diagnosis of ICANS is suspected after excluding alternative etiologies (see Section 9.2.3.4 and Appendix 5), site personnel must complete the dedicated ICANS CRF. Guidance regarding the management of ICANS events is provided in Table 13.

Table 13: ICANS Management Guidance

Event	Grade	Action with Linvoseltamab	Management Guide	Rechallenge
ICANS	2	Refer to Section 6.1.3.5 and temporarily pause study drug	Treat symptomatically as per local guidelines. Suspend study drug at least temporarily until symptoms resolve to grade 1 or baseline. Participants should undergo appropriate evaluation to determine the etiology of the symptoms.	Any decision to restart treatment upon resolution of symptoms must be discussed and agreed upon by the investigator and the sponsor.
	≥3	Refer to Section 8.3.3 and discontinue if grade 4 or recurrent grade 3 ICANS	Treat symptomatically as per local or institutional guidelines.	Participants who experience a single episode of grade 3 signs and symptoms of CNS toxicity (excluding headache) that is

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Event	Grade	Action with Linvoseltamab	Management Guide	Rechallenge
			Participants should undergo appropriate evaluation to determine the etiology of the symptoms.	transient and resolves to grade 1 or baseline within 24 hours may continue the study drug regimen. Any decision to restart treatment after resolution of symptoms of a grade 3 event must be discussed and agreed upon by the investigator and the medical monitor. Participants who experience grade 4 or recurrent grade 3 toxicity consistent with ICANS must permanently discontinue study drug treatment.
Recurrent seizure	Any grade	Permanently discontinue study drug	Treat symptomatically as per local guidelines.	Not applicable.

CNS=central nervous system; ICANS=immune effector cell-associated neurotoxicity syndrome

8.3.4. B-Cell Depletion and Risk of Infection

Treatment with linvoseltamab is anticipated to deplete mature plasma cells and some subsets of B cells. This expected primary pharmacology may increase the risk of infections. In addition, MGUS and SMM have been associated with immune system dysfunction, which may further contribute to immunodeficiency. Standard myeloma guidelines recommend increased vigilance and frequent reassessment of each individual participant for risk of infection as well as early signs of infection.

Participants with known active bacterial, viral, fungal, mycobacterial, or other infection, or any major episode of infection requiring hospitalization or treatment with IV anti-infectives within 4 weeks prior to administering the first step-up dose of linvoseltamab are excluded from the trial. Infections of grade ≥3 will be reported as AESI (Section 10.1.3).

Recommendations for infection prevention:

- Vaccination against COVID-19, varicella zoster (recombinant vaccine), seasonal influenza, and pneumococcus is strongly recommended for all participants, preferably prior to study enrollment. Local guidelines regarding COVID-19 boosters should also be followed.
- Antiviral prophylaxis with acyclovir or equivalent medication (per institutional standards) is recommended for all participants throughout the study duration to prevent herpes simplex virus and varicella zoster virus reactivation.
- PJP prophylaxis may be considered for participants with a history of recurrent infections or other signs of immune dysfunction.
- Monitoring for viral reactivation (eg, CMV, EBV, HHV-6, adenovirus) is recommended whenever new signs of infection or cytopenias are observed.

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- Immunoglobulin replacement with IV immunoglobulin should be considered for participants with IgG concentrations less than 400 mg/dL.

More comprehensive guidance on infection prevention is available from the IDSA, NCCN, and IMWG (Raje, 2022).

8.3.5. Tumor Lysis Syndrome

TLS is uncommonly encountered during the management of plasma cell dyscrasias. Experience with linvoseltamab among RRMM patients indicates that the risk of TLS is low, and study participants with HR-MGUS or NHR-SMM are considered less at risk than patients with MM. However, tumor lysis prophylaxis, such as with fluid hydration and allopurinol, may be considered on an individual basis at investigator discretion and per institutional guidelines for participants who present with either an especially high tumor burden and/or indicators of rapid proliferation based on clinical (eg, a high uric acid and/or lactate dehydrogenase), biological (eg, high Ki-67), or molecular criteria (eg, a c-myc translocation).

8.4. Method of Treatment Assignment

All participants who sign the ICF will be assigned a unique participant number by the Interactive Response Technology at screening. Treatment with linvoseltamab will be conducted in an unblinded fashion.

A minimum of 80 participants will be randomized in part 2 at a 1:1:1:1 ratio to receive either the low dose or the high dose according to either schedule 1 or schedule 2 (see Section 6.1.1) as assigned by a central randomization scheme through an interactive voice response system / interactive web response system. Randomization will be stratified by number of 20-2-20 risk factors (0 or 1) as specified in Section 7.2.2, exclusion criterion #1.

8.5. Blinding/Masking

This is an open-label study.

8.6. Treatment Logistics and Accountability

8.6.1. Packaging, Labeling, and Storage

Linvoseltamab (R5458, REGN5458) will be provided in a sterile solution in a vial within cartons. Vials and cartons will have country-compliant labels. Open-label study drug will display the product lot number on the label.

Storage instructions will be provided in the pharmacy manual. Storage conditions of all drugs are provided on drug labeling and supersede all other sources of information.

8.6.2. Supply and Disposition of Treatments

Study drug will be shipped per labeled conditions to the investigator or designee at regular intervals or as needed during the study. At specified time points during the study (eg, interim site monitoring visits), at the site close-out visit, and following drug reconciliation and documentation by the site monitor, all opened and unopened study drug will be destroyed/returned to the sponsor or designee.

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8.6.3. Treatment Accountability

All drug accountability records must be kept current.

The investigator must be able to account for all opened and unopened study drug. These records should contain the dates, quantity, and study medication:

- Dispensed to each participant,
- Returned from each participant (if applicable), and
- Disposed of at the site or returned to the sponsor or designee.

All accountability records must be made available for inspection by the sponsor and regulatory agency inspectors; photocopies must be provided to the sponsor at the conclusion of the study.

8.6.4. Treatment Compliance

All drug compliance records must be kept current and made available for inspection by the sponsor and regulatory agency inspectors.

8.7. Concomitant Medications and Procedures

Any treatment administered from the time consent is obtained until 90 days following the last dose of study drug or start of non-protocol cancer therapy, whichever occurs first, will be considered concomitant medication. This includes medications that were started before the study and are ongoing during the study.

All concomitant medications, including the generic name, dose, dose unit, frequency, start/stop date, and relative administration to linvoseltamab must be recorded in the study CRF. Any treatments that are started subsequent to the 90-day follow-up visit should not be recorded, unless the medication is started to treat AEs deemed related to study drug.

8.7.1. Coronavirus Disease 2019 Vaccination During the Study

Participants are encouraged to be fully vaccinated against COVID-19 according to local guidelines before study enrollment. If a participant intends to receive a COVID-19 vaccine (completion of initial series or booster) during screening, day 1 of the study and the events associated with day 1 should not occur until at least 1 week after any COVID-19 vaccination dosing.

For participants enrolled in the safety run-in (part 1), COVID-19 vaccination should not occur during the 35-day safety observation period when dose-decision rules are in effect. Instead, COVID-19 vaccination should be finished at least 1 week prior to the first infusion of linvoseltamab or delayed until after completion of the 35-day safety observation period.

For all participants that intend to receive a COVID-19 vaccination while on study, it is recommended to delay COVID-19 vaccination, either as an initial series or as a booster dose, until they are receiving and tolerating the full dose of study drug.

COVID-19 vaccination, either as an initial series or as a booster dose received during the study, will be treated as a concomitant medication. As noted, administration of a COVID-19 vaccination should be separated from the time of administration of the investigational product (at least

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72 hours, ideally by at least 1 week) to avoid confounding the effects (eg, adverse effects) of the vaccine/booster with the effects of study drug.

8.7.2. Prohibited Medications and Procedures

During the period defined in Section 8.7, unless otherwise specified in Section 8.7.3, a participant may not receive any of the following:

- Corticosteroid: Ongoing systemic treatment of >10 mg per day of prednisone or equivalent except as specified in the protocol, or for treatment of an AE
- Live or live attenuated vaccination with replicating potential
- During the treatment period, any non-study treatment directed against plasma cells

8.7.3. Permitted Medications and Procedures

The following medications and procedures will be permitted, under the following conditions:

- Any medication required to treat an AE, including systemic corticosteroid therapy.
- Corticosteroid therapy for participants requiring physiologic replacement therapy or a brief course for prophylaxis (eg, contrast dye allergy) or for treatment of non-autoimmune conditions (eg, delayed-type hypersensitivity reaction caused by contact allergen).
- G-CSF, platelet, or RBC transfusions may be permitted for participants who require hematopoietic support.
- CYP450 substrates. Transient release of cytokines has been observed predominantly during initial weeks of dosing, and cytokines may suppress CYP450 enzymes (Morgan, 2001). For participants who are receiving concomitant CYP450 substrates, particularly those with a narrow therapeutic index, monitoring of the effect (eg, warfarin) or drug concentration (eg, cyclosporine or theophylline) is recommended, especially during the initial weeks of therapy.

Any other medication that is considered necessary for the participant's welfare and that is not expected to interfere with the evaluation of the study drug may be given according to the physician's clinical judgment.

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9. STUDY SCHEDULE OF EVENTS AND PROCEDURES

9.1. Schedules of Events

Study assessments and procedures are presented by study period and visit in Table 14 (Dosing Schedule 1) and Table 15 (Dosing Schedule 2).

When multiple procedures are scheduled for the same visit/time point, the following guidance should be followed:

- PRO assessments should be completed before all other procedures and/or discussions about the participant's health
- ECGs should be performed before blood draws, PET imaging, and study drug administration
- Vital signs should be measured prior to blood draws and other clinical assessments
- All blood draws should be performed before FDG tracer administration as part of PET imaging and study drug administration. In addition:
  - PK/ADA samples should be obtained as soon as possible after an ECG
  - Safety and efficacy samples should be obtained before biomarker samples

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Table 14: Schedule of Events—Dosing Schedule 1 (Part 1 & Part 2)

	SCREENING	TREATMENT PERIOD												FOLLOW-UP PERIOD[1]		
		Step-up Period			Cycle 1				Cycles 2 to 6				EOT[1]	Post-Treatment Follow-Up		Long-Term Follow-Up
Visit Week (W)		**W1**	**W2**	**W3**	**W4**	**W5**	**W6**	**W7**	**W8 to W27**				**Time after Last Linvoseltamab Dose**			
Visit Day (D)	**-28 to -1**	**D1**	**D1**	**D1**	**D1**	**D8**	**D15**	**D22**	**D1**	**D8**	**D15**	**D22**	**D30**	**W8 & W12**	**Q16W**	**Q24W**
Visit Window (Days)		±0	±2	±2	±2	±2	±2	±2	±2	±2	±2	±2	±7	±7	±30	±30
Study Procedures[2]																
Inpatient monitoring[3]		X	X	X												
Screening Only																
Informed consent	X															
Informed consent for optional FBR substudy[4]	X															
Inclusion/exclusion criteria	X															
Medical/oncology history	X															
Demographics	X															
Complete physical examination	X															
Height	X															
β2-microglobulin	X															
Echo or MUGA[2]	X															
HIV, HBV, and HCV testing	X															
PT/INR and aPTT/PTT	X															
Treatment																
Premedication[5]		X	X	X	X											
Linvoseltamab[6]		X	X	X	X	X	X	X	X							
Efficacy/Response Assessment																
Overall response assessment					C1D1 and then Q4W								X	X	X	
SPEP and SIFE[7,8]	X	X			C1D1 and then Q4W								X	X	X	
24-hour urine for UPEP and UIFE[,7,8,9]	X	X			C1D1 and then Q4W								X	X	X	
Ig quantification (IgA, IgM, IgG, IgD, IgE)[7,8]	X	X			C1D1 and then Q4W								X	X	X	

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	SCREENING	TREATMENT PERIOD												FOLLOW-UP PERIOD[1]		
		Step-up Period			Cycle 1				Cycles 2 to 6				EOT[1]	Post-Treatment Follow-Up		Long-Term Follow-Up
Visit Week (W)		**W1**	**W2**	**W3**	**W4**	**W5**	**W6**	**W7**	**W8 to W27**				**Time after Last Linvoseltamab Dose**			
Visit Day (D)	**-28 to -1**	**D1**	**D1**	**D1**	**D1**	**D8**	**D15**	**D22**	**D1**	**D8**	**D15**	**D22**	**D30**	**W8 & W12**	**Q16W**	**Q24W**
Visit Window (Days)		±0	±2	±2	±2	±2	±2	±2	±2	±2	±2	±2	±7	±7	±30	±30
Study Procedures[2]																
IgG subclass testing (IgG1, IgG2, IgG3, IgG4)[7]		X			C1D1 and then Q4W								X	X	X	
Serum FLC[8,10]	X	X			C1D1 and then Q4W								X	X	X	
Bone marrow aspirate and/or core biopsy[2,11]	X				Time of suspected CR, 12 months after the first linvoseltamab dose, and then annually for up to 3 years after achievement of CR or at time of BP											
Whole-body imaging (PET/CT and/or MRI)[2,12]	X				As clinically indicated											
Long-term outcomes (eg, survival status, progression to MM)																X
Safety																
Weight[2]	X	X			X				X				X			
Vital signs[13]	X	X	X	X	X	X	X	X	X				X	X		
Limited physical examination[2,14]		X	X	X	X	X	X	X	X				X	X		
Brief neurologic examination/ICE score[15]		X														
ECG	X	As clinically indicated														
ECOG performance status[2]	X	X			X				X				X	X		
AEs		Continuous reporting													Only trSAEs	
Concomitant medications and procedures		Continuous reporting													Only trSAEs or anticancer therapy	
Laboratory Testing (Local)																
Hematology[2,16]	X	X	X	X	X	X	X	X	X				X	X		
CD4/CD8 lymphocyte panel[16]	X				C1D1, then Q12W for the first 12 months											
Blood chemistry[2,16]	X	X	X	X	X	X	X	X	X				X	X		
Urinalysis[16]	X	As clinically indicated														

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	SCREENING	TREATMENT PERIOD												FOLLOW-UP PERIOD[1]		
		Step-up Period			Cycle 1				Cycles 2 to 6				EOT[1]	Post-Treatment Follow-Up		Long-Term Follow-Up
Visit Week (W)		**W1**	**W2**	**W3**	**W4**	**W5**	**W6**	**W7**	**W8 to W27**				**Time after Last Linvoseltamab Dose**			
Visit Day (D)	**-28 to -1**	**D1**	**D1**	**D1**	**D1**	**D8**	**D15**	**D22**	**D1**	**D8**	**D15**	**D22**	**D30**	**W8 & W12**	**Q16W**	**Q24W**
Visit Window (Days)		±0	±2	±2	±2	±2	±2	±2	±2	±2	±2	±2	±7	±7	±30	±30
Study Procedures[2]																
Pregnancy test[2,17]	X (serum)				X				X							
CRP and ferritin		For CRP and ferritin sample collection time points, refer to Table 16														
Pharmacokinetics and Immunogenicity Samples[6]																
PK/Drug concentration sample[18]		For PK/drug concentration sample collection time points, refer to Table 16														
ADA sample[19]		For ADA sample collection time points, refer to Table 16														
Biomarker Samples[7]																
DNA sample for pharmacogenomics		X														
Immune monitoring, tumor cell, and RNA analysis		X	X	X	X		X		C3D1, C4D1, suspected CR, and BP				X	X (part 1 only)	BP	
Serum cytokine profiling		For serum cytokine profiling sample collection time points, refer to Table 16														
Serum for exploratory biomarker		X	X	X	X		X		C3D1, C4D1, suspected CR, and BP				X	X (part 1 only)	BP	
Plasma for exploratory biomarker		X	X	X	X		X		C3D1, C4D1, suspected CR, and BP				X	X (part 1 only)	BP	
Antibody titer testing[21]		X							C3D1, EOT, and 12 months after the first linvoseltamab dose							
Microbiome stool sample	X				C1D1, C3D1, C6D1, and 12 months after the first linvoseltamab dose											
Serum for FBR use (optional)[4]		X	X	X	X		X		C3D1, C4D1, suspected CR, EOT, and BP							
Plasma for FBR use (optional)[4]		X	X	X	X		X		C3D1, C4D1, suspected CR, EOT, and BP							
Blood MRD[11]	X				Time of suspected CR, 12 months after the first linvoseltamab dose, and then annually for up to 3 years after achievement of CR to assess MRD											
Biomarker Samples (Bone Marrow Aspirate and Biopsy)[11]																
Bone marrow MRD[7]	X				Time of suspected CR, 12 months after the first linvoseltamab dose, and then annually for up to 3 years after achievement of CR to assess MRD											
Bone marrow biopsy for IHC/spatial transcriptomics	X				Time of CR and progression to MM											

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	SCREENING	TREATMENT PERIOD												FOLLOW-UP PERIOD[1]		
		Step-up Period			Cycle 1				Cycles 2 to 6				EOT[1]	Post-Treatment Follow-Up		Long-Term Follow-Up
Visit Week (W)		W1	W2	W3	W4	W5	W6	W7	W8 to W27				Time after Last Linvoseltamab Dose			
Visit Day (D)	-28 to -1	D1	D1	D1	D1	D8	D15	D22	D1	D8	D15	D22	D30	W8 & W12	Q16W	Q24W
Visit Window (Days)		±0	±2	±2	±2	±2	±2	±2	±2	±2	±2	±2	±7	±7	±30	±30
Study Procedures[2]																
Patient-Reported Outcome Assessments[20]																
EORTC QLQ-C30	X	X			X				X				X	X	X	
EORTC QLQ-MY20	X	X			X				X				X	X	X	
EQ-5D-5L	X	X			X				X				X	X	X	
FACIT Item GP5	X	X	X	X	X	X	X	X	X				X			

ADA=antidrug antibody; AE=adverse event; aPTT=activated partial thromboplastin time; BP=biochemical progression; C=cycle; CD4=cluster of differentiation 4; CD8=cluster of differentiation 8; CR=complete response; CRP=C-reactive protein; CT=computed tomography; D=day; DNA=deoxyribonucleic acid; Echo=echocardiogram; ECG=electrocardiogram; ECOG=Eastern Cooperative Oncology Group; EORTC QLQ-C30=European Organization for Research and Treatment of Cancer Quality of Life Questionnaire – Core 30; EORTC QLQ-MY20=EORTC Quality of Life Questionnaire-Multiple Myeloma module 20; EOT=end of treatment; EQ-5D-5L=EuroQol 5 Dimensions 5 Levels; FACIT Item GP5=Functional Assessment of Chronic Illness Therapy Item GP5; FBR=future biomedical research; FLC=free light chain; HBV=hepatitis B virus; HCV=hepatitis C virus; HIV=human immunodeficiency virus; ICE=immune effector cell-associated encephalopathy; IgA=immunoglobulin A; IgD=immunoglobulin D; IgE=immunoglobulin E; IgG=immunoglobulin G; IgG1= immunoglobulin G subclass 1; IgG2=immunoglobulin G subclass 2; IgG3=immunoglobulin G subclass 3; IgG4=immunoglobulin G subclass 4; IHC=immunohistochemistry; INR=international normalized ratio; MM=multiple myeloma; MRD=minimal residual disease; MRI=magnetic resonance imaging; MUGA=multigated acquisition scan; PET-CT=positron emission tomography-computed tomography; PK=pharmacokinetic; PT=prothrombin time; PTT=partial thromboplastin time; Q4W=once every 4 weeks; Q12W=once every 16 weeks; Q24W=once every 24 weeks; RNA=ribonucleic acid; SAE=serious adverse event; SIFE=serum immunofixation electrophoresis; SPEP=serum protein electrophoresis; trSAE=treatment-related serious adverse event; UIFE=urine immunofixation; UPEP=urine protein electrophoresis; W=week

Table 15: Schedule of Events—Dosing Schedule 2 (Part 2)

	SCREENING	TREATMENT PERIOD								FOLLOW-UP PERIOD[1]		
		Step-up Period			Cycles 1 to 3				EOT[1]	Post-Treatment Follow-up		Long-Term Follow-up
Visit Week (W)		W1	W2	W3	W4 to W16				Time after Last Linvoseltamab Dose			
Visit Day (D)	-28 to -1	D1	D1	D1	D1	D8	D15	D22	D30	W8 & W12	Q16W	Q24W
Visit Window (Days)		±0	±2	±2	±2	±2	±2	±2	±7	±7	±30	±30
Study Procedures[2]												
Inpatient monitoring[3]		X	X	X								
Screening Only												
Informed consent	X											
Informed consent for optional FBR substudy[4]	X											
Inclusion/exclusion criteria	X											
Medical/oncology history	X											

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	SCREENING	TREATMENT PERIOD								FOLLOW-UP PERIOD[1]		
		Step-up Period			Cycles 1 to 3				EOT[1]	Post-Treatment Follow-up		Long-Term Follow-up
Visit Week (W)		W1	W2	W3	W4 to W16				Time after Last Linvoseltamab Dose			
Visit Day (D)	-28 to -1	D1	D1	D1	D1	D8	D15	D22	D30	W8 & W12	Q16W	Q24W
Visit Window (Days)		±0	±2	±2	±2	±2	±2	±2	±7	±7	±30	±30
Study Procedures[2]												
Demographics	X											
Complete physical examination	X											
Height	X											
β2-microglobulin	X											
Echo or MUGA[2]	X											
HIV, HBV, HCV testing	X											
PT/INR and aPTT/PTT	X											
Treatment												
Premedication[5]		X	X	X	C1							
Linvoseltamab[6]		X	X	X	X		X					
Efficacy/Response Assessment												
Overall response assessment					C1D1 and then Q4W				X	X	X	
SPEP and SIFE[7,8]	X	X			C1D1 and then Q4W				X	X	X	
24-hour urine for UPEP and UIFE[7,8,9]	X	X			C1D1 and then Q4W				X	X	X	
Ig quantification (IgA, IgM, IgG, IgD, IgE)[7,8]	X	X			C1D1 and then Q4W				X	X	X	
IgG subclass testing (IgG1, IgG2, IgG3, IgG4)[7]		X			C1D1 and then Q4W				X	X	X	
Serum FLC[8,10]	X	X			C1D1 and then Q4W				X	X	X	
Bone marrow aspirate and/or biopsy[2,11]	X				Time of suspected CR, 12 months after treatment initiation, and then annually for up to 3 years after achievement of CR or at time of BP							
Whole-body imaging (PET/CT and/or MRI)[2,12]	X				As clinically indicated							
Long-term outcomes (eg, overall survival, progression to MM)												X
Safety												
Weight[2]	X	X			X				X			
Vital signs[13]	X	X	X	X	X		X		X	X		
Limited Physical examination[2,14]		X	X	X	X		X		X	X		
Brief neurologic examination/ICE score[15]		X										
ECG	X	As clinically indicated										

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	SCREENING	TREATMENT PERIOD								FOLLOW-UP PERIOD[1]		
		Step-up Period			Cycles 1 to 3				EOT[1]	Post-Treatment Follow-up		Long-Term Follow-up
Visit Week (W)		W1	W2	W3	W4 to W16				Time after Last Linvoseltamab Dose			
Visit Day (D)	-28 to -1	D1	D1	D1	D1	D8	D15	D22	D30	W8 & W12	Q16W	Q24W
Visit Window (Days)		±0	±2	±2	±2	±2	±2	±2	±7	±7	±30	±30
Study Procedures[2]												
ECOG performance status[2]	X	X			X				X	X		
AEs		Continuous reporting									Only for trSAEs	
Concomitant medications and procedures		Continuous reporting									Only for trSAEs or anticancer therapy	
Laboratory Testing (Local)												
Hematology[2,16]	X	X	X	X	X		X		X	X		
CD4/CD8 lymphocyte panel[16]	X				C1D1, then Q12W for the first 12 months							
Blood chemistry[2,16]	X	X	X	X	X		X		X	X		
Urinalysis[16]	X	As clinically indicated										
Pregnancy test[2,17]	X (serum)				X							
CRP and ferritin		For CRP and ferritin sample collection time points, refer to Table 16										
Pharmacokinetics and Immunogenicity Samples[6]												
PK/Drug concentration sample[18]		For PK/drug concentration sample collection time points, refer to Table 16										
ADA sample[19]		For ADA sample collection time points, refer to Table 16										
Biomarker Samples[7]												
DNA sample for pharmacogenomics		X										
Immune monitoring, tumor cell, and RNA analysis		X	X	X	C1D1, C1D15, suspected CR, EOT, and BP							
Serum cytokine profiling		For serum cytokine profiling sample collection time points, refer to Table 16										
Serum for exploratory biomarker		X	X	X	C1D1, C1D15, suspected CR, EOT, and BP							
Plasma for exploratory biomarker		X	X	X	C1D1, C1D15, suspected CR, EOT, and BP							
Antibody titer testing[21]	X				C3D1 and 12 months after the first linvoseltamab dose							
Microbiome stool sample	X				C1D1, C3D1, and 12 months after the first linvoseltamab dose							
Serum for FBR use (optional)[4]		X	X	X	C1D1, C1D15, suspected CR, EOT, and BP							
Plasma for FBR use (optional)[4]		X	X	X	C1D1, C1D15, suspected CR, EOT, and BP							
Blood MRD[11]	X				Time of suspected CR, 12 months after first linvoseltamab dose, and then annually for up to 3 years after achievement of CR to assess MRD							

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	SCREENING	TREATMENT PERIOD								FOLLOW-UP PERIOD[1]		
		Step-up Period			Cycles 1 to 3				EOT[1]	Post-Treatment Follow-up		Long-Term Follow-up
Visit Week (W)		**W1**	**W2**	**W3**	**W4 to W16**				**Time after Last Linvoseltamab Dose**			
Visit Day (D)	**-28 to -1**	**D1**	**D1**	**D1**	**D1**	**D8**	**D15**	**D22**	**D30**	**W8 & W12**	**Q16W**	**Q24W**
Visit Window (Days)		±0	±2	±2	±2	±2	±2	±2	±7	±7	±30	±30
Study Procedures[2]												
Biomarker Samples (Bone Marrow Aspirate and Biopsy)[11]												
Bone marrow MRD[7]	X				Time of suspected CR, 12 months after first linvoseltamab dose, and then annually for up to 3 years after achievement of CR to assess MRD							
Bone marrow biopsy for IHC/spatial transcriptomics	X				Time of CR and progression to MM							
Patient-Reported Outcomes Assessments[20]												
EORTC QLQ-C30	X	X			X				X	X	X	
EORTC QLQ-MY20	X	X			X				X	X	X	
EQ-5D-5L	X	X			X				X	X	X	
FACIT Item GP5	X	X	X	X	X		X		X			

ADA=antidrug antibody; AE=adverse event; aPTT=activated partial thromboplastin time; BP=biochemical progression; C=cycle; CD4=cluster of differentiation 4; CD8=cluster of differentiation 8; CR=complete response; CRP=C-reactive protein; CT=computed tomography; D=day; DNA=deoxyribonucleic acid; Echo=echocardiogram; ECG=electrocardiogram; ECOG=Eastern Cooperative Oncology Group; EORTC QLQ-C30=European Organization for Research and Treatment of Cancer Quality of Life Questionnaire – Core 30; EORTC QLQ-MY20=EORTC Quality of Life Questionnaire-Multiple Myeloma module 20; EOT=end of treatment; EQ-5D-5L=EuroQol 5 Dimensions 5 Levels; FACIT Item GP5=Functional Assessment of Chronic Illness Therapy Item GP5; FBR=future biomedical research; FLC=free light chain; HBV=hepatitis B virus; HCV=hepatitis C virus; HIV=human immunodeficiency virus; ICE=immune effector cell-associated encephalopathy; IgA=immunoglobulin A; IgD=immunoglobulin D; IgE=immunoglobulin E; IgG=immunoglobulin G; IgG1= immunoglobulin G subclass 1; IgG2=immunoglobulin G subclass 2; IgG3=immunoglobulin G subclass 3; IgG4=immunoglobulin G subclass 4; IHC=immunohistochemistry; INR=international normalized ratio; MM=multiple myeloma; MRD=minimal residual disease; MRI=magnetic resonance imaging; MUGA=multi-gated acquisition scan; PET-CT=positron emission tomography-computed tomography; PK=pharmacokinetic; PT=prothrombin time; PTT=partial thromboplastin time; Q4W=once every 4 weeks; Q16W=once every 16 weeks; Q24W=once every 24 weeks; RNA=ribonucleic acid; SAE=serious adverse event; SIFE=serum immunofixation electrophoresis; SPEP=serum protein electrophoresis; trSAE=treatment-related serious adverse event; UIFE=urine immunofixation; UPEP=urine protein electrophoresis; W=week

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9.1.1. Footnotes for the Schedule of Events Tables

1. EOT: participants who discontinue study treatment as per Section 8.2.3 but who do not withdraw from the study must return to the clinic for an EOT visit (Section 9.1.3). The follow-up period starts after the EOT visit and consists of (1) post-treatment follow-up, where safety monitoring continues for up to 90 days after the last linvoseltamab dose and efficacy assessments continue until confirmed biochemical progression, start of non-protocol cancer therapy, or death; and (2) long-term follow-up, which starts after the end of post-treatment follow-up, to collect survival status as well as other relevant long-term outcomes. The follow-up period will continue for up to 5 years after the last dose of linvoseltamab, unless death or study termination by the sponsor occurs earlier.
2. Clinical assessments (ie, physical exam, weight, ECOG performance status, hematology, blood chemistry, and pregnancy test) required on W1D1 prior to administering the first step-up dose of linvoseltamab, may be performed up to 3 days before the visit, unless otherwise specified. Clinical assessments during treatment may be performed within 24 hours prior to study drug administration, unless otherwise specified. The following assessments do not need to be repeated if they have been performed within 8 weeks (56 days) prior to W1D1: whole body imaging, Echo or MUGA, or bone marrow biopsy. Archival tissue from a core biopsy performed within the 8-week window must be provided for central laboratory analysis if used in lieu of fresh tissue. If archival tissue is unavailable, a repeat bone marrow biopsy must be performed.
3. All participants must be observed for at least 24 hours in a monitored inpatient setting after the EOI of each linvoseltamab step-up dose.
4. Informed Consent for Optional FBR substudy: for participants who consent to participate in a separate optional FBR substudy.
5. Premedications include dexamethasone, an antihistamine, and acetaminophen (paracetamol). See Section 8.1.2 for further details.
6. Dosing days must be at least 5 days apart. C1D1 begins when the first full dose of linvoseltamab is administered after completion of the step-up schedule.
7. Samples will be collected predose and submitted to the central laboratory, or Dana-Farber Cancer Institute as indicated in the lab manual (US sites only), for analysis. Pharmacogenomic sample collections are mandatory.
8. A second response assessment will be required to confirm response or biochemical progression. This may occur at the next scheduled visit or an unscheduled visit using a separate efficacy/response assessment kit.
9. If the screening sample is collected within 14 days of W1D1, then there is no requirement to repeat collection of 24-hour urine before the W1D1 time point.
10. FLCs may be tested as an exploratory biomarker measure using exploratory serum samples at additional time points other than those prespecified in Table 16.
11. A bone marrow aspirate and core biopsy sample will be collected for all participants at screening. The screening bone marrow aspirate sample will be used to assess chromosomal abnormalities, immunophenotypes by flow cytometry, baseline clonality to assess MRD,

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and other exploratory analyses. Subsequent bone marrow samples (aspirate and/or core biopsy) will be collected within 30 days after confirmation of suspected CR, 12 months after treatment initiation, and then annually for up to 3 years after achievement of CR (if ongoing response). If biochemical progression occurs beyond 12 weeks from the date when the first dose of study drug is administered, a bone marrow aspirate and/or core biopsy must be collected within 30 days.

12. Local assessment for myeloma-defining radiologic findings will be completed at screening. See Section 9.2.2.2 for recommended imaging procedures. Repeat whole-body imaging should be performed with the same imaging modality used at screening and adhere to the imaging manual specifications.
13. Step-up doses (week 1 to week 3): vital signs will be taken predose/infusion, every 30 minutes (±15 minutes) for the first 4 hours until EOI, then hourly (±15 minutes) until 4 hours following EOI, and subsequently every 4 hours (± 1 hour) until 24 hours after EOI. From first full dose on C1D1 to end of therapy: vital signs will be taken pre-infusion, every 30 minutes (±15 minutes) for the first hour from start of infusion and then hourly (±15 minutes) until EOI, as applicable. Vital signs on days without dosing may be taken at any time during the visit, and pulse oximetry is not required.
14. A complete physical examination may be completed, if clinically indicated.
15. A brief neurologic examination and ICE score assessment are required before administration of the first step-up dose of linvoseltamab on W1D1. Subsequent assessments should be performed as clinically indicated during treatment to monitor for neurotoxicity and/or ICANS (Appendix 5).
16. Hematology, blood chemistry, and urinalysis: see Section 9.2.3.6 for list of analytes.
17. Applicable to WOCBP only: a serum β-hCG level must be measured during screening within 72 hours of administering the first step-up dose. After screening, subsequent tests may be performed on urine.
18. PK/Drug concentration sample: sBCMA may be evaluated at time points when PK will be assessed (Table 16).
19. During the treatment period, ADA serum samples will be collected predose on day 1 at the visits listed in Table 16. In the event of AESI, samples may be collected for ADA analysis at or near the event when possible.
20. All PRO questionnaires will be administered and completed during the scheduled visit prior to any other study procedures. The baseline EORTC QLQ-C30, EORTC QLQ-Y20, EQ-5D-5L, and FACIT Item GP5 assessments will be collected on day 1 of the step-up period prior to the first step-up dose of linvoseltamab. Where multiple questionnaires are administered, questionnaires shall be administered in the following order: EORTC QLQ-C30, EORTC QLQ-MY20, EQ-5D-5L, and FACIT Item GP5.
21. Antibody titer testing to assess immunity against pneumococcus and diphtheria should be performed locally. All sample time points must be assessed using the same assay to allow serial comparison.

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9.1.2. Detailed Sample Collection Time Points

Table 16: Detailed Sample Collection Time Points for Dosing Schedules 1 & 2

Visit		Time Point	Serum Cytokines	PK	ADA	CRP and Ferritin
Screening		N/A				
Step-Up Doses	**Day 1 (Week 1)**	Predose	X	X	X	X
		EOI (T0)	X	X		
		T4H	X	X		X
	Day 8 (Week 2)	Predose	X	X		X
		EOI (T0)	X	X		
		T4H	X	X		X
	Day 15 (Week 3)	Predose	X	X		X
		EOI (T0)	X	X		
		T4H	X	X		X
Cycle 1	**Day 1**	Predose	X	X		X
		EOI (T0)	X	X		
	Day 8[a]	Predose		X		X
		EOI (T0)	X	X		
	Day 15	Predose		X		X
		EOI (T0)	X	X		
	Day 22[a]	Predose		X		X
		EOI (T0)		X		
Cycle 2	**Day 1**	Predose		X		
		EOI (T0)		X		
Cycle 3	**Day 1**	Predose		X	X	
		EOI (T0)		X		
Cycle 4	**Day 1[a]**	Predose		X		
		EOI (T0)		X		
Cycle 5	**Day 1[a]**	Predose		X		
		EOI (T0)		X		
Cycle 6	**Day 1[a]**	Predose		X	X	
		EOI (T0)		X		
30 days after last dose		Any time during visit		X	X	
Weeks 8 and 12 after last dose		Any time during visit		X	Week 12 after last dose	

[a] Only applicable to dosing schedule 1
ADA=antidrug antibody; CRP=C-reactive protein; D=day; EOI=end of infusion; PK=pharmacokinetic; T0=time 0 (EOI); T4H=time 4 hours (4 hours post-EOI)

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9.1.3. End of Treatment Visit

Participants who finish treatment or discontinue study treatment as per Section 8.2.3 but who do not withdraw from the study must return to the clinic for an EOT visit (30 days after last dose or prior to the start of non-protocol cancer therapy, whichever occurs earlier) and will be expected to complete all applicable follow-up assessments (eg, efficacy and/or safety monitoring) as per the Schedules of Events (Table 14 and Table 15). If a participant is unable to return for an EOT visit, every effort should be made to contact the participant and collect AE information.

A participant who withdraws consent to continue participation in the study will not be followed for any reason after consent has been withdrawn.

9.1.4. Unscheduled Visits

All attempts should be made to keep participants on the study schedule. Unscheduled visits may be necessary for repeat testing following abnormal laboratory results, for management of ongoing AEs, or other safety-related reasons, as warranted. Additional step-up doses or treatment resumption at a dose less than the assigned full dose should also be documented as unscheduled visits (see Section 8.2.2). Relevant assessments that occur during unscheduled visits should be reported in the respective CRF for the visit date.

9.2. Study Procedures

9.2.1. Procedures Performed Only at the Screening/Baseline Visit

The following procedures will be performed for the sole purpose of determining study eligibility or characterizing the baseline population:

- Medical/oncology history
- Demographics
- Complete physical examination
- Height
- β2-microglobulin
- Echo or MUGA (cardiac ejection fraction)
- HIV, HBV, and HCV testing
- PT/INR and aPTT/PTT

The participant's clinical laboratory assessments must be repeated prior to administering the first step-up dose if they were collected >3 days prior to dosing, and the participant must continue to meet all eligibility criteria prior to administering the first step-up dose of linvoseltamab.

9.2.2. Efficacy Procedures

All efficacy procedures will be conducted at the time points indicated in the Schedule of Events (Table 14 and Table 15). Efficacy/disease assessments should be performed regularly at the prescribed interval (eg, Q4W while on treatment) regardless of interruptions in treatment. If a participant is unable to undergo a disease assessment within the allowed window due to logistical

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or medical reasons, the disease assessment should be obtained at the next available date and before subsequent study treatment.

9.2.2.1. Overall Response Assessment

Disease assessments by investigators will be performed during the treatment and follow-up periods (ie, overall response assessment) according to IMWG criteria based on central laboratory analysis of bone marrow and peripheral blood samples. Characterization of serum and urine monoclonal protein will be performed by SPEP/SIFE and UPEP/UIFE, respectively. Serum FLCs and the corresponding ratio will be measured as well. Bone marrow core biopsy and aspirate will be assessed as described in Section 9.2.2.2 to enumerate plasma cell number at baseline and post-treatment. In addition, exploratory testing to evaluate any potential interference of linvoseltamab may be performed if an appropriate assay is available. This test may be performed on unused serum from standard MM disease assays or using exploratory research serum samples. Immunoglobulin quantification of various immunoglobulin subtypes (ie, IgG, IgA, IgM, IgD, IgE) and IgG subclass testing (Section 9.2.3.6) will also be performed at the time points indicated in the Schedule of Events (Table 14 and Table 15).

Two consecutive assessments will be required to confirm all responses. If a central laboratory result is not available, a result obtained locally and entered into the EDC may be used.

9.2.2.2. Bone Marrow Aspirate and/or Biopsy

All participants will require a bone marrow aspirate and/or core biopsy at screening to confirm a diagnosis of HR-MGUS or NHR-SMM. Subsequent bone marrow aspirates and/or biopsies will be collected according to the Schedule of Events (Table 14 and Table 15) and the below guidance:

- Upon confirmation of suspected CR, a bone marrow aspirate must be collected within 30 days to assess MRD.
- If biochemical progression occurs beyond 12 weeks from the date when first dose of study drug is administered, a bone marrow aspirate and/or core biopsy must be collected within 30 days.
- All participants who remain on study without evidence of biochemical progression must undergo bone marrow aspirate collection and/or core biopsy at 12 months after the first linvoseltamab dose for assessment of MRD and evaluation of other molecular readouts.

Study-related bone marrow biopsy and/or aspirate samples must be shipped as described in the study laboratory manual. Evaluation methods may include but are not limited to IHC, flow cytometry, FISH, MRD assessment, and nucleotide sequencing (Section 9.2.7.3). Detailed instructions for sample collection and preparation are provided in the laboratory manual.

9.2.2.3. Whole-Body Imaging

Whole-body imaging must be performed during screening for all participants to determine the presence of osteolytic skeletal lesions, intramedullary focal lesions, and/or plasmacytomas that would disqualify a participant from enrollment in the study (Hillengass, 2019). The preferred first imaging method is whole-body ^{18}F-FDG-PET/CT, although whole-body low-dose CT is an acceptable alternative. If the CT component is negative, whole-body MRI (or MRI of the spine

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and pelvis if whole-body MRI is not available) is recommended to exclude focal lesions as myeloma-defining events. Image acquisition must conform to the quality standards specified in the imaging manual.

9.2.2.4. Minimal Residual Disease

MRD status is an important prognostic marker and achievement of MRD negativity is associated with favorable outcomes in patients with MM (Martinez-Lopez, 2014). MRD will be assessed according to IMWG guidelines by either flow cytometry (eg, EuroFLOW™) and/or NGS methods (clonoSEQ assay cleared by FDA for MM only). Samples may be evaluated by either NGS or by flow cytometry, and in some cases both methods may be used to compare concordance as neither method has been fully validated in this patient population. Next-generation sequencing will be performed longitudinally if a baseline sample establishing clonotypes is available for participants at the time of suspected CR or better. If the baseline sample is not evaluable, then only flow cytometry may be used to evaluate MRD at suspected CR or better. Samples for both methods will be collected at screening, at suspected CR or better, and at longitudinal time points to assess MRD. MRD at 10^{-5} sensitivity will be analyzed.

Sample collection time points are noted for bone marrow sample collection in the Schedules of Events (Table 14 and Table 15). The collection procedures for each method will be detailed in the laboratory manual.

9.2.2.5. Overall Survival Status Data Collection

Every effort will be made to collect survival status Q24W for all participants according to the Schedule of Events (Table 14 and Table 15), including participants who withdraw from the study for any reason but have not withdrawn consent to collect survival status. If the death of a participant is not reported, the date of last participant contact in this study will be used for determination of the participant's last known date alive. If death is reported, efforts must be taken to determine the cause of death. If follow-up occurs remotely, the investigator may choose to collect survival status for all participants discontinued from their site on a fixed biannual basis for practical reasons.

9.2.3. Safety Procedures

All safety procedures including ECOG performance status will be conducted at the time points indicated in the Schedule of Events (Table 14 and Table 15). Monitoring for safety including AEs (Section 10.1) and associated concomitant medications and procedures will be continuous.

9.2.3.1. Body Weight

Body weight will be assessed using calibrated scales. Participants should void (empty bladder) prior to weight assessment. Participants should be wearing undergarments only and no shoes during weight assessments. Body weight will be recorded to the nearest 0.1 kg.

9.2.3.2. Vital Signs

Vital signs, including temperature, blood pressure, pulse, pulse oximetry (on dosing days), and respiration rate will be collected at study visits indicated in the Schedules of Events (Table 14 and Table 15). Blood pressure should be measured after the participant has been resting quietly for at least 5 minutes. Blood pressure may be obtained from a seated or recumbent position and should

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be done consistently throughout the study. When scheduled at the same time as other procedures, vital signs should be measured prior to clinical laboratory assessments, PK, or exploratory sample collection.

9.2.3.3. Physical Examination

A physical examination will be conducted at study visits as indicated in the Schedule of Events (Table 14 and Table 15).

Care should be taken to examine and assess any abnormalities that may be present, as indicated by the participant's medical history. Screening requires a complete physical examination; a limited physical examination may be performed at subsequent visits.

9.2.3.4. Neurologic Examination

A brief neurologic examination including elements of the ICE score will be performed and recorded at baseline and as clinically indicated thereafter according to the Schedule of Events (Table 14 and Table 15) to screen and monitor participants for potential symptoms of neurotoxicity and ICANS.

The CARTOX-10-based assessment tool proposed by the ASTCT, termed the ICE score, grades neurotoxicity by evaluating alterations in speech, orientation, handwriting, and concentration that are highly suggestive of the encephalopathy observed in participants with ICANS (Lee, 2019). Symptoms of ICANS may manifest as delirium, encephalopathy, aphasia, lethargy, difficulty concentrating, agitation, seizures, motor weakness, depressed level of consciousness, and cerebral edema. Symptoms excluded because they are less specific and are usually managed symptomatically are headaches, tremor, myoclonus, asterixis, and hallucinations (Lee, 2019). See Appendix 5 for additional information about the scoring method.

9.2.3.5. Electrocardiogram

ECGs should be performed before blood is drawn during visits requiring blood draws. A standard 12-lead ECG will be performed at visits according to the Schedules of Events (Table 14 and Table 15). Heart rate will be recorded from the ventricular rate, and QRS, RR, and QT (identify QTcF) intervals will be recorded. The ECG strips or reports will be retained with the source.

9.2.3.6. Laboratory Testing

Hematology, CD4 and CD8 lymphocyte panel, blood chemistry, urinalysis, CRP, ferritin, and pregnancy testing samples will be analyzed by a local laboratory. Detailed instructions for blood sample collection are in the laboratory manual provided to study sites.

If a specific laboratory test is not routinely performed at a center, and instead, the center performs a different but equivalent test (eg, carbon dioxide instead of bicarbonate or urea instead of BUN), the equivalent test should be performed, and results collected. At centers where the bicarbonate (or equivalent) test is not performed as part of the routine chemistry panel, the test may be omitted.

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Samples for laboratory testing will be collected at visits according to Schedule of Events (Table 14 and Table 15). Tests will include:

Blood Chemistry

Sodium	Total protein, serum	Total bilirubin
Potassium	Creatinine	Uric acid
Chloride	BUN	
Carbon dioxide	AST	
Calcium	ALT	
Glucose	Alkaline phosphatase	
Albumin	LDH	

Hematology

Hemoglobin	Differential:
Hematocrit	Neutrophils
RBCs	Lymphocytes
WBCs	Monocytes
Platelet count	Basophils
	Eosinophils
	Plasma cells (if applicable)

Urinalysis

Color	Glucose	RBC
Clarity	Blood	Hyaline and other casts
pH	Bilirubin	Bacteria
Specific gravity	Leukocyte esterase	Epithelial cells
Ketones	Nitrite	Crystals
Protein	WBC	Yeast

Other Laboratory Tests

CD4+ and CD8+ T-Cell Panel: CD4+ and CD8+ will be measured at screening and other time points as noted in the Schedule of Events (Table 14 and Table 15).

Pregnancy Test: Serum β-hCG will be measured at screening (≤72 hours prior to administering the first step-up dose of linvoseltamab), and urine β-hCG will be measured at all other time points for WOCBP.

CRP and Ferritin: These inflammatory markers will be measured at the time points specified in the Schedule of Events (Table 14 and Table 15).

Serum Calcium Corrected for Albumin (derived): Levels of unbound/free calcium (ie, biologically active calcium) will be calculated by adjusting for abnormal albumin levels ("corrected serum calcium").

Abnormal Laboratory Values and Laboratory Adverse Events

All laboratory values must be reviewed by the investigator or authorized designee.

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Significantly abnormal test results that occur after start of treatment must be repeated to confirm the nature and degree of the abnormality. When necessary, appropriate ancillary investigations should be initiated. If the abnormality fails to resolve or cannot be explained by events or conditions unrelated to the study medication or its administration, the Medical/Study Director must be consulted.

The clinical significance of an abnormal test value, within the context of the disease under study, must be determined by the investigator.

Criteria for reporting laboratory values as an AE are provided in Section 10.1.1.

9.2.4. Drug Concentration and Measurements

Blood samples for the assessment of linvoseltamab concentration in serum will be collected after the first linvoseltamab dose and throughout the treatment period, as listed in Table 16. Actual sample collection times will be recorded. Detailed instructions for blood sample collection are provided in the laboratory manual. The sBCMA concentration may be measured from selected PK time points.

Any unused samples may be used for exploratory research including the possible evaluation of sBCMA (Section 9.2.7.2) and assay development and/or validation, biomarker research (Section 9.2.7), and/or FBR (Section 9.2.10).

9.2.5. Immunogenicity Measurements and Samples

Samples for ADA assessment will be collected at the time points listed in Table 16. Detailed instructions for blood sample collection are provided in the laboratory manual.

Any unused samples may also be used for exploratory research including assay development and/or validation, biomarker research (Section 9.2.7), and/or FBR (Section 9.2.10).

9.2.6. Patient-Reported Outcome Assessment Procedures

PRO measurements will be assessed according to the Schedule of Events (Table 14 and Table 15).

PRO measurements will include the validated instruments EORTC QLQ-C30 (Aaronson, 1993a) and EORTC QLQ-MY20 (Cocks, 2007) (Stead, 1999), commonly used in MM clinical trials. In addition, the FACIT Item GP5 will be used to assess overall impact of treatment side effects.

- The EORTC QLQ-C30 is a 30-item, generic questionnaire used to assess HRQoL in patients with cancer (Cocks, 2007) (Stead, 1999). It is one of the most frequently used generic measures of HRQoL in cancer clinical trials. The EORTC QLQ-C30 assesses HRQoL across multiple domains, including global health status, global quality of life, functioning (physical, role, emotional, cognitive, and social functioning), symptom scales (fatigue, nausea and vomiting, and pain), and single items (dyspnea, insomnia, constipation, diarrhea, financial impact, and appetite loss).
- The EORTC QLQ-MY20 is a 20-item, disease-specific module which was developed to be used in conjunction with the EORTC QLQ-C30 to further assess HRQoL in patients with MM (Cocks, 2007) (Stead, 1999). The EORTC QLQ-MY20 assesses quality of life across multiple domains, including future perspectives, body image, MM-specific symptoms, and treatment side effects.

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- The EQ-5D-5L is a self-administered, generic, standardized, health status measure, consisting of an EQ-5D descriptive system and an EQ-VAS (visual analog scale) (EuroQol., 2017). The EQ-5D-5L descriptive system assesses 5 dimensions of health: mobility, self-care, usual activities, pain/discomfort, and anxiety/depression. Each dimension is rated on a 5-level scale: no problems, slight problems, moderate problems, severe problems, and extreme problems. The EQ-VAS component is a vertical visual analog scale used by patients to rate their health.
- FACIT Item GP5 is a recommended item by FDA in its recent draft guidance for cancer trials to assess the overall impact of treatment side effect as a core PRO (FDA, 2021). It uses a single item "I am bothered by side effects of treatment" on a 5-point scale (0 = not at all, 1 = a little bit, 2 = somewhat, 3 = quite a bit, 4 = very much).

Non-English-speaking participants will complete the questionnaire using validated language translations developed and recommended for each instrument.

9.2.7. Pharmacodynamic and Exploratory Procedures

In this study, biomarker assessments will be performed to explore the mechanism of action of linvoseltamab in HR-MGUS and NHR-SMM and to evaluate the predictive value of specific biomarkers with respect to safety, drug exposure, and efficacy.

The analyses performed in the sections below may or may not be reported in the CSR. The collected samples will be stored for up to 15 years after the final date of the database lock. The samples may be utilized for exploratory research of HR-MGUS and NHR-SMM/MM, the mechanism of linvoseltamab action, and the biological systems targeted by linvoseltamab. After 15 years, any residual samples will be destroyed.

Biomarker samples will be collected at time points according to the Schedule of Events (Table 14 and Table 15). These samples are distinct from the optional FBR samples as outlined in Section 9.2.10.

Biomarker measurements may be performed on blood and bone marrow cells, serum, plasma, stool, and/or urine samples to evaluate the effects of linvoseltamab on the natural history of HR-MGUS and NHR-SMM or relevant physiological and pathogenic processes. Serum for additional testing will be collected for exploratory assays (such as mass spectrometry and Olink®) to evaluate changes in proteins following treatment with linvoseltamab.

The biomarkers studied are believed to be relevant to the pathophysiology of HR-MGUS and NHR-SMM, target engagement, mechanism of action of linvoseltamab, and/or possible AEs.

9.2.7.1. Peripheral Blood Biomarkers

Samples for peripheral blood (plasma, serum, and blood for cell phenotyping, RNA, and DNA analyses) biomarker assessments will be collected at time points listed in the Schedule of Events (Table 14 and Table 15).

Biomarker assays may include, but will not be limited to, the following:

- Immunophenotyping of T-cell and B-cell subsets and other cell populations (including monocytes and NK cells) in peripheral blood (by flow cytometry or similar methods) will be performed to assess baseline and post-treatment changes.

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- Additional exploratory analyses requiring blood for DNA or RNA isolation may include but are not limited to WES, bulk/single-cell RNA sequencing analysis, microsatellite instability, T-cell and B-cell clonality, and immune receptor repertoire sequencing.
- Peripheral blood MRD monitoring to evaluate the level of circulating tumor reduction by linvoseltamab.
- Antibody titers against pneumococcus and diphtheria to evaluate the effects of linvoseltamab treatment on post-vaccination immunity.

9.2.7.2. Serum Cytokines and Other Biomarkers

Serum samples will be collected for cytokine analysis according to Table 16.

Serum may also be used to evaluate additional biomarkers of interest (Table 14 and Table 15) relating to HR-MGUS and NHR-SMM and the mechanism of action of linvoseltamab, including the evaluation of exploratory FLC testing. Cytokine changes triggered by binding of linvoseltamab to T cells may reflect T-cell activation. Understanding the magnitude and time course of pharmacodynamic changes in serum cytokines may inform dosing and help understand safety and efficacy.

9.2.7.3. Bone Marrow Exploratory Analyses

If sufficient bone marrow aspirate or core biopsy tissue is available after assessing tumor burden and confirming CR and MRD, samples may be stored for analysis of DNA, RNA, or protein to quantify the following:

- B-cell and T-cell immune receptor repertoires by sequencing
- Number and distribution of marrow-infiltrating lymphocytes relative to malignant plasma cells such as CD8+ T cells, CD4+ T cells, T regulatory cells, and, tissue permitting, other immune cell types including B cells, myeloid-derived cells, NK cells, etc
- Expression levels (messenger RNA and/or protein) including but not limited to checkpoint modulators (analysis may include nucleic sequencing and flow cytometry methods as well as additional methodologies as required)
- Mutations in known oncogenes and potential tumor neoantigens
- Expression of other B-cell antigens to monitor resistance or response such as CD38, GPRC5D, and SLAMF7
- MRD monitoring to evaluate the depth of tumor reduction by linvoseltamab in bone marrow tissue
- Any bone marrow core biopsy may be evaluated by IHC/immunofluorescence techniques to better understand the tumor microenvironment composition

Tissue permitting, additional biomarkers may be measured using techniques that evaluate the tumor exome, single-cell genetics, microsatellite instability, T-cell and B-cell clonality, and immune receptor repertoire.

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9.2.8. Microbiome Assessment

Participants are required to collect stool samples at screening for the evaluation of the bacteria that reside in the intestine. Subsequent stool samples will be taken throughout treatment according to the Schedule of Events (Table 14 and Table 15). Study-related stool samples will be analyzed at the Dana-Farber Cancer Institute as described in the study laboratory manual. Evaluation methods may include but are not limited to DNA and RNA sequencing of the bacteria by assays including but not limited to shotgun DNAseq and/or metatranscriptomics to identify if specific bacteria associate with clinical response or resistance to linvoseltamab by the participant.

9.2.9. Pharmacogenomic Analysis

Whole-blood samples for DNA and RNA extraction will be collected from peripheral blood and bone marrow cells for pharmacogenomic analyses to understand the genetic determinants of efficacy and safety associated with the treatments in this study and the molecular basis of HR-MGUS/NHR-SMM and related diseases (Table 14 and Table 15).

These samples will be single coded as defined by the ICH guideline E15. Samples will be stored for up to 15 years after the final date of the database lock. If there are specific site or country requirements involving the pharmacogenomic analyses that the sponsor is unable to comply with, samples will not be collected at those sites.

The purpose of the pharmacogenomic analyses is to identify genomic associations with clinical or biomarker response to linvoseltamab in HR-MGUS/NHR-SMM and possible AEs. In addition, associations between genomic variants and prognosis or progression of HR-MGUS/NHR-SMM to MM may also be studied. These data may be used or combined with data collected from other studies to identify and validate genomic markers related to the study drug, target pathway, or HR-MGUS/NHR-SMM and related diseases.

Analyses may include sequence determination or single nucleotide polymorphism studies of candidate genes and surrounding genomic regions. Other methods, including whole-exome sequencing, whole-genome sequencing, T-cell receptor sequencing, DNA copy number variation, and transcriptome sequencing (or other methods for quantitating RNA expression) may also be performed. The list of methods may be expanded to include novel methodology that may be developed during this study or sample storage period. Results from the genomic analyses will not be reported in the CSR.

9.2.10. Future Biomedical Research (Optional)

Participants who agree to participate in the FBR substudy will be required to consent to this optional substudy before samples are banked for FBR.

Additional peripheral blood samples will also be collected from participants who agree to participate in this optional substudy (see Table 14 and Table 15 for collection times).

Residual biomarker samples for study-related research, as well as unused PK and ADA samples, will be stored for up to 15 years after the final date of the database lock (or for a shorter time if required per regional laws and regulations). The samples may be utilized for FBR that may or may not be directly related to the study, including being used as reference samples and assay development or validation. The results of these FBR analyses will not be presented in the CSR.

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10. SAFETY EVALUATION AND REPORTING

10.1. Recording and Reporting Adverse Events

10.1.1. General Guidelines

The investigator must promptly record all clinical events occurring during the study data collection per the Schedule of Events (Table 14 and Table 15). Medical conditions that existed or were diagnosed prior to the signing of the ICF will be recorded as part of medical history. Abnormal laboratory values and vital signs observed at the time of informed consent should also be recorded as medical history. Any subsequent worsening (ie, any clinically significant change in frequency and/or intensity) of a preexisting condition that is temporally associated with the use of the study drug should also be recorded as an AE.

At each visit, the investigator will determine whether any AEs have occurred by evaluating the participant. AEs may be directly observed, reported spontaneously by the participant, or by questioning the participant at each study visit. Participants should be questioned in a general way without asking about the occurrence of any specific symptoms. The investigator must assess all AEs to determine seriousness, severity, and causality in accordance with the definitions in Section 10.2. The investigator's assessment must be clearly documented in the site's source documentation with the investigator's signature. The investigator should follow up on SAEs (and AESI) until they have resolved or are considered clinically stable; AEs should be followed until they are resolved or last study visit, whichever comes first. Assessment of AEI is described in Section 6.1.3.1, and assessment of trigger events in Section 6.1.3.5.

Always report the diagnosis as the AE or SAE term. When a diagnosis is unavailable, report the primary sign or symptom as the AE or SAE term with additional details included in the narrative until the diagnosis becomes available. If the signs and symptoms are distinct and do not suggest a common diagnosis, report them as individual entries of AE or SAE.

Laboratory results, vital signs, and other diagnostic results or findings should be appraised by the investigator to determine their clinical significance. Isolated abnormal laboratory results, vital-sign findings, or other diagnostic findings (ie, not part of a reported diagnosis) should be reported as AEs if they are symptomatic, lead to study drug discontinuation, dose reduction, require corrective treatment, or constitute an AE in the investigator's clinical judgment.

For events that are serious due to hospitalization, the reason for hospitalization must be reported as the SAE (diagnosis or symptom requiring hospitalization). A procedure is not an AE or SAE, but the reason for the procedure may be an AE or SAE. Preplanned (prior to signing the ICF) procedures, treatments requiring hospitalization for preexisting conditions that do not worsen in severity, and admission for palliative or social care should not be reported as SAEs (see Section 10.2 for definitions).

For deaths, the underlying or immediate cause of death should always be reported as an SAE.

Diagnosis of malignancy will not be considered an AE or SAE if it is clearly consistent with the typical progression pattern of the underlying condition (including time course, affected organs, etc). Clinical symptoms of progression may be reported as AEs if the symptom cannot be determined as exclusively due to progression of the underlying malignant process or does not fit

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the expected pattern of progression for the condition under study. If there is any uncertainty about an AE being attributable to the development of malignancy, it should be reported as an AE or SAE.

During the follow-up period, any SAE deemed treatment related by an investigator that occurs after the safety monitoring window ends (90 days after last dose of study drug) and during post-treatment follow-up should also be reported.

All AEs, SAEs, AESI, and pregnancy reports are to be reported according to the procedures in Section 10.1.3.

10.1.2. Reporting Procedure

All SAEs and AESI must be reported to the sponsor (or designee) by completing the AE CRF immediately but no later than 24 hours after learning of the event. For SAEs and AESI, a detailed narrative summarizing the course of the event, including its evaluation, treatment, and outcome should be provided on the AE CRF. Specific or estimated dates of event onset, treatment, and resolution should be included, when available. Medical history, concomitant medications, and laboratory data that are relevant to the event should also be summarized in the narrative. For fatal events, the narrative should state whether an autopsy was or will be performed and include the results if available. Information not available at the time of the initial report must be documented in a follow-up report. Source documents (including hospital or medical records, diagnostic reports, etc) will be summarized in the narrative on the AE CRF, retained at the study center, and made available upon request.

Urgent safety queries must be followed up and addressed promptly. Follow-up information and response to non-urgent safety queries should be combined for reporting to provide the most complete data possible within each follow-up.

10.1.3. Events that Require Expedited Reporting to Sponsor

The following events also require reporting to the sponsor (or designee) within 24 hours of learning of the event:

- **SAEs**
- AESI (serious and nonserious) for this study include the following:
  - Grade 2 or greater CRS (Lee, 2019)
  - Grade 2 or greater ICANS
  - Grade 3 or greater TLS
  - Grade 3 or greater IRR
  - Grade 3 or greater allergic reaction
  - Grade 3 or greater infection
  - HBV reactivation

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The following events also require reporting to the sponsor (or designee) within 24 hours of learning of the event:

- **Overdose of study drug:** Accidental or intentional overdose of at least 2 times the intended dose of linvoseltamab should be reported as an AE, even if not associated with signs or symptoms. If the overdose is associated with signs or symptoms, the signs or symptoms should be reported separately according to AE or SAE requirements.
- **Pregnancy:** Although pregnancy is not considered an AE, it is the responsibility of the investigator to report to the sponsor (or designee) within 24 hours of identification any pregnancy occurring in a female participant or female partner of a male participant during the study or within 6 months of the last dose of study drug. Any complication of pregnancy affecting a female study participant or female partner of a male study participant and/or fetus and/or newborn that meets the SAE criteria must be reported as an SAE. Outcomes for all pregnancies should be reported to the sponsor.

10.1.4. Other Adverse Events of Special Interest that Do Not Require Expedited Reporting to Sponsor

Although these AESI do not require expedited reporting to the sponsor, the following events are of interest and will involve specific data collection to characterize better:

- Grade 1 CRS
- Grade 1 ICANS
- Grade 1 or 2 IRR

10.1.5. Abnormal Vital Signs or Laboratory or Electrocardiogram Results

The criteria for determining whether an abnormal objective test finding should be reported as an AE include:

- The test result is associated with accompanying symptoms, and/or
- The test result requires additional diagnostic testing or medical/surgical intervention, and/or
- The test result leads to a change in dosing (outside of protocol-stipulated dose adjustments), discontinuation of study drug, significant additional concomitant drug treatment, or other therapy.

Contact the medical/study director in the event the investigator feels that an abnormal test finding should be reported as an AE, although it does not meet any of the above criteria.

Repeating an abnormal test in the absence of any of the above conditions does not constitute an AE. Any abnormal test result that is determined to be an error does not require reporting as an AE.

Evaluation of severity of laboratory abnormalities will be assessed according to the scale outlined in Section 10.2.5.

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10.1.6. Follow-up

Information for any non-SAE that starts during study treatment or within 90 days after the last linvoseltamab dose will be collected from the time of the event until resolution of the event or until the participant's last study visit, whichever comes first.

SAE information will be collected from the time of the event until the event is considered chronic and/or stable or when the participant initiates a new anticancer treatment regimen, whichever comes first. SAEs thought to be related to study treatment may be followed longer.

10.2. Definitions

10.2.1. Adverse Event

An AE is any untoward medical occurrence in a participant administered a study drug that may or may not have a causal relationship with the study drug. Therefore, an AE is any unfavorable and unintended sign (including abnormal laboratory finding), symptom, or disease which is temporally associated with the use of a study drug, whether or not considered related to the study drug (ICH, 1994).

PRO data are generally not reportable as individual AEs and thus will not be reported or reconciled as such.

10.2.2. Serious Adverse Event

An SAE is any untoward medical occurrence that at any dose:

- Results in **death** – includes all deaths, even those that appear to be completely unrelated to study drug (eg, a car accident in which a participant is a passenger).
- Is **life-threatening** – in the view of the investigator, the participant is at immediate risk of death at the time of the event. This does not include an AE that had it occurred in a more severe form might have caused death.
- Requires inpatient **hospitalization** or **prolongation of existing hospitalization** – Inpatient hospitalization is defined as a hospital admission (any duration) or an emergency room visit for longer than 24 hours. Prolongation of existing hospitalization is defined as a hospital stay that is longer than was originally anticipated for the event or is prolonged due to the development of a new AE as determined by the investigator or treating physician.
- Results in persistent or significant **disability/incapacity** (substantial disruption of one's ability to conduct normal life functions).
- Is a **congenital anomaly/birth defect.**
- Is an **important medical event** – Important medical events may not be immediately life-threatening or result in death or hospitalization but may jeopardize the participant or may require intervention to prevent one of the other serious outcomes listed above (eg, intensive treatment in an emergency room or at home for allergic bronchospasm; blood dyscrasias or convulsions that do not result in hospitalization; or development of drug dependency or drug abuse).

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Hospitalization or death due solely to manifestations consistent with typical progression of underlying malignancy will not be considered an SAE.

Criteria for reporting SAEs must be followed for these events.

10.2.3. Adverse Events of Special Interest

An AESI (serious or non-serious) is one of scientific and medical interest specific to the sponsor's product or program for which ongoing monitoring and rapid communication by the investigator to the sponsor can be appropriate. Such an event might warrant further investigation to characterize and understand it.

10.2.4. Infusion Reactions

Infusion reactions are defined in Section 8.3. All infusion reactions must be reported as AEs as described in Section 10.1.1. Depending on the nature of the event, rapid communication by the trial sponsor to other parties (eg, regulators) might also be warranted (Section 10.1.3).

10.2.5. Severity

The severity of AEs, infusion reactions, and test findings classified as AEs will be graded according to the NCI-CTCAE v5.0 grading system. CRS and ICANS are unique toxicities that require special consideration according to the guidance provided in Section 8.3.2 and Section 8.3.3, respectively.

AEs not listed in the NCI-CTCAE v5.0 or specifically described in this protocol will be graded according to the scale provided in Table 17.

Table 17: Grading System for Adverse Events[a]

Grade	Severity	Description
1	Mild	Asymptomatic or mild symptoms; clinical or diagnostic observations only; intervention not indicated
2	Moderate	Minimal, local or noninvasive intervention indicated, limiting age-appropriate instrumental ADL[b]
3	Severe	Severe or medically significant but not immediately life-threatening; hospitalization or prolongation of hospitalization indicated; disabling; limiting self-care ADL[c]
4	Life-Threatening	Life-threatening consequences: urgent intervention indicated
5	Death	Death related to AE

[a] AEs not listed in the NCI-CTCAE v5.0 or not specifically described in this protocol
[b] Instrumental ADL refer to preparing meals, shopping for groceries or clothes, using the telephone, managing money, etc.
[c] Self-care ADL refer to bathing, dressing and undressing, feeding self, using the toilet, taking medications, and not bedridden.
ADL=activities of daily living; AE=adverse event; NCI-CTCAE=National Cancer Institute-Common Terminology Criteria for Adverse Events

10.2.6. Causality

The investigator must provide causality assessment as to whether or not there is a reasonable possibility that the drug caused the AE, based on evidence or facts, the investigator's clinical

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judgment, and the following definitions. The causality assessment must be made based on the available information and can be updated as new information becomes available.

The following factors should be considered when assessing causality:

- Temporal relationship: time to onset vs time drug was administered
- Nature of the reactions: immediate vs long-term
- Clinical and pathological features of the events
- Existing information about the drug and same class of drugs
- Concomitant medications
- Underlying and concurrent illnesses
- Response to dechallenge (drug discontinuation) or dose reduction
- Response to rechallenge (re-introduction of the drug) or dose increase, when applicable
- Participant's medical and social history

Causality to the study drug (including study drug administration):

- Related:
  - The AE follows a reasonable temporal sequence from study drug administration and cannot be reasonably explained by the nature of the reaction, participant's clinical state (eg, disease under study, concurrent diseases, concomitant medications), or other external factors.

  or

  - The AE follows a reasonable temporal sequence from study drug administration and is a known reaction to the drug under study or its class of drugs or is predicted by known pharmacology.
- Not Related:
  - The AE does not follow a reasonable sequence from study drug administration or can be reasonably explained by the nature of the reaction, participant's clinical state (eg, disease under study, concurrent diseases, and concomitant medications) or other external factors.

Causality to the study conduct (protocol-specified procedure):

- Related:
  - The AE follows a reasonable temporal sequence from a protocol-specified procedure and cannot be reasonably explained by the nature of the reaction, participant's clinical state (eg, disease under study, concurrent diseases, concomitant medications), or other external factors.
- Not Related:
  - The AE does not follow a reasonable sequence from a protocol-specified procedure or can be reasonably explained by the nature of the event, participant's clinical state

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(eg, disease under study, concurrent diseases, and concomitant medications), or other external factors.

10.3. Safety Monitoring

The investigator will monitor the safety of the study participant at the investigator's site(s) as per the requirements of this protocol and consistent with current GCP. Any questions or concerns should be discussed with the sponsor in a timely fashion. The sponsor will monitor the safety data from across all study sites. The medical/study director will have primary responsibility for the emerging safety profile of the compound, but will be supported by other departments (eg, Global Patient Safety; Biostatistics and Data Management). Safety monitoring will be performed on an ongoing basis (eg, individual review of SAEs) and on a periodic cumulative aggregate basis.

10.4. Notifying Health Authorities, Institutional Review Board /Ethics Committee, and Investigators

During the study, the sponsor and/or the CRO will inform health authorities, IECs, and the participating investigators of any SUSARs occurring in other study centers or other studies of the active study drug (linvoseltamab), as appropriate per local reporting requirements. In addition, the sponsor and/or CRO will comply with any additional local safety reporting requirements.

Upon receipt of the sponsor's notification of a SUSAR that occurred with the study drug, the investigator will inform the EC unless delegated to the sponsor.

Event expectedness for study drug (linvoseltamab) is assessed against the Reference Safety Information section of the IB that is effective for expedited safety reporting.

At the completion of the study, the sponsor will report all safety observations made during the conduct of the trial in the CSR to health authorities and IECs as appropriate.

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11. STATISTICAL PLAN

Details of data summaries, data conventions, and plans for statistical analyses will be provided in the SAP document for the study.

11.1. Statistical Hypothesis

This is an exploratory study to evaluate the safety and preliminary clinical activity of linvoseltamab, and therefore, no formal statistical hypothesis testing is planned.

The primary endpoint in part 1 is frequency of AEIs and the frequency and severity of TEAEs assessed during the 35-day safety observation period.

The primary endpoint for the expansion portion (part 2) is achievement of CR as determined by the investigator using IMWG criteria.

11.2. Justification of Sample Size

Up to 24 participants will be enrolled during the safety run-in and dose-escalation component (part 1) of the study, and a minimum of 80 participants will be enrolled in the expansion part with the selected dosing regimens for expansion (part 2). These will be combined to reach a total sample size of approximately 104 participants (with a minimum of 80 participants in part 2) treated with the selected dosing regimens and who have at least one post-baseline response assessment.

Results from a minimum of 20 participants per treatment arm, treated with the selected dosing regimen and having at least 1 post-baseline response assessment, can estimate the CR rate with reasonable precision and explore preliminary clinical activity, safety, PK properties, and biomarker changes associated with the dosing regimens.

Table 18 provides 2-sided, 95% binomial exact CIs calculated for different potential achievement of CR that may be observed in 20 participants per treatment arm.

Table 18: The 95% Binomial Exact Confidence Intervals for Possible Observed Complete Response Rates with 20 Participants per Treatment Arm

Number of Participants with Responses (%)	Lower Confidence Limit (%)	Upper Confidence Limit (%)
4 (20.0)	5.7	43.7
6 (30.0)	11.9	54.3
8 (40.0)	19.1	63.9
10 (50.0)	27.2	72.8
12 (60.0)	36.1	80.9
14 (70.0)	45.7	88.1
16 (80.0)	56.3	94.3
18 (90.0)	68.3	98.8
19 (95.0)	75.1	99.9

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11.3. Analysis Sets

11.3.1. Efficacy Analysis Sets

The FAS will include all participants who receive at least 1 dose (any amount) of study drug and had at least 1 post-baseline response assessment. Efficacy endpoints will be analyzed using the FAS. Participants will be classified according to the dosing regimen assigned at study entry.

11.3.2. Safety Analysis Set

The SAF will include all participants who received any amount of study drug. Treatment compliance/administration and all clinical safety variables will be analyzed using the SAF. Participants will be classified according to the dosing schedule assigned at study entry.

11.3.3. Pharmacokinetic Analysis Sets

The PK analysis set is a subset of the FAS comprising participants who have at least 1 non-missing PK result following the first dose of study drug.

11.3.4. Immunogenicity Analysis Sets

The immunogenicity analysis set is a subset of the FAS comprising participants who have at least 1 non-missing ADA result following the first dose of study drug.

11.3.5. Soluble BCMA Analysis Sets

The sBCMA analysis set is a subset of the FAS comprising participants who have an available baseline and at least 1 post-dose sample collected.

11.3.6. Patient-Reported Outcome Analysis Sets

The PRO analysis set is a subset of the FAS comprising participants who have at least 1 non-missing PRO result following the first study dose.

11.3.7. Biomarker Analysis Set

The biomarker analysis set is a subset of the FAS comprising participants who have an available baseline biomarker result.

11.4. Statistical Methods

For continuous variables, descriptive statistics will include the following information: the number of participants reflected in the calculation (n), mean, standard deviation, Q1, median, Q3, minimum, and maximum.

For categorical or ordinal data, frequencies and percentages will be displayed for each category.

11.4.1. Participant Disposition

The number and percentage of participants screened and randomized, and the primary reasons for screening failure and discontinuation will be summarized.

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11.4.2. Demography and Baseline Characteristics

Demographic variables (eg, age, race, and gender), baseline characteristics, medical/oncology history, and prior and concomitant mediations and therapies will be summarized by treatment group. See Section 5.1 for a full list of demographic and baseline variables. No statistical hypothesis tests will be performed on these characteristics.

11.4.3. Efficacy Analyses

11.4.3.1. Primary Efficacy Analysis

The primary endpoint for the expansion portion (part 2) is achievement of CR as determined by the investigator using IMWG criteria (Appendix 1).

The CR rate is defined as the proportion of participants with a best response of CR by IMWG criteria (Kumar, 2016). For the primary analysis, participants with inevaluable responses will be considered as non-responders. The point estimate and exact 95% CI will be provided using the Clopper-Pearson method.

11.4.3.2. Secondary Efficacy Analysis

The MRD negativity rate is defined as the proportion of participants who achieve post-baseline MRD negativity (assessed in bone marrow aspirate at 10^{-5} sensitivity) among participants with CR or better, based on FAS. The MRD negativity rate at 12 months after the first linvoseltamab dose will be summarized.

ORR is defined as the proportion of treated participants who achieve a best overall response of PR or better. The ORR and the corresponding 95% CI will be calculated using the Clopper-Pearson method.

DOR is defined for responders (participants who achieve PR or better) only. It is the time from the date of the first documented response (at least PR) until the date of the first documented biochemical progression or death due to any cause, whichever occurs first. For participants who are alive and have not experienced biochemical progression at the time of analysis, DOR will be censored at their latest available adequate disease assessments.

Biochemical PFS is defined as the time from first dose date until the first date of biochemical progression or death due to any cause, whichever occurs first. For participants who are alive and have not experienced biochemical progression at the time of analysis, biochemical PFS will be censored at their latest available adequate disease assessments.

The Kaplan-Meier curve and estimated median (or an appropriate percentile if the median is not reached) along with its 95% CI will be provided for each of the secondary time-to-event endpoints described above.

11.4.4. Control of Multiplicity

This is not applicable since no formal statistical hypothesis testing is planned.

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11.4.5. Safety Analysis

11.4.5.1. Adverse Events

Definitions

TEAEs are defined as those AEs with either initial onset after the first linvoseltamab dose or that represent an exacerbation of a preexisting condition after the first linvoseltamab dose.

Analysis

The primary analysis of TEAEs will focus on events that occur up to 30 days after the last treatment administration or the day prior to the start of non-protocol cancer therapy, whichever comes first. Supplemental analyses will include all TEAEs up to 90 days after the last treatment administration or the day prior to start of non-protocol cancer therapy. Treatment-related TEAEs will be analyzed separately.

The number (n) and percentage (%) of participants who reported any of the following up to 30 days after last treatment administration will be summarized by dosing regimen:

- TEAEs by SOC and PT
- TEAEs by severity (according to the grading scale outlined in Section 10.2.5), SOC, and PT
- Treatment-emergent AESI
- TEAEs leading to permanent treatment discontinuation
- SAEs, including deaths, by SOC and PT

The number and percentage of patients with treatment-emergent AEI observed during the 35-day safety observation period in part 1, as defined in Section 6.1.3.1, will be summarized by dose level.

11.4.5.2. Other Safety

Vital Signs

Vital signs (temperature, pulse rate, pulse oximetry [on dosing dates], blood pressure, and respiration rate) at baseline and change from baseline to each scheduled assessment time will be summarized with descriptive statistics.

Laboratory Tests

Laboratory test results (blood chemistry, hematology, urinalysis, and other laboratory tests [Section 9.2.3.6]) at baseline and change from baseline at each scheduled assessment time will be summarized with descriptive statistics.

Number and percentage of participants with a PCSV will be tabulated.

11.4.5.3. Treatment Exposure

Treatment duration and relative dose intensity will be summarized.

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11.4.6. Pharmacokinetics

11.4.6.1. Analysis of Drug Concentration Data

The PK parameters may include but are not limited to:

- C_{max}
- C_{eoi}
- C_{trough}

The concentrations of linvoseltamab in serum over time and selected PK parameters will be summarized using descriptive statistics for the purpose of estimating PK exposures in participants with HR-MGUS or NHR-SMM. This descriptive statistical assessment will include the geometric means and ratios of the geometric means for selected PK parameters, as deemed appropriate.

No formal statistical hypothesis testing will be performed.

11.4.7. Analysis of Immunogenicity Data

The immunogenicity variables will be summarized using descriptive statistics. Immunogenicity will be characterized by ADA status, ADA category, and maximum titer observed in participants in the ADA analysis set. For samples confirmed as drug-specific ADA positive but found negative at the lowest titer dilution, the lowest dilution in the titer assay is imputed.

The ADA status of each participant may be classified as one of the following:

- Positive
- Preexisting – If the baseline sample is positive and all post-baseline ADA titers are reported as less than 9-fold the baseline titer value
- Negative – If all samples are found to be negative in the ADA assay

The ADA category of each ADA-positive participant is classified as:

- Treatment-boosted – A positive result at baseline in the ADA assay with at least 1 post-baseline titer result 9-fold the baseline titer value
- Treatment-emergent – A negative result or missing result at baseline with at least 1 positive post-baseline result in the ADA assay. ADA responses that are treatment-emergent will be further subcategorized as follows:
  - Persistent – A positive result in the ADA assay detected in at least 2 consecutive post-baseline samples separated by at least a 16-week post-baseline period (based on normal sampling time) with no ADA-negative results in-between regardless of any missing samples
  - Transient – Not persistent or indeterminate regardless of any missing samples
  - Indeterminate – A positive result in the ADA assay at the last collection time point only regardless of any missing samples

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The maximum titer category of each participant is classified as:

- Low (titer <1,000)
- Moderate (1,000 ≤ titer ≤10,000)
- High (titer > 10,000)

The following will be summarized by treatment group and ADA titer level:

- Number (n) and percent (%) of ADA-negative participants
- Number (n) and percent (%) of preexisting participants
- Number (n) and percent (%) of treatment-emergent ADA-positive participants
  - Number (n) and percent (%) of persistent treatment-emergent ADA-positive participants
  - Number (n) and percent (%) if indeterminate treatment-emergent ADA-positive participants
  - Number (n) and percent (%) of transient treatment-emergent ADA positive participants
  - Number (n) and percent (%) of treatment-boosted ADA-positive participants

A listing of all ADA titer levels will be provided for participants with preexisting, treatment-emergent, and treatment-boosted ADA responses.

11.4.7.1. Immunogenicity with Exposure

Potential association between immunogenicity and systemic exposure to linvoseltamab will be explored by treatment groups. Plots of individual linvoseltamab concentration time profiles may be provided to examine the potential impact of ADA category and maximum titer category on these profiles.

11.4.7.2. Immunogenicity, Safety, and Efficacy

Potential association between immunogenicity variables and safety may be explored with a primary focus on the following safety events during the TEAE period:

- Infusion reactions
- Hypersensitivity (SMQ: Hypersensitivity [Narrow])
- Anaphylaxis (SMQ: Anaphylactic Reaction [Narrow])

Potential association between immunogenicity variables and efficacy endpoints may be explored (eg, scatter plot or spaghetti plot).

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The safety and efficacy analyses mentioned above will be conducted using the following categories:

- ADA Positive
  - Treatment-emergent
  - Treatment-boosted
- Maximum post-baseline titer category

11.4.8. Analysis of Pharmacodynamic and Exploratory Biomarker Data

The following variables will be explored for their relationship with IMWG response:

- Analysis of pharmacodynamic variables include the change in peripheral-blood major T-cell subsets (CD3, CD4, CD8) over time.
- Baseline BCMA expression detected by IHC on the bone marrow core biopsy will be analyzed descriptively as a continuous variable using intensity of staining in a percentage of cells as well as using the H score of cytoplasmic and membrane staining.
- Analysis of sBCMA in serum at baseline (by medians and quantiles such as tertiles) will be analyzed descriptively as subgroups relating to IMWG response. The concentrations of sBCMA in serum over time will be summarized by descriptive statistics.
- Analysis of baseline bone marrow T-cell percentages and major subsets (CD3, CD4, CD8) will be described.

Additional exploratory biomarkers such as changes in circulating B cells, monocytes, and NK cells may be described in a separate report.

11.4.9. Analysis of Patient-Reported Outcome Data

For each treatment group, PRO data, as measured by the validated instruments EORTC QLQ-MY20, EORTC QLQ-C30, EQ-5D-5L, and FACIT Item GP5 will be described, including mean scores at each assessment and change from baseline scores over time.

Additional analyses of PRO data will be performed. Time to deterioration in global health status/QoL (per EORTC QLQ-C30) will be summarized using the Kaplan-Meier method and will include the median and 95% CI. Longitudinal mixed-effects model analyses will be used to assess change from baseline in global health status/QoL. The model will include time point as a fixed-effect categorical factor, the baseline PRO score as a fixed-effect covariate, and the baseline vs time interaction. Evaluation on general health status, patient-reported function, and symptoms (per EORTC QLQ-C30 and QLQ-MY20) regarding time to deterioration and change from baseline will be summarized using methods mentioned above. Further details regarding these analyses will be included in the SAP. Missing items will be handled per the scoring manuals of each questionnaire administered.

PRO instruments will be scored according to their respective scoring manuals and user's guides. Participant disposition and PRO completion rates will be summarized by visit.

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Summary statistics of absolute scores and change from baseline by visit will be reported for each scale/subscale of all PRO instruments. Line charts depicting the mean and mean change from baseline will be provided.

11.5. Interim Analysis

No interim analysis will be performed.

11.6. Statistical Considerations Related to Premature Study Termination

If the study is terminated prematurely, only those parameters required for the development program and/or reporting to regulatory authorities will be summarized. Investigator and sponsor responsibilities surrounding the premature termination of a study are presented in Section 15.1.

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12. QUALITY CONTROL AND QUALITY ASSURANCE

In accordance with ICH E6, the sponsor is responsible for quality assurance to ensure that the study is conducted, and the data generated, recorded, and reported in compliance with the protocol, GCP, and any applicable regulatory requirement(s). The planned quality assurance and quality control procedures for the study are described in this section.

12.1. Data Management and Electronic Systems

12.1.1. Data Management

A data management plan specifying all relevant aspects of data processing for the study (including data validation [quality checking], cleaning, correcting, and releasing) will be maintained and stored at Regeneron (sponsor).

AEs and medical history will be coded using MedDRA. A medical coding plan will specify the processes and the dictionary used for coding. All data coding (eg, AEs, baseline findings, medication, medical/oncology/surgical history) will be done using internationally recognized and accepted dictionaries.

The CRF data for this study will be collected with an EDC tool.

12.1.2. Electronic Systems

Electronic systems that may be used to process and/or collect data in this study will include the following:

- Interactive Response Technology participant assignment, study drug supply
- EDC system – data capture – Medidata Rave
- Electronic Clinical Outcome Assessment Consortium (eCOA) for collection of PRO data
- SAS – statistical review and analysis
- Pharmacovigilance safety database

12.2. Study Monitoring

12.2.1. Monitoring of Study Sites

Regeneron uses a study-specific risk-based approach to study monitoring and oversight, aligned with risk-based quality principles, outlined in ICH E6 (R2) Guideline for Good Clinical Practice. RBQM methodology focuses on employing a fit-for-purpose monitoring strategy, supported either directly by Regeneron as sponsor or via our CRO partners. RBQM strategies may include reduced SDV, targeted SDR, the use of onsite and remote monitoring visits (as allowed by local laws), and centralized monitoring to identify site level risks and study level trends. The investigator must allow study-related monitoring activities to occur.

The study monitors may perform targeted SDR to verify that data recorded in the CRF by authorized site personnel are accurate and verifiable from source documents, that the safety and rights of participants are being protected, and that the study is being conducted in accordance with

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the current approved protocol version and any other study agreements, ICH GCP, and all applicable regulatory requirements.

12.2.2. Source Document Requirements

Investigators are required to prepare and maintain adequate and accurate participant records (source documents). The site is responsible to ensure quality within their records and systems and are accountable for ensuring that all source data and CRF data are timely, accurate, and complete.

The investigator must keep all source documents on file with the CRF. (Throughout this protocol, CRF refers to either a paper or electronic CRF.) CRFs and source documents must be always available for inspection by authorized representatives of the sponsor and regulatory authorities.

12.2.3. Case Report Form Requirements

Study data obtained during the clinical study will be recorded on CRFs within the EDC system by trained site personnel. All required CRFs must be completed for each participant enrolled in the study. The investigator must ensure the accuracy, completeness, and timeliness of the data reported to the sponsor in the CRFs. After review of the clinical data for each participant, the investigator must provide an electronic signature. A copy of each participant CRF casebook is to be retained by the investigator as part of the study record and must always be available for inspection by authorized representatives of the sponsor and regulatory authorities.

A paper CRF for each participant enrolled in the study must be completed and signed by the study investigator or authorized designee. The CRF must be typed or filled out using indelible ink. The writing must be legible. Errors should be crossed out but not obliterated, the correction inserted, and the change initialed and dated by the investigator or authorized designee. The investigator must ensure the accuracy, completeness, legibility, and timeliness of the data reported to the sponsor in the CRFs. CRFs must be always available for inspection by authorized representatives of the sponsor and regulatory authorities.

Corrections to the CRF will be entered into the CRF by the investigator or an authorized designee. All changes, including date and person performing corrections, will be available via the audit trail, which is part of the EDC system. For corrections made via data queries, a reason for any alteration must be provided.

12.3. Audits and Inspections

This study may be subject to a quality assurance audit or inspection by the sponsor or regulatory authorities. Should this occur, the investigator is responsible for:

- Informing the sponsor of a planned inspection by the authorities as soon as notification is received, and authorizing the sponsor’s participation in the inspection
- Providing access to all necessary facilities, study data, and documents for the inspection or audit
- Communicating any information arising from inspection by the regulatory authorities to the sponsor immediately
- Taking all appropriate measures requested by the sponsor to resolve the problems found during the audit or inspection

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Documents subject to audit or inspection include but are not limited to all source documents, CRFs, medical records, correspondence, ICFs, EC files, documentation of certification and quality control of supporting laboratories, and records relevant to the study maintained in any supporting pharmacy facilities. Conditions of study material storage are also subject to inspection. In addition, representatives of the sponsor may observe the conduct of any aspect of the clinical study or its supporting activities both within and outside of the investigator's institution.

In all instances, the confidentiality of the data must be respected.

12.4. Study Documentation

12.4.1. Certification of Accuracy of Data

A declaration assuring the accuracy and content of the data recorded on the CRF (electronic or paper) must be signed electronically by the investigator. This signed declaration accompanies each set of participant's final CRFs that will be provided to the sponsor.

12.4.2. Retention of Records

The investigator must retain all essential study documents, including ICFs, source documents, investigator copies of CRFs, and drug accountability records for at least 15 years following the completion or discontinuation of the study, or longer, if a longer period is required by relevant regulatory authorities. The investigator must obtain written approval from the sponsor before discarding or destroying any essential study documents during the retention period following study completion or discontinuation. Records must be destroyed in a manner that ensures confidentiality.

If the investigator's personal situation is such that archiving can no longer be ensured, the investigator must inform the sponsor (written notification), and the relevant records will be transferred to a mutually agreed-upon destination.

12.4.3. Recruitment Strategy

Potential study participants may be identified by the investigator through a variety of means, such as publicizing the trial or using existing patient lists. Recruitment resources may include but are not limited to recruitment flyers, brochures, social media ads, newspaper ads, radio ads, etc. A third-party vendor may assist in recruitment efforts, such as the development of recruitment materials. All patient-facing recruitment material, including media advertising and receptionist scripts, will be reviewed and approved by the appropriate authority prior to use. Resources may be in paper or electronic form. Regeneron will not access any patient-identifiable information as part of recruitment efforts.

12.4.4. Data Protection

Participants will be assigned a unique identifier by the sponsor. Any participant records or datasets that are transferred to the sponsor will contain the identifier only; participant names or any information that would make the participant identifiable will not be transferred.

The participant must be informed that their personal study-related data will be used by the sponsor in accordance with local data protection law. The level of disclosure must also be explained to the

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participant who will be required to give consent for their data to be used as described in the informed consent.

The participant must be informed that their medical records may be examined by Clinical Quality Assurance auditors or other authorized personnel appointed by the sponsor, by appropriate IEC members, and by inspectors from regulatory authorities.

The contract between sponsor and study sites specifies responsibilities of the parties' related data protection, including handling of data security breaches and respective communication and cooperation of the parties.

Information technology systems used to collect, process, and store study-related data are secured by technical and organizational security measures designed to protect such data against accidental or unlawful loss, alteration, or unauthorized disclosure or access.

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13. ETHICAL AND REGULATORY CONSIDERATIONS

13.1. Good Clinical Practice Statement

It is the responsibility of both the sponsor and the investigator(s) to ensure that this clinical study will be conducted in accordance with the ethical principles that have their origin in the Declaration of Helsinki, and that are consistent with the ICH guidelines for GCP and applicable regulatory requirements.

13.2. Informed Consent

The principles of informed consent are described in ICH guidelines for GCP.

The ICF used by the investigator must be reviewed and approved by the sponsor prior to submission to the appropriate EC. A copy of the EC-approved ICF and documentation of approval must be provided to the sponsor before study drug will be shipped to the study site.

It is the responsibility of the investigator or designee (if acceptable by local regulations) to obtain written informed consent from each participant prior to study participation and after the aims, methods, objectives, and potential hazards of the study have been explained to the participant in language that the participant can understand. The ICF should be signed and dated by the participant and by the investigator or authorized designee who reviewed the ICF with the participant.

- Participants who can write but cannot read will have the ICF read to them before signing and dating the ICF.
- Participants who can understand but who can neither write nor read will have the ICF read to them in the presence of an impartial witness, who will sign and date the ICF to confirm that informed consent was given.

The original ICF must be retained by the investigator as part of the participant’s study record, and a copy of the signed ICF must be given to the participant.

If new safety information results in significant changes in the risk/benefit assessment, or if there are significant changes to the study procedures, the ICF must be reviewed and updated appropriately. All study participants must be informed of the new information and provide their written consent if they wish to continue in the study. The original signed revised ICF must be maintained in the participant’s study record, and a copy must be given to the participant.

13.3. Participant Confidentiality and Data Protection

The investigator must take all appropriate measures to ensure that the anonymity of each study participant will be maintained. Participants should be identified by a participant identification number only on CRFs or other documents submitted to the sponsor. Documents that will not be submitted to the sponsor (eg, signed ICF) must be kept in strict confidence.

The participant's and investigator's personal data, which may be included in the sponsor database, will be treated in compliance with all applicable laws and regulations. The sponsor shall take all appropriate measures to safeguard and prevent access to this data by any unauthorized third party.

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13.4. Ethics Committee

An appropriately constituted EC, as described in ICH guidelines for GCP, must review and approve:

- The protocol, ICF, and any other materials to be provided to the participants (eg, advertising) before any participant may be enrolled in the study
- Any amendment or modification to the study protocol or ICF before implementation, unless the change is necessary to eliminate an immediate hazard to the participant, in which case the EC should be informed as soon as possible
- Ongoing studies on an annual basis or at intervals appropriate to the degree of risk

In addition, the EC should be informed of any event likely to affect the safety of participants or the continued conduct of the clinical study.

A copy of the EC approval letter with a current list of the EC members and their functions must be received by the sponsor prior to shipment of drug supplies to the investigator. The approval letter should include the study number and title, the documents reviewed, and the date of the review.

Records of the EC review and approval of all study documents (including approval of ongoing studies) must be kept on file by the investigator.

13.5. Clinical Study Data Transparency

Final study results will be published on public clinical trial registries according to applicable local guidelines and regulations. For data integrity, scientific, and statistical reasons, published results from all participants will be disclosed following End of Study (Section 6.1.4).

Treatment codes will be disseminated to each investigation site thereafter.

For purposes of data disclosure, interim analyses will be available for disclosure when required by local regulations. If the integrity of the ongoing study cannot be ensured or study blinding cannot be maintained, then only final results will be disclosed.

14. PROTOCOL AMENDMENTS

The sponsor may not implement a change in the design of the protocol or ICF without an EC-approved amendment. Where required per local legislation, regulatory authority approval will also be sought.

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15. PREMATURE TERMINATION OF THE STUDY OR CLOSE-OUT OF A SITE

15.1. Premature Termination of the Study

The sponsor has the right to terminate the study prematurely. Reasons may include efficacy, safety, or futility, among others. Should the sponsor decide to terminate the study, the investigator(s) will be notified in writing.

15.2. Close-out of a Site

The sponsor and the investigator have the right to close-out a site prematurely.

Investigator's Decision

The investigator must notify the sponsor of a desire to close-out a site in writing, providing at least 30 days' notice. The final decision should be made through mutual agreement with the sponsor. Both parties will arrange the close-out procedures after review and consultation.

Sponsor's Decision

The sponsor will notify the investigator(s) of a decision to close-out a study site in writing. Reasons may include the following, among others:

- The investigator has received all items and information necessary to perform the study, but has not enrolled any participant within a reasonable period
- The investigator has violated any fundamental obligation in the study agreement, including but not limited to, breach of this protocol (and any applicable amendments), breach of the applicable laws and regulations, or breach of any applicable ICH guidelines
- The total number of participants required for the study are enrolled earlier than expected

In all cases, the appropriate EC and Health Authorities must be informed according to applicable regulatory requirements, and adequate consideration must be given to the protection of the participants' interests.

16. CONFIDENTIALITY

Confidentiality of information is provided as a separate agreement.

17. FINANCING AND INSURANCE

Financing and insurance information is provided as a separate agreement.

18. PUBLICATION POLICY

Publication rights and procedures will be outlined in a separate clinical study agreement.

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19. REFERENCES

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20. INVESTIGATOR'S AGREEMENT

I have read the attached protocol: "Phase 2 Dose-Ranging Interception Study of Linvoseltamab in Patients with High-Risk Monoclonal Gammopathy of Undetermined Significance or Non-High-Risk Smoldering Multiple Myeloma" and agree to abide by all provisions set forth therein.

I agree to comply with the current International Council for Harmonisation Guideline for Good Clinical Practice and the laws, rules, regulations, and guidelines of the community, country, state, or locality relating to the conduct of the clinical study.

I also agree that persons debarred from conducting or working on clinical studies by any court or regulatory agency will not be allowed to conduct or work on studies for the sponsor or a partnership in which the sponsor is involved. I will immediately disclose it in writing to the sponsor if any person who is involved in the study is debarred, or if any proceeding for debarment is pending, or, to the best of my knowledge, threatened.

This document contains confidential information of the sponsor, which must not be disclosed to anyone other than the recipient study staff and members of the EC. I agree to ensure that this information will not be used for any purpose other than the evaluation or conduct of the clinical investigation without the prior written consent of the sponsor.

_____ _____

(Signature of Investigator) (Date)

_____

(Printed Name)

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APPENDIX 1. INTERNATIONAL MYELOMA WORKING GROUP RESPONSE CRITERIA

Category	Criteria
sCR	• CR as defined below, **plus** • Normal FLC ratio, **and** • Absence of clonal plasma cells in bone marrow by immunohistochemistry or 2- to 4-color flow cytometry[a]
CR[b]	• Negative immunofixation of the serum and urine[c], **and** • Disappearance of any soft tissue plasmacytomas, **and** • <5% plasma cells in bone marrow
VGPR[b]	• Serum and urine M-protein detectable by immunofixation but not on electrophoresis, **or** • ≥90% reduction in serum M-protein plus urine M-protein level <100 mg/24 hours • In addition to the above criteria, if present at baseline, a ≥90% reduction in size of soft tissue plasmacytomas compared with baseline
PR	• ≥50% reduction of serum M-protein plus reduction in urine M-protein by ≥90% or to <200 mg/24 hours • If serum and urine M-protein are not measurable, a ≥50% decrease in the difference between involved and uninvolved FLC levels is required in place of the M-protein criteria • If serum and urine M-protein are not measurable, and serum free light assay is also not measurable, ≥50% reduction in plasma cells is required in place of M-protein, provided baseline bone marrow plasma cell percentage was ≥30% • In addition to the above criteria, if present at baseline, a ≥50% reduction in the size of soft tissue plasmacytomas is also required
MR	• ≥25% to ≤49% reduction of serum M-protein **and** reduction in 24-hour urine M-protein by 50 to 89% • In addition to the above criteria, if present at baseline, a ≥25% to ≤49% reduction in the size of soft tissue plasmacytomas is also required
SD	• Not meeting criteria for CR, VGPR, PR, or PD
PD	• Increase of 25% from lowest confirmed response value in one or more of the following: – Serum M-protein (absolute increase must be ≥0.5 g/dL)[d] – Urine M-protein (absolute increase must be ≥200 mg/24 hours) – Only in patients without measurable serum and urine M-protein levels, the difference between involved and uninvolved FLC levels (absolute increase must be >10 mg/dL) – Only in patients without measurable serum and urine M-protein levels and without measurable involved FLC levels, BMPC percentage (absolute increase must be ≥10%) • Appearance of new osteolytic bone lesions or soft tissue plasmacytomas

BMPC=bone marrow plasma cell; CR=complete response; FLC=free light chain; IHC=immunohistochemistry; IMWG=International Myeloma Working Group; MR=minimal response; PD=progressive disease; PR=partial response; sCR=stringent complete response; SD=stable disease; VGPR=very good partial response

[a] Presence or absence of clonal plasma cells is based upon the kappa/lambda ratio. An abnormal kappa/lambda ratio by IHC or flow cytometry requires a minimum of 100 plasma cells for analysis. An abnormal ratio reflecting the presence of an aberrant clone is a kappa/lambda ratio of >4:1 or <1:2.

[b] Clarifications to IMWG criteria for coding CR and VGPR in subjects in whom the only measurable disease is by serum FLC levels: CR in such subjects indicates a normal FLC ratio of 0.26 to 1.65 in addition to CR criteria listed above. VGPR in such subjects requires a ≥90% decrease in the difference between involved and uninvolved FLC levels.

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[c] No evidence of residual myeloma-associated monoclonal protein isotype(s) on immunofixation of the serum and urine. New positive immunofixation results after achieving CR, when the isotype is different, can be secondary to therapeutic monoclonal antibodies or monoclonal/oligoclonal immune reconstitution and should be carefully reviewed to prevent confusion with relapse.
[d] M-protein increase of ≥1 g/dL is sufficient to define progression if baseline M-protein is ≥5 g/dL.

Notes:

- All response categories require two consecutive assessments of the applicable disease parameter (serum M-protein, urine M-protein, or serum FLC), performed at any time (no minimum interval is required, it can be done the same day); however, to confirm response or progressive disease, two discrete samples are required; testing cannot be based upon the splitting of a single sample. Radiographic and bone marrow assessments need not be confirmed.
- VGPR and CR categories require serum and urine studies regardless of whether disease at baseline was measurable on serum, urine, both, or neither.
- Patients will be considered to have progressive disease if they meet the criteria for progression by a variable that was not considered measurable at baseline; however, for patients who had a measurable serum or urine M-spike at baseline, progression cannot be defined by increases in serum FLC alone.
- Serum FLC levels should only be used for response assessment when both the serum and urine M-component levels are deemed not measurable at baseline.
- For IgA myeloma, quantitative immunoglobulin measurements are preferred for disease assessments; the same percentage changes apply as for serum M-spike.
- The lowest confirmed value before suspected progression will be used as baseline for calculation of progression; if a serum and/or urine spike is considered too low to quantitate, this value can be assigned as zero as a baseline for documentation of subsequent progressive disease.

Source: Adapted from (Kumar, 2016) and (Rajkumar, 2011)

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APPENDIX 2. SLIM CRAB DIAGNOSTIC CRITERIA FOR A MYELOMA-DEFINING EVENT

Multiple Myeloma-Defining Biomarkers	
S	Greater than or equal to Sixty percent clonal plasma cells in the bone marrow
Li	Involved/uninvolved free Light chain ratio of ≥100 with the involved FLC being ≥100 mg/L
M	MRI with > 1 focal marrow lesion (at least 5 mm in size)
End Organ Damage	
C	HyperCalcemia (serum calcium >1 mg/dL above ULN or >11 mg/dL)
R	Renal insufficiency (creatinine >2 mg/dL or creatinine clearance <40 mL/min) (Appendix 4)
A	Anemia (hemoglobin >2 g/dL below LLN or <10 g/dL)
B	Presence of lytic Bone lesions (presence of one or more osteolytic lesions seen on skeletal radiography, whole-body MRI, or whole-body FDG-PET/CT)

Source: (Rajkumar, 2014) (The Myeloma Clinical Trials, 2021)
FDG-PET/CT=^{18}F-fluorodeoxyglucose positron emission tomography/computed tomography; FLC=free light chain; LLN=lower limit of normal; MRI=magnetic resonance imaging; ULN=upper limit of normal

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APPENDIX 3. CHANGE FROM BASELINE IN SCALE SCORES FOR EORTC QLQ-C30 AND EORTC QLQ-MY20

Change from Baseline and Visit Responses for EORTC QLQ-C30

EORTC QLQ-C30 Scales	Clinically Meaningful Threshold	Sensitivity Threshold	Visit Response
Symptom scales	≤ -10	≤ -sensitivity threshold	Improved
	(-10, 10)	(-sensitivity threshold, sensitivity threshold)	Stable
	≥ +10	≥ +sensitivity threshold	Worsened
Functional scales and global health status	≥ +10	≥ +sensitivity threshold	Improved
	(-10, 10)	(-sensitivity threshold, sensitivity threshold)	Stable
	≤ -10	≤ -sensitivity threshold	Worsened

EORTC QLQ-C30=European Organization for Research and Treatment of Cancer Quality of Life Questionnaire – Core 30

Change from Baseline and Visit Responses for EORTC QLQ-MY20

EORTC QLQ-MY20 Scales	Clinically Meaningful Threshold	Sensitivity Threshold	Visit Response
Disease symptoms scale	≤ -10	≤ -sensitivity threshold	Improved
	(-10, 10)	(-sensitivity threshold, sensitivity threshold)	Stable
	≥ +10	≥ +sensitivity threshold	Worsened
Side effects of treatment scale	≥ +10	≤ -sensitivity threshold	Improved
	(-10, 10)	(-sensitivity threshold, sensitivity threshold)	Stable
	≤ -10	≥ +sensitivity threshold	Worsened
Future perspective scale	≤ -9	≤ -sensitivity threshold	Improved
	(-9, 9)	(-sensitivity threshold, sensitivity threshold)	Stable
	≥ +9	≥ +sensitivity threshold	Worsened
Body image scale	≥ +13	≤ -sensitivity threshold	Improved
	(-13, 13)	(-sensitivity threshold, sensitivity threshold)	Stable
	≤ -13	≥ +sensitivity threshold	Worsened

EORTC QLQ-MY20=European Organization for Research and Treatment of Cancer Quality of Life Questionnaire – Multiple Myeloma Module 20

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APPENDIX 4. ESTIMATED GLOMERULAR FILTRATION RATE

Modification of Diet in Renal Disease (MDRD) Formula

Using standardized S_{Cr} in **mg/dL**, the glomerular filtration rate (GFR) can be estimated using the following MDRD equation:

$$\text{GFR (mL/min/1.73 m}^2) = 175 \times (S_{Cr})^{-1.154} \times (\text{age})^{-0.203} \times (1.212 \text{ if black}) \times (0.742 \text{ if female})$$

If the serum creatinine is obtained using the International System of Units (SI) (ie, micromol/L), use the following formula to convert SI units to conventional (mg/dL) units:

$$\frac{\text{serum creatinine (micromol/L)}}{88.4} = \text{serum creatinine (mg/dL)}$$

Source: (Levey, 2006)

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APPENDIX 5. IMMUNE EFFECTOR CELL–ASSOCIATED ENCEPHALOPATHY SCORE TO ASSESS FOR IMMUNE EFFECTOR CELL–ASSOCIATED NEUROTOXICITY SYNDROME

This CARTOX-10–based assessment tool, termed the ICE score, grades neurotoxicity by evaluating alterations in speech, orientation, handwriting, and concentration that are highly suggestive of the encephalopathy observed in participants with ICANS (Lee, 2019). The ICE score is derived as follows:

- Orientation: orientation to year, month, city, hospital: 4 points
- Naming: ability to name 3 objects (eg, point to clock, pen, button): 3 points
- Following commands: ability to follow simple commands (eg, "Show me 2 fingers" or "Close your eyes and stick out your tongue"): 1 point
- Writing: ability to write a standard sentence (eg, "Our national bird is the bald eagle"): 1 point
- Attention: ability to count backwards from 100 by 10: 1 point.

Total ICE score corresponds to the following ICANS grading:

- 10, no impairment
- 7 to 9, grade 1 ICANS
- 3 to 6, grade 2 ICANS
- 0 to 2, grade 3 ICANS
- 0 due to participant unarousable and unable to perform ICE assessment, grade 4 ICANS

See Section 8.3.3 for management of ICANS.

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APPENDIX 6. LINVOSELTAMAB MODIFIED FULL DOSE LEVELS

Modified full dose levels will be enrolled if emerging safety issues warrant a smaller dose increment following a completed dose level.

Dose Level	Linvoseltamab	Dose Increment Percentage
DL0	25 mg	
DL0a	35 mg	DL0 +40%
DL1	50 mg	DL0 +100%
DL1a	65 mg	DL1 +30%
DL1b	75 mg	DL1 +50%
DL1c	85 mg	DL1a +30%
DL2	100 mg	DL1 +100%

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SIGNATURE OF SPONSOR'S RESPONSIBLE OFFICERS

(Medical/Study Director, Regulatory Representative, Clinical Study Lead, and Biostatistician)

To the best of my knowledge, this report accurately describes the planned conduct of the study.

Study Title: Phase 2 dose-ranging and interception study of linvoseltamab in patients with high-risk monoclonal gammopathy of undetermined significance or non-high-risk smoldering multiple myeloma

Protocol Number: R5458-ONC-2257 Original

See appended electronic signature page

Sponsor's Responsible Medical/Study Director

See appended electronic signature page

Sponsor's Responsible Regulatory Liaison

See appended electronic signature page

Sponsor's Responsible Clinical Study Lead

See appended electronic signature page

Sponsor's Responsible Biostatistician

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Approval/eSignature	Onyekachukwu Ogbonnaya onyekachukwu.ogbonn@regeneron.com Clinical 07-Aug-2023 20:30:48 GMT+0000

Approval/eSignature	Arijit Sinha arijit.sinha@regeneron.com Biostatistics / Data Management 08-Aug-2023 07:53:10 GMT+0000

Approval/eSignature	Gianina Meltzer gianina.meltzer@regeneron.com Regulatory 08-Aug-2023 13:09:10 GMT+0000

Approval/eSignature	Shawn Sarkaria shawn.sarkaria@regeneron.com Clinical 08-Aug-2023 14:27:08 GMT+0000

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